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DEPARTMENT OF
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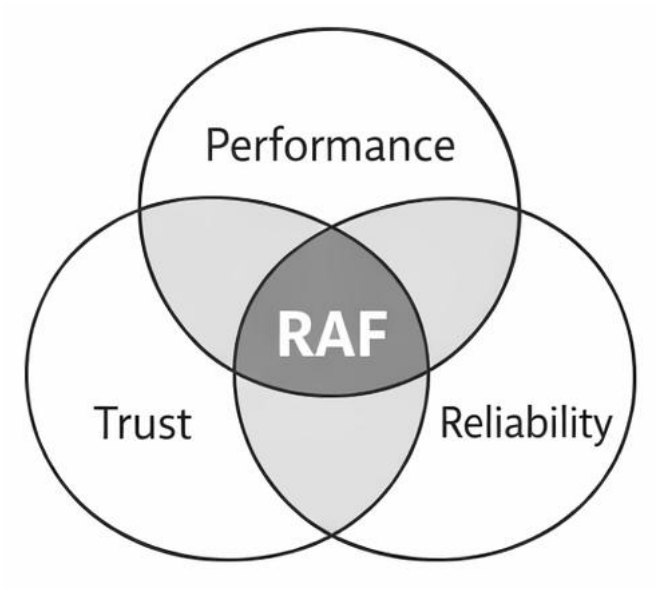
RAF: A Modular, Multi-Layered Reliability Assessment Framework for AI Trustworthiness in Safety-Critical Contexts

Integrating Predictive Performance, System Reliability, and End-User Trust from a Human-Centered Perspective



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Abstract

The safe deployment of AI in safety-critical domains demands more than high predictive performance—it requires systematic, transparent, and actionable reliability assessment. This thesis introduces the RAF—a modular, multi-layered reliability assessment framework that addresses three critical dimensions, or pivot points, for safe and effective AI deployment: predictive performance, system reliability, and end-user perceived trust—and integrates them into a unified approach to AI trustworthiness. Using radar-based heartbeat detection as a controlled case study, the RAF operationalizes deep learning-based modeling, multi-source uncertainty quantification, and reliability-aware system design to examine how these pivot points can be jointly addressed within a single system architecture.

Experimental evaluation using deep learning models on synthetic radar data showed that while pointwise accuracy remained high due to class imbalance, event-based metrics such as precision, recall, and F1-score were low. Although the RAF's applied reliability scoring and threshold-based flagging mechanisms did not improve event-detection performance in this instantiation, they enabled principled identification and communication of low-confidence predictions without degrading baseline behavior. Together with the theoretical analysis, these results demonstrated the feasibility of integrating reliability assessment as an explicit system layer, highlighted the limitations of performance-only evaluation, and established the RAF as a structured foundation for risk-aware and transparent AI system design, with ethical implications for safety-critical contexts.

Keywords

reliability assessment, trustworthy AI, safety-critical applications, uncertainty quantification, human-centered AI, deep learning

RAF: Ett Modulärt, Flerskiktat Ramverk för AI-Trustworthiness i Säkerhetskritiska Sammanhang

Med Fokus på Prediktiv Prestanda, Systemtillförlitlighet och Användartillit ur ett Människocentrerat Perspektiv

Sammanfattning

En säker användning av AI i säkerhetskritiska kontexter kräver mer än hög prediktiv prestanda—den förutsätter systematisk, transparent och handlingsbar tillförlitlighetsbedömning. Denna uppsats introducerar RAF (Reliability Assessment Framework), ett modulärt och flerskiktat ramverk för tillförlitlighetsbedömning som adresserar tre kritiska dimensioner (pivot points) för säker och effektiv AI-implementering: prediktiv prestanda, systemtillförlitlighet och användarupplevd tillit. Dessa dimensioner integreras i ett samlat angreppssätt för AI-trustworthiness. Med radarbaserad hjärtslagsdetektion som kontrollerad fallstudie operationaliserar RAF osäkerhetskvantifiering och tillförlitlighetsmedveten systemdesign för att undersöka hur dessa pivot points kan hanteras gemensamt inom en och samma systemarkitektur.

Experimentell utvärdering med hjälp av deep learning och syntetisk radardata visade att pointwise accuracy förblev hög till följd av kraftig klassobalans i datan, medan event-baserade mått såsom precision, recall och F1-score var låga. Även om RAF:s tillämpade mekanismer för tillförlitlighetsbedömning och tröskelbaserad flaggning inte förbättrade detektionsprestandan i denna instans, möjliggjorde de en principiell identifiering och kommunikation av låg-tillförlitliga prediktioner utan att försämra baseline-beteende. Tillsammans med den teoretiska analysen visade resultaten att tillförlitlighetsbedömning kan integreras som ett explicit systemlager, belyste begränsningarna med prestandabaserad utvärdering i isolering och etablerade RAF som en strukturerad grund för riskmedveten och transparent AI-systemdesign med etiska implikationer för säkerhetskritiska sammanhang.

Nyckelord

tillförlitlighetsbedömning, trustworthy AI, säkerhetskritiska tillämpningar, osäkerhetskvantifiering, människocentrerad AI, deep learning

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1 Introduction

1.1 Context

This thesis addresses a fundamental challenge in contemporary AI system development: the absence of integrated, rigorous, systematic and human-centered reliability assessment. This gap is substantiated in the following sections through evidence from both technical reliability research and human-centered AI literature. While advances in machine learning have yielded remarkable gains in predictive performance across a wide range of applications, successful real-world deployment depends on more than statistical performance alone. In high-stakes operational contexts, AI systems must not only perform well under controlled conditions, but also operate reliably, transparently, and safely when confronted with uncertainty, variability, and evolving environments.

A key motivation for this work is the recognition that optimizing for technical performance alone is insufficient for safe and effective deployment, as machine learning systems must also support appropriate human understanding, verification, and trust (Amershi et al., 2019; Doshi-Velez & Kim, 2017). Trust and human-centered design are essential for successful AI deployment, as systems that support transparency, human control, and alignment with human goals are more likely to achieve acceptance and effective (Shneiderman, 2022). At the same time, high-profile failures of opaque and poorly governed AI systems have demonstrated how technical shortcomings can translate into real-world harm and unaccountable decision-making (O'Neil, 2016). Historically, periods of declining institutional confidence and reduced funding for AI research—commonly referred to as "AI winters" (Hendler, 2008)—have reflected broader shifts in expectations, perceived progress, and confidence in the field. These historical patterns illustrate that sustained progress in AI depends not only on technical capability, but also on maintaining confidence among institutions and the users and stakeholders who interact with these systems. Together, these findings highlight the importance of approaches that address reliability, interpretability, and trustworthiness as essential factors for ensuring continued adoption and long-term support of AI systems. In contemporary contexts, the increasing visibility and deployment of AI in public-facing applications, combined with documented cases of unreliable or misleading system behavior, raises the possibility that erosion of user and public trust could influence broader institutional and societal support for AI, extending beyond the primarily institutional and funding-driven factors associated with previous AI winters.

These dynamics are particularly consequential in safety-critical domains, where unreliable predictions may carry material, ethical, or regulatory consequences. Ensuring reliability is therefore not merely a technical objective but a structural requirement for sustainable and responsible AI deployment. Reliability, in this thesis, is understood not simply as high and stable predictive accuracy under optimal conditions, but as the ability of a system to produce safe, robust, and well-signaled outputs under both expected and unexpected operating conditions.

Despite substantial research on robustness, uncertainty estimation, auditing, and explainability, existing approaches often address isolated aspects of reliability rather than integrating them within a unified, system-level architecture. This fragmentation motivates the need for a structured framework capable of systematically coordinating technical safeguards, governance mechanisms, and human-centered interaction principles.

To address this gap, this thesis proposes a reliability assessment framework (the RAF) designed to operationalize reliability as a multi-dimensional and lifecycle-spanning property. The framework is grounded in three critical pivot points for AI system viability: predictive performance, system-level trustworthiness, and end-user trust. Together, these dimensions define the conditions under which AI

systems can be considered technically effective, trustworthy, and sustainable in real-world deployment.

1.2 Objectives and Research Questions

Building on the preceding motivation, this thesis proposes and evaluates the RAF, designed to integrate all relevant factors for system viability under one unified structure. The objective is not to introduce a new prediction method, but to investigate how reliability assessment can be operationalized as a structured and actionable system function. To guide this investigation, the following research questions are posed:

1.2.1 Primary Research Queston (PRQ)

In what ways, and to what extent, can a modular, multi-layered reliability assessment framework maintain or improve AI system performance while systematically increasing both system-level trustworthiness and end-user perceived trust in safety-critical domains?

This question reflects the central hypothesis of the thesis: that reliability assessment can be architecturally integrated into AI systems without degrading predictive performance, while simultaneously supporting broader dimensions of trustworthiness and user trust.

1.2.2 Secondary Research Questions (SRQ)

To operationalize the primary question, three secondary research questions are formulated:

SRQ1: Limitations of Performance-Only Evaluation. *To what extent does predictive performance alone suffice for successful AI deployment in safety-critical domains, and what are the limitations of performance-driven evaluation without reliability assessment or human-centered principles?*

SRQ2: System Trustworthiness via Reliability Assessment. *How does the integration of layered reliability assessment methods (such as ensemble modeling, uncertainty quantification, and redundancy mechanisms) affect the statistical performance and operational safety of the system? To what extent do these methods enhance system trustworthiness?*

SRQ3: End-User Perceived Trust. *How do user-facing reliability outputs (e.g., per-sample confidence scores, actionable visual alerts) influence perceived trust and support informed decision-making by end-users in real-world scenarios?*

Together, these research questions correspond to three critical pivot points identified in this thesis: predictive performance, system trustworthiness, and end-user trust. These pivot points provide the conceptual foundation for the RAF and structure both the theoretical and empirical contributions of the work. By structuring the investigation around these pivot points, the thesis aims to not merely evaluate isolated reliability techniques, but to examine how technical, systemic, and human-centered factors must interact for AI systems to become viable in real-world contexts.

1.3 Significance and Normative Positioning

By addressing the interconnected aspects of model performance, system trustworthiness, and end-user trust, this study highlights potential implications for a broad range of stakeholders. For example, in healthcare, reliability-aware AI systems may better support patient monitoring by enabling uncertainty-aware interpretation of model outputs. In the automotive domain, similar principles could contribute to safer in-cabin health monitoring systems by supporting risk-aware decision logic. From a regulatory perspective, structured reliability assessment frameworks may offer conceptual tools for reasoning about compliance, auditability, and human oversight in safety-critical AI systems.

Importantly, while this framework aims to make reliability assessment more accessible and actionable in safety-critical AI contexts, its primary objective is not to accelerate or encourage wider AI deployment per se. Rather, the intent is to strengthen the conditions under which AI systems can be responsibly considered for use. The guiding philosophy of the RAF and this thesis is that improved safeguards should precede expanded deployment. Increased adoption of AI should be the consequence of demonstrable reliability and trustworthiness—not an objective in itself.

1.4 Thesis Structure Overview

The thesis is structured as follows:

Chapter 2: Background and Related Work. Provides the necessary technical and conceptual context for the study. The chapter reviews foundational concepts in reliability assessment, uncertainty estimation, and trust in AI systems, together with relevant perspectives from Human-Centered and Explainable AI. Rather than serving as a critical contribution itself, this chapter establishes the theoretical and methodological context that motivates the need for a more systematic and integrated reliability assessment framework.

Chapter 3: RAF (Reliability Assessment Framework). Introduces and formalizes the proposed framework, which constitutes the primary contribution of this thesis. The chapter presents the framework's guiding principles, layered architecture, and core components, and explains how technical, system-level, and human-centered considerations are integrated into a modular reliability-oriented design. The chapter defines the RAF independently of any specific application domain.

Chapter 4: Case Study – Practical Implementation and Evaluation. Applies the RAF to a concrete use case: AI-driven radar-based heartbeat detection. This chapter describes the experimental setup, the data used, model architectures, reliability metrics, and evaluation methodology. The purpose of the case study is to empirically instantiate and examine the RAF in a realistic signal-analysis pipeline, not to advance the state of the art in heartbeat detection itself.

Chapter 5: Discussion. Interprets the experimental findings in relation to the research questions. The discussion examines how the RAF performed under the evaluated conditions, analyzes limitations and threats to validity, and reflects on the implications for system trustworthiness, uncertainty management, and safety-critical AI deployment.

Chapter 6: Conclusion. Summarizes the contribution of the thesis, explicitly answers the research questions, and reflects on the implications and limitations of the proposed framework. The chapter reiterates the role of reliability assessment as a foundational system property for trustworthy AI in high-stakes domains and briefly identifies avenues for future investigation beyond the scope of the present study.

2 Related Work and Theoretical Foundations

Chapter 2 introduces the theoretical and conceptual foundations necessary to frame the problem addressed in this thesis. Relevant concepts from reliability assessment, uncertainty estimation, Human-Centered AI (HCAI), and Explainable AI (XAI) are discussed to clarify how technical performance, system behavior, and human trust are interrelated.

Rather than offering a critical comparison or exhaustive survey of existing approaches, this chapter serves to establish the conceptual background and shared terminology that motivates the need for a more systematic and integrated reliability assessment framework. The chapter then identifies gaps in how reliability is currently operationalized in AI systems, which directly inform the design choices presented in Chapter 3.

In this chapter, reliability is treated as a foundational technical property of AI systems, while trustworthiness is understood as an emergent system-level property that arises when reliable behavior is combined with appropriate transparency, governance, and human-centered design, enabling outputs to be safely interpreted, acted on, and overseen by humans.

2.1 Conceptualization and Importance of AI Reliability

The reliability of AI systems has emerged as a central concern as these techniques are increasingly deployed in high-stakes and dynamic domains. Unlike traditional deterministic software, ML models are probabilistic: their output depends on statistical patterns learned from data and may vary under shifting input conditions. This probabilistic nature introduces new challenges for ensuring that system behavior remains safe and reliable beyond controlled development settings.

Aggregate predictive performance metrics (such as accuracy, precision, recall, and F1-score) are commonly used to assess model performance. While useful during development, these metrics provide only partial insight into system behavior under real-world deployment conditions. A model that performs well on benchmark data may degrade when deployed in conditions that differ from those encountered during training, including changes in input characteristics or operating environment. Moreover, aggregate metrics alone do not reveal which specific predictions should (or should not) be trusted, limiting their usefulness without additional mechanisms for assessing uncertainty and reliability at the level of individual system outputs.

From a broader perspective, reliability extends beyond technical properties to include lifecycle oversight, accountability, and auditability. Reliable AI deployment therefore depends on traceability that enable system behavior and decisions to be documented and reviewed across the development and deployment phases (Raji et al., 2020). Reliability further depends on mechanisms that support meaningful human oversight, enabling operators to monitor, evaluate, and maintain system behavior, which is essential for ensuring appropriate human supervision and system governance, and enables trustworthiness and user trust (Shneiderman, 2022). Failures of opaque or insufficiently governed systems can undermine confidence in algorithmic decision-making and obscure responsibility for harmful outcomes (O'Neil, 2016).

Realizing the potential benefits of AI requires assurances that systems operate reliably across the full system lifecycle, including development, deployment, and post-deployment monitoring (Ashmore et al., 2021), and requires addressing safety risks arising from epistemic uncertainty (uncertainty arising from incomplete knowledge, such as limited data or model limitations) and harmful outcomes in real-world applications (Varshney & Alemzadeh, 2017). In high-risk applications, even a single unreliable output can lead to severe consequences, as reflected by safety engineering principles emphasizing the importance of detecting, signaling, and mitigating hazardous system states before harm propagates (Leveson, 2011).

Accordingly, reliability is a multi-dimensional construct and should not be treated as a performance characteristic or static property achieved at training time, but an ongoing system characteristic requiring continuous monitoring, assessment, feedback loops, and controlled responses to anomalous or degraded conditions. Without such mechanisms, systems may continue operating despite degraded or unsafe behavior, undermining safety, accountability, and operational viability.

In this thesis, reliability is therefore defined as the degree to which an AI system, assuming a minimum level of predictive validity, produces safe, robust, appropriately signaled outputs under both expected and unexpected operating conditions. Reliability concerns behavioral stability, resilience to perturbation and changing operating conditions, calibrated uncertainty estimation, and the system's ability to detect and communicate potential failure states. This definition forms the technical foundation for the broader notion of system trustworthiness developed in later sections.

2.2 Challenges to Reliable AI

The reliable operation of AI systems is far from guaranteed. Their behavior is shaped by the data on which they are trained, the contexts in which they are applied, and the unforeseen or adversarial factors they may encounter in practice. These factors, amplified by increasingly complex deployment environments, create a diverse landscape of technical and systemic vulnerabilities.

At the technical level, reliability can be undermined by weaknesses in data quality and model generalization. Task-irrelevant, incomplete, imbalanced, or inaccurate training data may produce fragile models that fail under realistic conditions (Ashmore et al., 2021). In complex AI pipelines, errors or deficiencies introduced in earlier lifecycle or system stages can propagate and affect reliability and safety (Ashmore et al., 2021). When deployment data differs from training distributions performance can degrade unpredictably, a phenomenon known as distributional shift (Ovadia et al., 2019). Such shifts may arise from environmental changes, sensor drift, demographic shifts, or evolving task conditions. Models are also vulnerable to adversarial manipulation, where deliberately constructed, often imperceptible perturbations to input data can cause incorrect predictions (Szegedy et al., 2014). Over time, additional instability may emerge through concept drift and outdated models, as underlying data distributions and task requirements evolve (Gama et al., 2014).

These vulnerabilities manifest in characteristic failure modes—recurring patterns in how systems fail. Some are overt and catastrophic, such as failures triggered by rare corner cases or adversarial inputs. Others are subtle but equally concerning, including silent failures that occur without triggering alerts, leaving operators unaware of degraded system performance (Amodei et al., 2016). Miscalibration of confidence estimates represents another critical issue: when reported confidence signals do not correspond to observed accuracy (Guo et al., 2017), this discrepancy can mislead users into over- or under-trusting system outputs. Maintaining calibrated uncertainty estimates is essential for reliable operation, as it enables downstream systems or operators to appropriately interpret and respond to unreliable predictions (Lakshminarayanan et al., 2017; Ovadia et al., 2019), as is maintaining resilience against specification and optimization failures, where imperfectly specified objectives can lead to unintended or harmful behavior (Amodei et al., 2016).

Socio-technical challenges further complicate reliability, as AI systems often operate within organizational, regulatory, and human decision-making contexts. Biased or unreliable systems may also disproportionally affect marginalized groups, reinforcing existing inequalities and producing unjust outcomes (O'Neil, 2016). Unmitigated system failures can also produce harmful downstream effects across critical societal domains, reinforcing inequality, undermining accountability, increasing societal polarization along various dimensions, and posing serious risk to justice and democracy (O'Neil, 2016).

The scale and speed of AI-driven decisions amplify these risks compared to conventional software systems, particularly in safety-critical domains. Across technical, organizational, and societal levels, these challenges illustrate that AI unreliability does not arise from a single weakness but from interacting vulnerabilities distributed across data, models, infrastructure, and human oversight. Addressing these failure modes therefore requires structured assessment strategies capable of detecting, monitoring, and mitigating risk across multiple dimensions.

2.3 Existing Approaches

A variety of methodologies have been developed to assess and strengthen AI reliability and trustworthiness. These approaches operate at different technical and organizational levels and typically focus on distinct aspects of the reliability problem. For analytical clarity, they can be grouped into model-centric, data-centric, uncertainty- and robustness-oriented, explainability-focused, HITL and oversight-focused, and broader framework-level initiatives.

2.3.1 Model-Centric Approaches

Model-centric approaches focus on the internal properties and behavior of learning algorithms. These methods attempt to improve reliability by modifying model architecture, training procedures, or formal verification mechanisms.

Adversarial testing and stress testing are used to expose vulnerabilities under perturbation or edge cases (Madry et al., 2018; Szegedy et al., 2014). Regularization and ensembling techniques can improve generalization and stability across unseen data (Lakshminarayanan et al., 2017). Additionally, uncertainty-aware modeling approaches, such as Bayesian inference and dropout-based approximate Bayesian methods, explicitly represent predictive uncertainty within the model, enabling more reliable decision-making and appropriate handling of uncertain predictions (Gal & Ghahramani, 2016).

These methods primarily strengthen model robustness and predictive stability, and uncertainty reliability at the algorithmic level.

2.3.2 Data-Centric Approaches

Data-centric approaches locate reliability challenges in dataset quality, representativeness, and governance.

Pre-training validation and dataset analysis aim to identify bias and imbalance and other data-level weaknesses before model training (Ashmore et al., 2021). Techniques such as data augmentation (Shorten & Khoshgoftaar, 2019), synthetic data generation (Nikolenko, 2021), and curriculum learning (Bengio et al., 2009) aim to improve generalization across varying complexity levels. Data monitoring approaches promote tracking operational data and system behavior, enabling detection of distributional changes, data degradation, and conditions requiring model maintenance (Ashmore et al., 2021), and continuous real-time monitoring of input data and system behavior is required to remain reliable under changing environmental conditions (Sculley et al., 2015).

These methods address reliability at the level of input distributions and data lifecycle management.

2.3.3 Uncertainty- and Robustness-Oriented Approaches

Because machine learning predictions are inherently probabilistic, even systems reporting high accuracy can still make erroneous predictions without indicating which specific cases require human review. This limitation has driven growing interest in uncertainty quantification—methods that allow models that attempt to measure and communicate model confidence in their predictions.

Bayesian Neural Networks—BNNs (Blundell et al., 2015), deep ensembles (Lakshminarayanan et al., 2017), and Monte Carlo Dropout—MCD (Gal & Ghahramani, 2016) provide probabilistic estimates of predictive uncertainty. BNNs represent the traditional Bayesian approach to uncertainty, but they are often computationally expensive and difficult to scale to modern deep architectures. Ensembles offer strong performance at the cost of compute, since multiple models process the data in parallel. MCD, by contrast, provides a lightweight approximation of uncertainty by introducing stochasticity at inference time, though it can suffer from high variance in its estimates and degraded predictive performance under distributional shift—that is, when deployment data diverge from training distributions. Further, calibration techniques improve the alignment between predicted probabilities and empirical frequencies (Guo et al., 2017), and empirical studies have demonstrated that many uncertainty estimation methods degrade under distributional shift, reducing their ability to reliably signal prediction uncertainty when deployment conditions differ from training data (Ovadia et al., 2019).

These techniques enhance operational reliability by enabling risk-aware decision-making, including the possibility of deferring or flagging high-uncertainty predictions for human review.

2.3.4 Explainability-Focused Approaches

As ML models become deeper and more complex, tracing and understanding the processes behind the model outputs become increasingly difficult for humans. Explainability and interpretability methods seek to make model behavior transparent or understandable to developers, auditors, and users.

Post-hoc explanation tools such as Local Interpretable Model-Agnostic Explanations (LIME; Ribeiro et al., 2016) and Shapley Additive exPlanations (SHAP; Lundberg & Lee, 2017) highlight influential features, enabling developers and auditors to examine model reasoning and support debugging, validation, and oversight. Interpretability approaches provide broader mechanisms for analyzing and explaining model decision processes, particularly in complex or opaque systems. Gradient-based attribution methods such as saliency maps identify the input features or regions that most strongly influence model predictions (Simonyan et al., 2013).

2.3.5 HITL and Oversight Mechanisms

Human-AI collaborative systems integrate algorithmic decision-making with structured human oversight, allowing human operators to supervise and intervene in automated processes.

HITL designs allow operators to review, correct, or override system outputs, supporting active human oversight and intervention in automated decision-making (Amershi et al., 2019). HCAI emphasize maintaining human responsibility and meaningful oversight, ensuring that automated decisions remain transparent, governable, and subject to human supervision and control (Shneiderman, 2022). Research in automation psychology highlights trade-offs between automation level, operator workload, and situational awareness, including increased risks of automation complacency at higher levels of automation (Parasuraman et al., 2000).

These approaches aim to ensure that automated decision-making remains transparent, controllable, and subject to human judgement, particularly in safety-critical or uncertain contexts.

2.3.6 Framework-Level and Governance Initiatives

Beyond individual techniques, higher-level frameworks structure reliability and trustworthiness at organizational, governance, and regulatory levels. These initiatives define processes, documentation practices, and accountability structures intended to ensure responsible development, deployment, and oversight across the system lifecycle.

Safety-oriented initiatives emphasize identification and mitigation of unintended risk (Amodei et al., 2016). Auditing and governance frameworks emphasize lifecycle accountability through structured documentation and audit processes that enable traceability and effective review (Raji et al., 2020). The NIST AI Risk Management Framework (RMF) provides structured approaches for identifying, assessing, and mitigating system risks (NIST, 2023). In regulated domains, system failures may trigger legal and compliance consequences under regulatory regimes such as the EU AI Act, reinforcing the need for structured reliability assessment, documentation, and governance mechanisms (European Parliament & Council, 2024). Documentation standards like Model Cards promote transparency by systematically reporting model characteristics, limitations, and intended uses (Mitchell et al., 2019). Human-centered design frameworks, such as ISO 9241-210, address usability, user involvement, and interaction quality across the system lifecycle, but do not target AI-specific reliability or uncertainty handling (ISO, 2019).

These initiatives shape how reliability considerations are embedded across system lifecycles in a slightly broader way. Table 2.1 summarizes a representative subset and highlights their primary intent and inherent scope limitations.

Table 2.1 *Overview of Influential Framework-Level Initiatives Related to AI Trustworthiness*

Initiative	**Type & domain**	**Scope & limitations**
Amodei et al. (2016)	Safety-oriented research initiative (AI safety)	Emphasizes identification of safety risks and failure modes. Limited guidance on system-level integration, deployment, and human-facing reliability mechanisms.
Raji et al. (2020)	Auditing framework (governance & accountability)	Focuses on accountability and lifecycle audits. Provides limited support for runtime reliability assessment and decision logic.
NIST AI RMF	Risk management framework (organizational trustworthiness)	Provides high-level risk management processes and trustworthiness guidance. Not prescriptive about technical implementation or runtime reliability operationalization.
Model Cards	Documentation / reporting standard (transparency)	Improves model-level transparency. Does not address system behavior, deployment dynamics, or runtime reliability.
EU AI Act	Regulatory framework (legal compliance)	Defines legal obligations. Does not specify how technical reliability should be operationalized.
ISO 9241-210	Human-centered design framework (usability & human interaction)	Provides process guidance for user-centered system design. Does not address AI-specific reliability, uncertainty, or runtime decision logic.

In addition to the initiatives summarized above, a range of related efforts address similar aspects of trustworthy AI. These include high-level ethical guidelines and principles, for example OECD AI

Principles, UNESCO AI Ethics Recommendations, and EU HLEG on AI. While influential, such initiatives are either intentionally high-level and normative in nature, focused on individual techniques or practices, or overlap substantially with the categories already represented in Table 2.1. The overview presented here is therefore not intended as a complete survey of all existing frameworks or methods, but rather as a representative selection of influential framework-level initiatives relevant to reliability assessment and system-level trustworthiness.

2.4 Limitations and the Need for Integration

Despite progress in developing techniques and frameworks to improve AI reliability and trustworthiness, existing approaches remain limited in scope and operational integrations. As shown in Section 2.3, current methods address specific aspects of reliability but typically focus on individual components or stages of the system rather than providing unified, system-level reliability assurance. While these methods provide important advances within their respective domains, they do not by themselves ensure that reliability is operationalized as a continuous and coordinated system-level property. These limitations are analyzed below, across key reliability dimensions.

A first limitation concerns *metric-level incompleteness*. Many approaches rely on performance metrics such as accuracy, calibration, or uncertainty estimates to assess model stability and reliability. These measures provide valuable insight into model behavior but capture only specific statistical properties of predictions. As discussed in Section 2.3.1 and Section 2.3.3, model-centric and uncertainty-oriented methods improve predictive stability and quantify uncertainty, but their scope is largely confined to the model itself. As a result, high performance according to these metrics does not necessarily guarantee reliable system behavior under changing operational conditions, nor does it ensure that uncertainty information is effectively used within broader system decision processes.

A second limitation concerns *lifecycle assurance fragmentation*. Data-centric approaches, monitoring techniques, and governance initiatives emphasize the importance of dataset quality, documentation, and lifecycle management. These methods help identify data weaknesses, detect distributional changes, and support model maintenance. However, as shown in Section 2.3.2 and Section 2.3.6, these approaches typically operate as lifecycle-specific processes—such as dataset validation, periodic monitoring, or auditing—rather than as components of unified reliability architectures. Consequently, reliability assurance may occur intermittently or at specific lifecycle checkpoints, rather than being maintained as a persistent operational system property, and when system processes are inadequately audited or documented, system behavior and errors may become difficult to trace, obscuring accountability (Raji et al., 2020), which could prevent effective reliability oversight and assurance.

A third limitation concerns *human-centered operationalization*. Explainability methods and HITL designs improve transparency and enable human oversight. These approaches support user understanding, auditability, and accountability, as discussed in Section 2.3.4 and Section 2.3.5. However, they are typically implemented as auxiliary tools or interface-level features rather than as fully integrated components of reliability assessment pipelines. As a result, reliability-related information—such as uncertainty estimates or explanation outputs—may not be consistently incorporated into structured system-level decision-making, monitoring, or intervention processes.

A fourth limitation concerns *socio-technical and governance integration.* Framework-level initiatives—including auditing standards, risk management frameworks, regulatory instruments, and ethical guidelines—provide essential structures for accountability, documentation, and compliance. These initiatives play a critical role in promoting responsible AI deployment, as described in Section 2.3.6. However, they are primarily procedural, organizational, or regulatory in nature and do not directly define mechanisms for integrating technical reliability signals into runtime system behavior.

Together, these limitations reveal a pattern across existing approaches. Methods developed to improve reliability at the model, data, lifecycle, human, and governance levels each strengthen specific aspects of system behavior but remain confined to their respective domains. As summarized in Table 2.2, these approaches provide only partial coverage of reliability requirements, which leads to a fragmentation that prevents reliability signals generated at individual system layers from being consistently propagated, interpreted, and acted on across the full operational system.

This structural gap motivates the development of the RAF, introduced in Chapter 3. The RAF is designed to address these limitations by providing a modular, multi-layered framework that integrates reliability assessment across technical, lifecycle, human, and governance dimensions.

Table 2.2 *Comparative Overview of Current Methods vs. Trustworthiness Requirements*

Requirements	Current methods	Scope limitations
Modularity & multi-layered assessment	Robustness testing, isolated data validation, standalone explainability tools	Lack of integrated frameworks combining methods across data, model, system, and user dimensions
Multi-dimensional reliability metrics	Accuracy, F1, pointwise uncertainty scores	Metrics assess specific model properties but do not capture broader system-level reliability
Uncertainty quantification & calibration	Deep ensembles, MCD, calibration methods	Improve uncertainty awareness but operate at the model level and are not integrated into broader reliability architectures
Robustness & resilience to perturbations/faults	Stress testing, adversarial evaluation, data augmentation	Narrowly scoped, with limited system-level recovery, adaptation, or fail-safe mechanisms
Continuous monitoring & lifecycle assurance	Offline testing, periodic audits, basic drift detection	Limited real-time monitoring, weak integration with operational workflows
HITL integration	Manual review during testing and development, ad hoc operator alerts	Limited structured mechanisms for oversight, intervention, or override
User trust: explainability & uncertainty communication	Interpretability tools (LIME, SHAP)	Complex to interpret, often not actionable for end-users, and often not implemented to enhance user trust
Accountability, ethics & governance	Ethical guidelines, audits, checklists	Predominantly procedural or self-regulatory, limited operationalization at system level, or focus on technical reliability with ethical and societal factors neglected

2.5 Human-Centered Trustworthiness: HCAI and XAI

As highlighted in the preceding sections, many approaches to AI reliability emphasize technical aspects such as performance, robustness, or uncertainty estimation. However, these limitations reflect

a broader issue: achieving reliable AI behavior in real-world systems cannot be reduced to technical properties alone. It also depends on effective human-technology interaction. Feedback mechanisms, accountability, and transparency not only support technical performance, but also enable appropriate use, facilitate risk mitigation, and foster calibrated trust. Since AI systems are embedded in social and organizational contexts, attention to human-centered factors is essential for ensuring that reliability translates into real-world safety and benefit. Two overlapping but distinct research fields address this dimension: Human-Centered AI (HCAI) and Explainable AI (XAI).

HCAI emphasizes embedding human needs, values, and oversight across the full lifecycle of AI systems (Shneiderman, 2022). It promotes transparency, accountability, controllability, and user empowerment, ensuring that AI systems remain understandable, governable, and aligned with human and societal goals. These principles ensure that AI systems remain understandable, governable, and aligned with human and societal goals. Without mechanisms that allow users to assess system reliability and understand its limitations, end-users may either over-rely on automated outputs or disengage from them entirely. Both over- and under-trust can undermine safe and effective human-AI interaction (Shneiderman, 2022). When such principles are neglected, opaque systems may operate without sufficient accountability or oversight, increasing the risk of harmful outcomes. O'Neil (2016) documents such failures in domains including credit scoring, hiring, and criminal justice, where black-box models propagated bias and reinforced societal inequalities. By foregrounding human agency and accountability, HCAI provides a foundation for ensuring that AI systems serve human benefit rather than undermine it.

XAI complements this perspective by providing methods for making model behavior interpretable. Techniques such as SHAP, LIME, and saliency-based approaches aim to make opaque model decisions more transparent and analyzable. Interpretability frameworks provide systematic approaches for explaining black-box models, enabling transparency and validation necessary for understanding and trusting system behavior (Guidotti et al., 2018). These tools support developers in debugging, regulators in auditing, and user comprehension. However, explanations do not ensure that users receive a sound and complete account of model behavior, as explanation design involves trade-offs between soundness correctness, completeness, and the risk of overwhelming the user (Miller, 2019). XAI therefore contributes to—but do not replace—the need for technical reliability mechanisms.

Taken together, HCAI and XAI illustrate that trustworthiness requires human-centered consideration. Technical safeguards must be complemented by design practices that make system reliability visible, actionable, and comprehensible for operators and end-users. A central thesis of this work is therefore that both system reliability (accurate, robust, and well calibrated predictions) and subjective trust (user confidence, interpretability, and alignment with human expectations) are necessary conditions for safe and effective AI deployment.

This perspective motivates a broader understanding of system trustworthiness as an emergent property grounded in reliability assessment, transparent model behavior, and structured governance. Importantly, subjective user trust remains distinct from technical trustworthiness. A system may be robust, calibrated and resilient, yet still fail in practice if end-users do not understand or trust it. Conversely, user confidence without technical safeguards risks misplaced trust and unsafe outcomes. This thesis therefore treats trust as a complementary but separate condition for AI project success, complementing statistical performance and systemic trustworthiness. The following section formalizes these distinctions through three critical pivot points for AI system viability.

2.6 Critical Pivot Points for Successful AI Systems

As outlined in the preceding sections, current approaches to AI reliability provide valuable tools but remain fragmented, addressing isolated aspects rather than offering holistic solutions. Table 2.2 highlighted how no existing framework systematically integrates the requirements of overall system

viability. This fragmentation underscores the need for a comprehensive framework that unifies these dimensions of reliability to enable broader trustworthiness.

Through a synthesis of the literature and the identified gaps, this thesis proposes three critical pivot points that must all be satisfied for AI systems to be considered reliable, especially in safety-critical domains. The selection of these three criteria is grounded in their ability to collectively span the full lifecycle of an AI system—from technical aspects ("Does it work?"), through systemic trustworthiness ("Will it keep working safely, and signal when it does not?"), to human adoption and oversight ("Will people use and trust it?"). Each dimension addresses a distinct but essential layer of effective AI operations and a deficiency in either can compromise the whole.

(1) System Predictive Performance.

The foundation of any AI system is its ability to make correct and consistent predictions, classifications, or decisions based on input data. Predictive performance provides the minimum technical condition for system utility. High performance is essential, as it reflects the system's capacity to fulfil its intended function.

However, statistical performance under controlled conditions does not guarantee reliable behavior under real-world conditions. Even highly performing models produce errors that, if left unaddressed or uncommunicated, can have severe consequences in safety-critical domains.

Predictive performance must therefore be complemented with reliability assessment—metrics and mechanisms that identify when outputs may be unsafe, uncertain, or require human review. The RAF explicitly accounts for this by treating statistical performance as one of several criteria for assessing and ensuring overall system reliability and viability.

(2) System Trustworthiness.

System trustworthiness extends beyond predictive performance. In this thesis, it is defined as an emergent system-level property: the extent to which an AI system can be relied on to operate safely and appropriately in real-world contexts.

System trustworthiness is grounded in three interacting foundations. First, the system must incorporate structured reliability assessment and continuous monitoring, enabling it to detect uncertainty or anomalous behavior. Second, it must implement transparency and human-centered interaction and situational awareness mechanisms that make its operational state and limitations visible to relevant stakeholders. Third, it must operate within clearly defined governance and accountability structures that constrain its behavior and define action pathways when reliability thresholds are exceeded.

However, even trustworthy systems that also meet predictive goals may still fail in deployment if these qualities are not perceived and accepted by end-users. This insight motivates the third pivot point.

(3) End-User Trust and Subjective Experience.

Beyond model-level performance and reliability, and system trustworthiness, viable AI systems require well-calibrated subjective trust from end-users.

Subjective trust concerns how users perceive, interpret, and respond to system behavior and outputs. It is shaped by clarity of uncertainty communication, transparency of reasoning, usability of interfaces, and availability of meaningful intervention mechanisms. When this is in place, it brings two key benefits: it fosters well-placed trust while reducing misplaced trust, and it enables HITL feedback pathways that strengthen both system reliability, and user experience and control.

Within the RAF, end-user trust is treated as distinct from system trustworthiness. While trustworthiness reflects engineered system properties, subjective trust determines whether those properties translate into appropriate real-world use.

(*** INSERT SCHEMATIC??!?)

While elements of these domains have been examined separately in the literature, to the author's knowledge no prior work has systematically framed them as mutually dependent pivot points and operationalized them as interlocking gates for AI project viability. Together, they form a unified model of reliability that spans technical, systemic, and human-centered dimensions.

They can be understood both as pivot points—critical junctures where the outcome of an AI project may shift toward success or failure—and as gates, where progress from development to deployment depends on each condition being met. Whether viewed through the metaphor of pivot points or gates, their fulfillment is non-negotiable for the responsible and viable application of AI in real-world contexts.

Taken together, these pivot points establish the core requirements for next-generation AI system frameworks. Effective frameworks must be modular and extensible, employ multi-dimensional metrics that go beyond single-score evaluations, support continuous and real-time monitoring, and integrate human-centered design features such as transparency, explainability, and actionable feedback. These principles provide the foundation for the RAF presented in the following chapter, which operationalizes the pivot points into a structured, multi-layered framework.

2.7 Rationale for a Multi-Layered Approach

*** insert visualization of the swiss cheese layered architecture?!

In safety-critical domains such as healthcare, autonomous driving, and aviation, system failures can have severe consequences for individuals and society. To mitigate such risks, these fields employ layered safety architectures in which no single safeguard is relied upon exclusively. Instead, multiple, independent defenses are implemented so that the likelihood of a hazard passing through every layer—the "holes" in each—is minimized. This concept is captured in the Swiss Cheese Model of accident causation (Reason, 2000). This model has become a foundational principle in modern safety engineering and risk management, emphasizing that errors and failures are often the result of latent weaknesses aligning across system layers, rather than the breakdown of any single component in isolation. Its key insight is that while individual safeguards may be imperfect, their combination produces greater system resilience.

This layered safety principle is directly relevant to AI reliability. As AI systems become increasingly complex and operate in dynamic, high-stakes environments, the risk of unexpected failures—due to data shifts, adversarial manipulation, or unforeseen edge cases—increases accordingly. Strategies that rely on a single reliability method or validation procedure may be inadequate for capturing such multifaceted risks. Inspired by the Swiss Cheese Model, the RAF adopts a multi-layered design in which distinct and complementary mechanisms provide overlapping lines of defense. When the limitations of one mechanism are encountered, others can compensate, thereby reducing the likelihood of undetected or cascading errors. By transferring layered safety principles from other high-risk industries into AI system design, this thesis positions multi-layered reliability assessment as a structural prerequisite in safety-critical AI contexts.

2.7.1 Operationalizing Layered Reliability

Within the RAF, the layered reliability principle is operationalized by combining multiple, partially independent assessment mechanisms. In the experimental setup presented in Chapter 4, three such

mechanisms are employed as illustrative examples of layered reliability can be implemented in practice. Each method, without implying hierarchical ordering, targets different sources of uncertainty or failure and contributes to overall risk reduction.

Ensemble-based uncertainty estimation evaluates the degree of consensus among multiple independently trained models. This method is particularly effective at capturing predictive uncertainty and identifying cases where the system may be overconfident or predictions are unreliable. However, ensembles alone can still fail in scenarios where all models are similarly misled by correlated errors or by systemic data biases.

Latent space divergence leverages the learned feature representations within a variational autoencoder (VAE) as an additional uncertainty signal. By measuring deviations in latent representations relative to learned training distributions, this mechanism can identify atypical or OOD states that may not be captured by aggregate performance metrics. This approach adds a complementary perspective to ensemble uncertainty, as it does not rely solely on model outputs but instead analyzes internal representations. Nevertheless, latent space metrics can sometimes be insensitive to subtle but harmful distributional shifts if not properly calibrated.

MCD variance provides a Bayesian-inspired measure of prediction uncertainty by randomly dropping out units in the network during inference (or testing) and aggregating the resulting variability in predictions. MCD complements the other methods by offering probabilistic insights but can become overconfident in some regimes and may fail to capture all aspects of epistemic uncertainty.

Individually, each of these methods has inherent strengths and, crucially, unique weaknesses—or holes. By integrating multiple, partially independent assessment mechanisms into a multi-layered reliability architecture, the probability that an error or uncertainty will pass undetected through all layers is reduced. This layered approach reflects the core design philosophy of the RAF, emphasizing redundancy and diversity as essential safeguards in the assessment of AI system reliability.

The specific mechanisms used in this chapter are selected as illustrative and complementary examples of how the RAF's layered reliability principle can be operationalized in practice. The framework remains agnostic to the particular techniques employed—what is essential is the overlapping coverage of distinct uncertainty and failure modes.

2.8 Case Study: Radar-Based Physiological Monitoring

To ground the theoretical framework and methodological innovations of this thesis, a practical case study is presented involving radar-based cardiac monitoring in healthcare and automotive settings. This case study is motivated by ongoing collaboration with Emsense, a company specializing in non-invasive physiological monitoring technologies.

Radar-based systems, while promising for unobtrusive monitoring, generate complex, noise-prone data that challenge existing AI-driven analysis and reliability assessment. These issues are particularly acute in safety-critical environments, such as hospital patient monitoring and automotive in-cabin health detection, where failure to detect or act on physiological anomalies can have severe consequences.

In this thesis, the radar use-case serves two purposes: first, to illustrate how AI-driven reliability assessment can address the challenges of noisy, high-stakes physiological data; and second, to provide a practical testbed for evaluating the proposed RAF. It is therefore a contextual lens through which the framework is explored and demonstrated. The specific challenges and design requirements of this case are further explored in subsequent chapters, serving both as motivation and validation for the proposed methodological contributions.

2.9 Chapter Summary

This chapter surveyed the current state of AI reliability across technical and human-centered lines, identifying where existing methods succeed and where they fall short in practice. The review uncovered consistent gaps: fragmented, single-point initiatives; weak lifecycle integration from design to deployment; limited real-time monitoring and operational response; insufficient support for user-facing transparency and control; and generally a lack of holistic integration.

From this analysis, the chapter distilled what a more complete framework must provide: modular, multi-layered assessment; multi-dimensional metrics beyond aggregate performance; continuous monitoring and escalation pathways; and explicit human-centered mechanisms for oversight, interpretation, and action. It also provided the theoretical rationale for the components prioritized in the proposed assessment framework in this thesis.

The next chapter presents the RAF—an integrated architecture that formalizes these requirements and dimensions into a unified framework.

3 RAF (Reliability Assessment Framework)

This chapter introduces the RAF as a conceptual and methodological foundation rather than a finalized technical system or prescriptive standard. The RAF is intended as a structured starting point for the development of next-generation frameworks capable of addressing the full spectrum of trustworthiness requirements in AI systems. Its primary contribution lies in articulating what a comprehensive framework must be able to do—emphasizing integration and unification—rather than claiming to exhaustively specify every implementation detail. As such, the RAF is presented as a foundation for continued work towards trustworthy and viable AI systems.

3.1 Introduction and Overview

The RAF is a domain-agnostic framework that unifies complementary methods across five layers (data, model, system, human interaction, deployment) each contributing distinct yet interdependent aspects of trustworthiness. It ensures that the developed AI system is evaluated and maintained across the entire lifecycle, from data collection and model training to deployment, monitoring, and human oversight. By combining technical mechanisms (e.g., uncertainty quantification and redundancy) with systemic safeguards (e.g., continuous monitoring, auditability, and escalation) and human-centered measures (e.g., transparency and explainable interfaces), the RAF addresses all three critical pivot points identified in Chapter 2: predictive performance, system reliability, and end-user trust.

Figure 3.1 *Conceptual Illustration of the RAF and the Three Pivot Points*

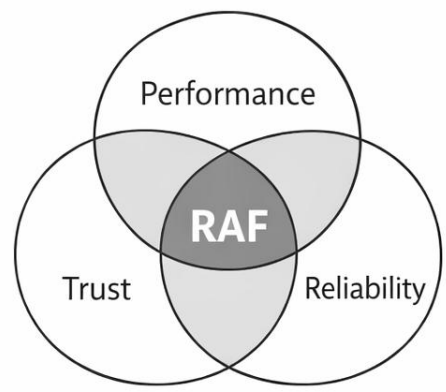
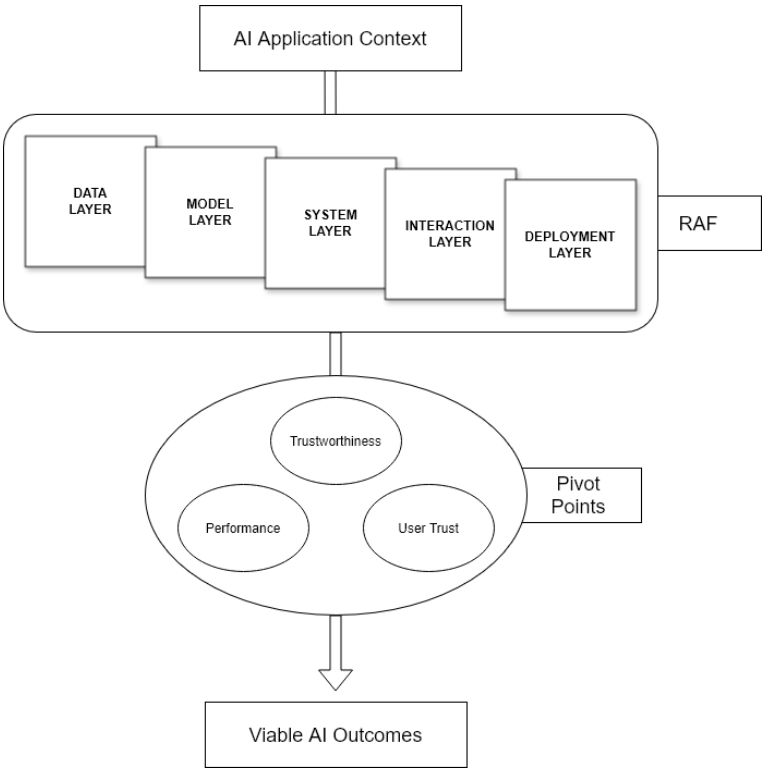


Figure 3.2 below provides a high-level conceptual overview of the RAF and its role in supporting viable AI deployment. The figure illustrates how the application of the RAF to concrete AI application contexts addresses the three critical pivot points identified in this thesis, which together mediate whether an AI system can be deployed safely and sustainable in practice. The layered structure of the RAF reflects the claim that no single layer or mechanism is sufficient to satisfy the full set of requirements for trustworthy AI. Instead, trustworthiness emerges from the coordinated integration of technical, systemic, and human-centered components within a unified framework.

Figure 3.2 *Conceptual Overview of the RAF's Role in Enabling Viable AI Outcomes*



To contextualize the RAF within the existing landscape of trustworthiness and reliability initiatives, Table 3.1 presents a high-level comparison with representative framework-level approaches discussed in Chapter 2. The comparison is grounded in the requirements and gaps synthesized in Section 2.4, and presented in Table 2.2, and focuses on the extent to which different initiatives provide operational support across key dimensions relevant for AI trustworthiness.

Table 3.1 *High-Level Comparison Between the RAF and Other Framework-Level Initiatives*

Initiative/framework	Reliability assessment capability	System-level & lifecycle integration	Human-centered trust & oversight	Governance & accountability
Amodei et al. (2016)	◐	◐	◐	○
Raji et al. (2020)	○	◐	◐	●
NIST AI RMF	◐	◐	◐	●
Model Cards	○	○	◐	◐
EU AI Act	○	◐	◐	●
ISO 9241-210	○	◐	●	○
RAF (this work)	●	●	●	◐

Note. Symbols indicate the extent to which each initiative provides operational coverage of each dimension: ● denotes explicit actionable mechanisms or requirements, ◐ denotes partial or indirect coverage (e.g., only high-level principles or reliance on external standards), and ○ denotes minimal or no coverage.

Table 3.1 helps clarify that, while existing approaches provide important contributions in their domains, none of them are designed to provide comprehensive coverage across all trustworthiness dimensions. The coverage categories in the table together represent the requirements identified in Chapter 2, while the coverage symbols reflect whether an initiative provides concrete, system-integrated mechanisms, rather than relying primarily on higher-level principles or complementary frameworks for full operational coverage. For example, the RAF provides explicit reliability assessment mechanisms, enabling reliability signals to be computed, acted on, and communicated

throughout the system lifecycle, justifying full coverage across the dimensions of reliability assessment capability, system-level & lifecycle integration, and human-centered trust & oversight. The RAF receives partial rather than full coverage for governance & accountability because it does not aim to replace existing regulatory or institutional frameworks. Instead, it complements them by translating governance requirements into actionable system-level mechanisms such as auditability, escalation policies, and role-based oversight.

Having positioned the RAF within the broader landscape of trustworthiness-related initiatives, the following sections describe the framework in detail. Section 3.2 details its internal principles. Section 3.3 describes the core technical and systemic components. Section 3.4 outlines the operational workflow across the AI lifecycle. Finally, section 3.5 provides a synthesis of the framework's contributions and its role within the broader thesis.

3.2 Overarching Philosophy and Guiding Principles

This section specifies how the RAF operationalizes the principles discussed in Chapter 2. The framework unites safety-engineering practices with HCAI and XAI to form an integrated reliability architecture. These principles serve as the conceptual backbone that informs all technical components described in Section 3.3.

As its foundation, the RAF is anchored in a set of guiding principles that permeate every aspect of its design and use: human-centeredness (ensuring that users remain informed, empowered, and in control), transparency and explainability (making system reasoning visible and auditable), ethical and regulatory alignment (upholding privacy, fairness, and accountability), and risk aversion (prioritizing safety and harm prevention over performance optimization). Together, these principles translate into a design philosophy where reliability is not merely a technical goal but a human, organizational, and ethical commitment. However, the principles define design constraints and capability expectations for RAF-aligned systems—specific technical implementation will vary depending on domain, regulatory environment, and risk profile.

The RAF further adopts a humanistic stance, asserting that all AI systems should deliver net positive outcomes for individuals, society, and the environment, and should never be deployed when significant risk of net harm exists. This principle of beneficial alignment ensures that technological advancement remains accountable to the broader well-being of humanity and the world it affects.

3.2.1 Human-Centeredness. The RAF adopts a human-centered design philosophy in which human oversight, interpretability, and operational control are treated as foundational requirements for reliable AI deployment. Within the RAF, human-centeredness is not treated as an optional interface feature but as an architectural design constraint that must be considered throughout the system lifecycle, from data design and model development to deployment and monitoring.

In practice, this principle implies that AI systems operating under the RAF must be designed to support meaningful human oversight. Reliability assessment is therefore coupled with escalation pathways that allow control to be transferred to human operators when system confidence or reliability falls below acceptable thresholds. Rather than assuming full automation, the RAF promotes shared human-AI decision models in which operators receive actionable alerts, contextual explanations, and traceable system status information.

Human centeredness also requires transparent communication of uncertainty and system state. The RAF therefore promotes user-facing mechanisms for communicating reliability, confidence, and model limitations. These may include reliability scores, uncertainty indicators, explanatory summaries, or domain-specific warning structures. The exact form of these mechanisms is domain-dependent, but their presence is treated as essential for supporting informed human decision-making and appropriate trust calibration.

Ethical safeguards and privacy protections are similarly treated as core reliability requirements rather than external compliance tasks. The RAF promotes data minimization, role-based access control, encrypted audit logging, and retention policies aligned with domain-specific regulatory and organizational requirements. The framework is designed to be compatible with established standards for trustworthy data handling, including GDPR (European Parliament & Council, 2016)–for example, data minimization and right to erasure, and ISO/IEC 27001 (ISO, 2022)—information security management), while allowing implementation details to be tailored to deployment context.

The RAF also adopts a "fail-acceptable" and risk-averse operational philosophy. When reliability cannot be established with sufficient confidence, systems designed under the RAF are expected to support conservative fallback behaviors, such as escalation to human review, suppression of automated action, or activation of predefined safe defaults. Where applicable, RAF-aligned systems may also integrate with existing institutional safety procedures to ensure coordinated responses across technical and organizational layers.

Through these mechanisms, the RAF operationalizes human-centeredness by ensuring that users remain informed, empowered, and positioned within meaningful decision-loops, while reducing likelihood of harm arising from automation uncertainty or failure.

3.2.2 Explainability, Accountability, and Ethical AI. The RAF embeds explainability and accountability as structural components, ensuring that system reasoning is both interpretable and auditable.

For all core models, the RAF not only supports XAI during development, but also supports integration of XAI methods, such as SHAP, LIME, and saliency mapping as components integrated in the user interface. Users and auditors can interrogate predictions at the individual and aggregate level, supporting transparency and trust, enhancing system trustworthiness.

Accountability within the RAF is enforced through several mechanisms. First, all predictions, interventions, and system decisions are logged securely to support traceability, auditability, and post hoc review. Second, automated escalation procedures are implemented so that low-confidence or ambiguous outputs are flagged and routed for review by authorized operators, with all actions being fully traceable. Additionally, role-based permissions ensure that only designated users have the authority to approve, modify, or override system recommendations, thereby maintaining a clear chain of responsibility.

Ethical AI is further promoted through rigorous compliance checks and privacy protections. Standardized fairness metrics are applied throughout the model's development and deployment lifecycle, with regular audits for bias and automated drift detection to ensure ongoing ethical system operation. Strict data minimization, de-identification, and access control are enforced at every stage of data handling, as outlined above. Finally, the RAF is continually updated to maintain compliance with evolving regulations, including the EU AI Act and other sector-specific standards.

Collectively, these mechanisms ensure that RAF outputs are never "black boxes" but are instead accompanied by clear and comprehensible rationales. Regular audits and escalation protocols support meaningful human oversight, while embedded privacy and fairness checks reinforce the ethical mandate for non-harm and user dignity. Through these technical solutions, the RAF enables end-users to make informed decisions and helps organizations meet stringent regulatory, legal and societal expectations for trustworthy AI.

3.2.3 Risk Aversion and Regulatory Alignment. The RAF adopts a risk-averse design philosophy grounded in the assumption that uncertainty, model degradation, and partial failure are inherent characteristics of real-world AI systems. From this perspective, the framework treats risk avoidance as a normative priority, explicitly prioritizing harm minimization in system decision-making

or action. Risk aversion therefore operates as a system-level design constraint, shaping when and how autonomy may be exercised by requiring sufficient and continuous evidence of reliable and safe operation. When such assurance cannot be established, the RAF favors restraint, deferral of actions, or human oversight rather than continued autonomous decision-making. This framing reflects a deliberate shift away from performance-centric optimization toward responsible, conservative system behavior appropriate for safety-critical and regulated domains.

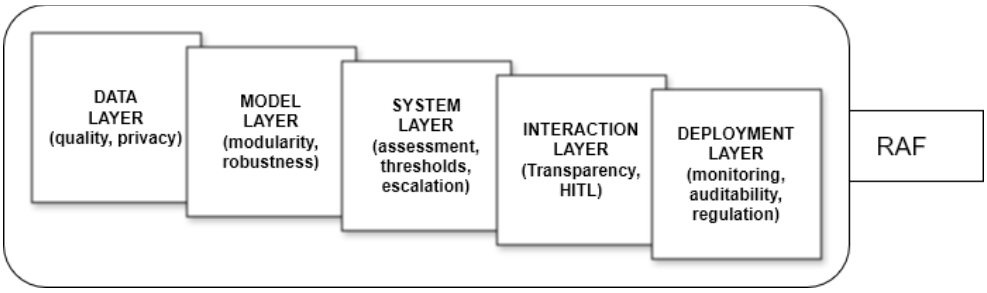
Regulatory alignment within the RAF is ensured by mapping these risk-averse design principles to established standards for trustworthy and accountable AI. Audit logging, access control, and information security measures are supported in accordance with frameworks such as ISO/IEC 27001, while data handling and privacy considerations are designed to meet GDPR requirements. Furthermore, the RAF explicitly supports meaningful human oversight, traceability, and accountability, aligning with both EU and international regulatory expectations for AI systems.

By embedding risk aversion and regulatory alignment as core design constraints, the RAF not only responds to existing legal and ethical requirements, but anticipates the increasing demand for conservative, transparent, and human-centered AI systems.

3.3 RAF Layers and Core Components

Building on the guiding principles outlined in Section 3.2, this section specifies the RAF's core components and how they are organized within a layered architecture. Components are grouped into five layers—data, model, system, interaction, and deployment—to reflect distinct responsibilities across the AI lifecycle. Each subsection describes the layer's primary role and the specific mechanisms it contains. Component selection is grounded in established theory, safety-critical best practices, and contemporary empirical research, and is described in terms of implementation, operational logic, and contribution to system reliability. Taken together, these layers define how the RAF operationalizes reliability as both a technical property and an operational, human-centered capability.

Figure 3.3 *Layered Architecture of the RAF and Representative Components*



3.3.1 Data Layer

The RAF's data-level components establish the foundation for all subsequent reliability assessment and modeling stages. They ensure that data entering the system is representative, consistent, and traceable throughout the AI lifecycle. Since the reliability of any AI system is fundamentally constrained by the quality and contextual validity of its data, the RAF emphasizes systematic practices to ensure that all inputs are both accurate, representative, and context appropriate.

By embedding these practices at the foundation of the framework, the RAF seeks to minimize the propagation of upstream errors and to ensure that all subsequent layers—model, system, and user—operate on data that is both trustworthy and interpretable. Through accuracy and transparency at the data-level, this foundation strengthens system reliability across all subsequent layers. The following subsections detail the specific mechanisms that realize these goals, including data generation, preprocessing, validation, augmentation, and drift detection.

3.3.1.1 Data Health and Validation. Ensuring data quality, representativeness, and continuous monitoring is foundational to the RAF. Input data must be rigorously validated before and during system operation. The framework recommends implementing an *automated data validation pipeline*, using both statistical and rule-based checks. For example, this can include schema validation, range and type checks, detection of missing or anomalous values, and validation against domain-specific constraints.

In addition, *EDA (Exploratory Data Analysis)* should be systematically conducted. This includes visual inspection of distributions, detection of outliers, assessment of class imbalance, and evaluation for potential sources of bias. The RAF suggests that all training, validation, and test datasets be re-examined after significant data updates or shifts in data acquisition processes to ensure ongoing validity.

Continuous data monitoring is implemented by tracking distributional drift and other key metrics throughout the system's lifecycle. Trigger thresholds for drift or anomaly detection should be set according to domain risk tolerance. For instance, population stability index (PSI), Kolmogorov-Smirnov (KS) statistic, or other drift metrics can be computed at regular intervals, with automated alerts sent to system admins or developers when significant change is detected. The RAF requires that these triggers be documented and their responses auditable for regulatory compliance.

Within RAF's modular architecture, the data layer forms the first link in a *continuous reliability feedback loop*. Detected issues—such as drift, bias, or missingness—automatically trigger reliability reassessment at the model and system layers, ensuring that deviations in data quality propagate upward as explicit reliability signals rather than silent failures. Data lineage tracking maintains end-to-end traceability of datasets and the processing and modification steps applied to them as data flows through the system.

3.3.1.2 Data Generation and Augmentation. In many applications, naturally occurring datasets may be incomplete, imbalanced, or insufficient to capture the full range of operational conditions that a system may encounter. The RAF therefore supports controlled data generation and augmentation as mechanisms to improve data representativeness and strengthen system robustness. These processes expand the effective coverage of the data distribution, enabling models to learn from a broader range of input conditions, including rare, edge-case, or degraded scenarios that may be underrepresented in collected datasets.

Data generation refers to the creation of new data instances that reflect plausible operational conditions, while *data augmentation* involves systematic transformation of existing data to introduce controlled variation without altering underlying semantic meaning. Augmentation techniques may include noise injection, signal perturbation, geometric transformation, or domain-specific modification, depending on the data modality. All generated or augmented data must be clearly labeled, version-controlled, and distinguishable from original observations to ensure full transparency and reproducibility.

When properly implemented, these techniques enhance reliability by reducing model sensitivity to noise, distributional variation, and previously unseen conditions. Exposure to controlled variation improves generalization and reduces the likelihood of brittle behavior when the system encounters unfamiliar or degraded inputs during deployment. In addition, structured data generation can support controlled evaluation of system behavior under known conditions, enabling systematic stress testing and validation of reliability mechanisms.

3.3.1.3 Privacy and Security Considerations. Privacy-preserving and security-oriented practices are embedded throughout all stages of data handling in the RAF. Data minimization policies are enforced during ingestion, ensuring that only features strictly necessary for model development

and operation are retained. De-identification routines—such as hashing, masking, or removal of direct identifiers—are applied as early as possible in the pipeline.

Access control follows role-based structure, restricting sensitive data views and audit log access to authorized personnel only. All data movement is logged, and sensitive records are encrypted, in compliance with established information security and data protection requirements. Security audits and penetration tests are recommended at least annually, or upon any major system update. Automated alerts should be triggered in the event of unauthorized access attempts or anomalous data export activity.

Through these combined mechanisms, the RAF embeds privacy, data security, and accountability as integral—not peripheral—elements of data reliability.

Collectively, these data-level components ensure that reliability is treated not as a static property but as a continuously maintained process. By enforcing quality, transparency, and security from the outset—and by linking data monitoring directly to higher-layer reliability assessments—the RAF establishes a dynamic foundation upon which model-, system-, and user-layer reliability can consistently build.

3.3.2 Model Layer

The RAF's model-level components ensure that architectures and training procedures are robust, adaptable, and aligned with the reliability objectives of the framework. Reliability is achieved through architectural modularity, ensemble diversity, and rigorous robustness testing. The model layer transforms validated data into stable, interpretable representations and produces uncertainty signals that inform system-level scoring and escalation. A modular architecture, using task-optimal models, enables targeted tuning or replacement as improved methods emerge. Baseline reference models anchor benchmarking and risk assessment.

This section outlines the modular signal modeling pipeline used in the reference implementation, explains the rationale for key architectural choices, and describes how ensemble diversity, baseline models, and robustness mechanisms jointly strengthen both predictive performance and overall reliability.

3.3.2.1 Model Pipeline Design. The RAF is model-agnostic and does not prescribe specific model architectures. Instead, it defines structural proposals for how models are integrated within reliability-aware systems. Selected models must be appropriate for the application domain and capable of producing stable, traceable outputs that can be evaluated using the system's reliability assessment and action mechanisms. The framework focuses on architectural properties—such as modularity, replaceability, and compatibility with monitoring and assessment strategies—rather than on specific model types or training methods.

Within the RAF, models are organized into modular pipelines consisting of one or more sequential processing stages. Each stage transforms inputs into intermediate or final representations while preserving compatibility with downstream reliability evaluation. This modular pipeline structure ensures that individual components can be independently evaluated, updated, or replaced without disrupting the overall system architecture.

In the reference implementation presented in Chapter 4, these principles are instantiated using a VAE (variational autoencoder) followed by a DAE (denoising autoencoder), combined with an ensemble strategy. The VAE→DAE stage learns structured latent representations of the input and then refines the reconstructed outputs by reducing residual noise. Multiple independently trained instances of this model combination form an ensemble to support uncertainty estimation and reliability scoring. This configuration serves as an illustrative example rather than a prescribed architecture. Alternative pipelines, using different models, can be integrated into the system without altering its overall

structure. This architectural flexibility ensures that the framework remains applicable across diverse domains and evolving modeling techniques.

3.3.2.2 Diversity Strategies. To strengthen reliability, the RAF promotes architectural and training diversity at the model level. Diversity may be introduced through the use of ensembles (training multiple independently initialized models), variation in initialization seeds (encouraging different optimization paths), alternate data splits (exposing models to slightly different training distributions), or augmentation policies (systematically altering input samples to improve robustness).

Diversity enables the system to reduce correlated errors and capture epistemic uncertainty more effectively. For example, when multiple independently trained models respond different in a substantial way to the same input, this variation signals areas of limited knowledge, ambiguous structure, or insufficient data coverage. Within the RAF, such disagreement functions as an early-warning mechanism: instead of committing to a potentially unstable prediction, the system surfaces uncertainty as an explicit reliability signal. By embedding diversity at the modeling stage, the RAF reduces the probability of silent, systematic failure and strengthens robustness against distributional shift and unforeseen edge cases.

In the reference implementation presented in Chapter 4, diversity is operationalized a deep ensemble consisting of independently trained but architecturally identical pipelines. Each ensemble member is exposed to different initialization conditions and training dynamics, ensuring meaningful variation in learned representations. Disagreement among members serves as an indicator of epistemic uncertainty. These diversity-driven signals are later quantified and propagated to the system layer, where they contribute to reliability scoring, threshold evaluation, and potential masking or escalation actions. In this way, diversity at the model level informs operational reliability decisions.

3.3.2.3 Robustness and Generalization Strategies. To strengthen model resilience under real-world conditions, the RAF supports robustness-enhancing strategies that improve generalization beyond training data. These may include adversarial training, in which deliberately perturbed samples are introduced during training to increase resistance to misleading inputs, as well as domain adaptation techniques such as fine-tuning on OOD data or applying transfer learning protocols to maintain performance across evolving environments.

The RAF also supports structured training strategies such as curriculum learning, defined as the progressive exposure of models to increasingly complex or realistic data conditions. Under high-variability or noisy conditions, models may first be trained on cleaner or simpler inputs before being incrementally challenged with more difficult or corrupted samples. In the Chapter 4 reference implementation, this principle is operationalized by generating synthetic radar signals with increasing levels of noise and artifact complexity and scheduling model exposure accordingly. Curriculum learning is not mandatory within the RAF but may enhance robustness and generalization when appropriate.

These strategies support stable behavior under distributional shift, sensor noise, and evolving operational contexts. By incorporating robustness-oriented training and adaptation mechanisms, the RAF strengthens the model layer's contribution to overall system reliability without prescribing specific techniques or algorithms

3.3.2.4 Baseline and Benchmarking Anchoring. For each application, the RAF recommends preserving a baseline version of the primary operational architecture, trained under identical conditions but without additional reliability-enhancing or assessment mechanisms (e.g., ensemble aggregation, uncertainty scoring, or action logic). This preserved baseline serves as a stable reference model against which reliability-oriented extensions can be evaluated. This anchoring ensures that reliability enhancements do not obscure or unintentionally degrade core predictive capability.

By explicitly preserving a baseline reference, the RAF supports transparent comparison, structured ablation analysis, and traceable evaluation of reliability-performance trade-offs. This benchmarking layer strengthens methodological rigor and enables clearer reasoning about whether reliability-oriented modifications genuinely improve system viability.

3.3.3 System Layer

The system layer operationalizes reliability within the RAF by transforming model outputs and associated uncertainty signals into structured reliability assessments and system-level decisions. While the model layer produces predictions and uncertainty-relevant information, the system layer is responsible for evaluating, interpreting, and acting on these signals in accordance with defined reliability policies. Through this, the RAF ensures that system behavior remains traceable, controllable, and aligned with domain-specific safety requirements. The system layer thereby serves as the central coordination point between predictive modeling and operational decision-making, enabling reliability-aware system operation throughout the system lifecycle.

3.3.3.1 Architectural Modularity. The RAF adopts modularity as a foundational design principle. Reliability assessment mechanisms are implemented as components that can be integrated, modified, or replaced. Each component operates as a discrete functional unit responsible for evaluating specific aspects of prediction stability, uncertainty, or system integrity. By separating these functions into independent modules, the framework enables flexible configuration and controlled evolution of reliability mechanisms.

A modular architecture also strengthens system resilience. Because reliability assessment is not dependent on any single mechanisms, the system maintains robustness against individual component limitations or failures. The ability to combine multiple independent assessment modules further supports comprehensive reliability evaluation and reduces the risk of undetected failures.

Finally, modularity ensures long-term maintainability and extensibility. As modeling techniques evolve, assessment components can be updated independently while preserving compatibility with the overall system structure. This modular and extensible design underpins the RAF's long-term viability, allowing the framework to evolve alongside advances in AI technology, shifting regulatory demands, and domain-specific practices, while maintaining consistent reliability assurance, resilience, transparency, and adaptability across diverse contexts.

3.3.3.2 Multi-Metric Reliability Assessment. In complex, real-world environments, no single reliability metric can adequately capture all relevant sources of uncertainty or failure. Different failure modes—such as epistemic uncertainty, distributional shift, or internal representational instability—may manifest in different parts of the system and remain invisible to any single assessment method. The RAF therefore adopts a *multi-metric reliability assessment principle,* according to which reliability is evaluated using multiple, complementary signals that provide distinct perspectives on model behavior and uncertainty.

In the experimental instantiation presented in this thesis, this principle is operationalized using three specific reliability metrics - ensemble standard deviation ($RS_{ensemble}$), VAE latent space divergence (RS_{latent}), and Monte Carlo dropout variance (RS_{MCD}). Here, these metrics serve as illustrative examples of how heterogeneous reliability signals can be combined within a layered assessment strategy. The RAF remains agnostic to the specific metrics employed, committing instead to the principles of multi-layered and multi-metric reliability assessment as a means of increasing robustness against errors, failures, and unstable uncertainty estimation.

Importantly, the value of this approach lies not in any individual metric, but in their combined use: each signal may detect failure modes that the others miss, and their joint consideration reduces the likelihood that unreliable predictions pass undetected. Here, ensemble deviation and MCD variance

both capture epistemic uncertainty, but from different perspectives (between-model and within-model variability, respectively), while latent space divergence targets OOD situations, identifying inputs that deviate from the model's learned latent structure, including cases where predictive confidence alone may be misleading. In this way, multi-metric assessment functions as a robust filtering strategy that increases overall system reliability through overlap, diversity, and redundancy in uncertainty estimation. Thresholds for individual reliability signals can be calibrated empirically or defined in a domain-specific manner, with stricter tolerances applied in safety-critical contexts such as healthcare or aviation. Formal definitions and calibration procedures for the reliability metrics used in this thesis are provided in Chapter 4.

To enable meaningful integration of heterogenous reliability signals, individual metrics needs to be *normalized* to a common scale prior to aggregation. Normalization ensures that differences in magnitude or distribution do not distort downstream reliability analysis. The specific normalization procedure is implementation-dependent.

Following normalization, multiple reliability signals may be aggregated into a *composite reliability score*. This composite metric provides a unified and operationally interpretable indicator of system reliability. Aggregation strategies—such as weighted averaging, rule-based combinations, or learned meta-models—are domain-specific and may reflect regulatory requirements or risk priorities. Combining complementary uncertainty signals into a single calibrated index enables simplified assessment, thresholding, and escalation logic while avoiding reliance on any individual metric.

3.3.3.3 Reliability-Driven Decision Logic. The composite reliability score serves as the operational trigger for reliability-based decisions within the RAF. Predictions exceeding predefined thresholds are processed normally, while those falling below initiate one or more safety responses—masking, flagging, or escalation—depending on assessed risk and context, thereby enabling or prompting human judgement where appropriate.

Thresholds for triggering these behaviors can be empirically determined—based on validation performance—or adaptively tuned during operation according to evolving data distributions, risk tolerance, or regulatory requirements. The flexibility to calibrate these parameters allows RAF-based systems to align with domain-specific safety norms, whether in clinical diagnostics, industrial automation, or other mission-critical applications. Adaptive thresholding can leverage real-time reliability indicators such as RS_{total} trends to respond dynamically to changing conditions. Thresholds may tighten during periods of instability or unfamiliar input patterns, and relax when conditions are stable and uncertainty remains low. This supports a controlled balance between operational efficiency and safety vigilance, and helps reduce user fatigue from excessive false positives or unnecessary alerts.

When a prediction or model behavior falls below a configured threshold, the system automatically triggers an appropriate action. *Masking* prevents unreliable outputs from propagating through downstream automated processes. When activated, the prediction is withheld from downstream modules or processes, ensuring that no low-confidence information contributes to subsequent system behavior. *Flagging* allows the prediction to be presented to users or subsystems but marks it with a visual alert or textual warning indicating elevated uncertainty or risk. Unlike masking or escalation, flagging is non-blocking, allowing continued system operation while supporting informed human judgement. *Escalation* transfers decision authority to a human operator or oversight mechanism. In these cases, the system defers autonomous action until reliability conditions are verified.

In safety-critical deployments, these mechanisms are reinforced through automated red-flag and hold-for-review pathways. Whenever the composite reliability score or an individual metric indicates high uncertainty, the system automatically initiates escalation protocols that alert designated operators or oversight systems. These alerts are integrated with existing infrastructure—such as dashboard

notifications—and link directly to workflow interfaces where human review and confirmation can occur. By ensuring that flagged predictions are promptly acknowledged and addressed, this automation bridges algorithmic assessment and human oversight, closing the reliability loop in real time.

All reliability evaluations and resulting actions are automatically recorded, including timestamps, model identifiers, and triggering conditions. This audit trail ensures traceability for compliance, debugging, and post-incident analysis.

Together these policies and automated pathways ensure that reliability assessment is not merely diagnostic but operationally active, continuously monitoring, moderating, and communicating the trustworthiness of the system.

3.3.4 Interaction Layer

The end-user and interaction layer of the RAF operationalizes human-centered reliability, ensuring that AI systems remain transparent, interpretable, and under effective human control. This layer bridges technical reliability mechanisms with the lived experience of human operators and decision makers, providing the tools, explanations, and safeguards necessary for safe and trustworthy human-AI collaboration.

Central objectives at this level include maintaining situational awareness (through transparency and real-time reliability feedback), ensuring control and accountability (via HITL mechanisms), and fostering trust and understanding (through XAI, visualizations, and user support). The RAF operationalizes these objectives through integrated mechanisms such as HITL control, XAI-driven interpretability tools, structured user training, and fail-acceptable operational guidance. Together, these components ensure that users remain informed, empowered, and able to interpret, challenge, or override system behavior when needed—transforming reliability from a technical metric into an experiential property of trustworthy human–AI collaboration.

3.3.4.1 HITL (Operational Implementation). HITL integration forms a central safeguard within the RAF, ensuring that humans retain ultimate oversight and decision authority in all safety-critical contexts. Reliability information—such as uncertainty scores, reliability flags, or escalation alerts—is clearly communicated to operators, enabling timely and informed intervention. These mechanisms act as a vital check on automated processes, allowing for escalation protocols, manual override, and operator feedback when ambiguity or elevated risk is detected.

When a prediction's composite reliability score falls below threshold, the system automatically initiates *escalation protocols*—suspending autonomous actions, issuing real-time alerts, and transferring operational control to a designated expert or operator. At any time, authorized users may exercise *manual override* to suspend or adjust automated decisions, and all such interventions are logged for transparency and auditability. Following intervention, *operator feedback* is collected through interaction and structured forms, and integrated into the system's continuous improvement loop, informing retraining and policy refinement.

To further support safety under uncertainty, the RAF integrates active user guidance and real-time operational support. In domains such as emergency healthcare or aviation, where cognitive load and timing are critical, the system provides clear, step-by-step instructions when reliability is compromised. This may include concise checklists, color-coded warnings, or audio-visual alerts prompting the safest next actions—such as seeking expert review or temporarily suspending automated control. By embedding such guidance, the RAF ensures fail-acceptable behavior, maintaining safety and efficiency even under degraded or partially failed conditions.

Importantly, the RAF embeds HITL not merely as a defensive measure but as a collaborative interface between humans and intelligent systems. By ensuring transparent communication and traceable

control pathways, HITL integration transforms oversight from a reactive safety measure into a proactive enabler of accountable and cooperative human-AI interaction in high-stakes environments.

3.3.4.2 Explainability, Visualization, and User Support. The RAF embeds explainability and user-centered visualization mechanisms throughout the system to ensure that reliability is not only measurable but also interpretable and actionable by end-users. User-facing interpretability is achieved through several components, that collectively strengthen user understanding, situational awareness, and well-founded confidence in system behavior.

Interactive dashboards provide real-time visualizations of model predictions, reliability metrics, and overall system health. Visual indicators such as color coding, confidence bands, and dynamic trend lines allow operators to monitor evolving reliability conditions. Context-aware tooltips and annotations help users interpret metric values and understand how system reliability changes over time or under different operational contexts. In line with human-centered design principles emphasizing clear feedback and appropriate signaling (Norman, 2013), and human-AI interaction guidelines supporting calibrated trust and appropriate reliance on system outputs through clear communication of system capabilities, limitations, and behavior (Amershi et al., 2019), the RAF avoids unconditional "green" or "reliable" signals to reduce the risk of inappropriate trust and to support informed user monitoring under uncertainty.

Complementing visualization, *explanation methods* are integrated directly into the interface to provide transparent insight into model reasoning. The RAF supports model-agnostic XAI techniques such as SHAP, LIME, and saliency mapping, allowing users to request both sample-specific and aggregate explanations on demand. These tools make it possible to trace predictions back to the most influential features, enabling developers, regulators, or clinicians to verify that the system's logic aligns with domain expertise and ethical expectations.

Because interpretability alone is insufficient without clear communication, the RAF implements *uncertainty communication* protocols that translate technical reliability measures into plain-language statements and actionable guidance. When reliability decreases or anomalies are detected, users receive intuitive explanations, along with prioritized next steps. This bridges the gap between technical uncertainty and user comprehension, reducing the likelihood of misinterpretation or overreliance.

To promote effective interaction regardless of user expertise, the RAF includes structured onboarding and continuous training resources. Interactive tutorials, guided walkthroughs, and context-sensitive help-features ensure that users understand reliability metrics, interpret visualizations, and respond appropriately to alerts. Targeted training and ongoing theoretical support further reduce misuse or misinterpretation, reinforcing the foundation of reliable human-AI collaboration.

Together, these mechanisms transform explainability from a post-hoc diagnostic tool into a continuous, user-centered process that promotes transparency, learning and trust. By combining visual clarity, interpretive depth, and actionable guidance, the RAF ensures that end-users are not merely informed observers but empowered collaborators in maintaining the system's reliability and safety.

3.3.4.3 Trust and Transparency Features. The RAF reinforces trust and transparency as foundational conditions for reliable and ethical AI. Achieving both regulatory compliance and genuine user confidence requires that AI systems provide clear, interpretable justifications for their decisions and reliability assessments. The commitment extends beyond satisfying external audit requirements, it is about empowering users—whether clinicians, engineers, or operators—to understand, interrogate, and meaningfully engage with systems. By making both model reasoning and reliability estimation transparent, the RAF strengthens accountability, promotes user trust, and aligns human understanding with system intent.

Each system output is accompanied by a *prediction rationale*—a concise explanation derived from the active XAI method that summarizes the most influential input features and the reliability factors underpinning the result. These rationales allow users to trace system logic, verify alignment with domain knowledge, and detect potential sources of bias or misclassification.

Complementing this interpretability are *system status indicators*, which communicate real-time information on system health, model drift, and update activity. Visual cues such as color-coded warnings, progress markers, or timestamped update logs provide operators with continuous situational awareness, helping to ensure that any degradation in model reliability or data integrity is immediately visible.

The RAF also incorporates *structured error reporting*, enabling users to flag questionable predictions or operational anomalies directly through the interface. Each report automatically generates an entry in the audit log, creating a traceable feedback channel that feeds into retraining cycles, quality control processes, and compliance audits. This ensures that transparency is maintained not only in system outputs but also in the handling of user-identified issues.

By combining prediction rationale, system status indicators, and structured error reporting within a unified, auditable framework, the RAF ensures that end-users are not passive recipients of AI outputs but informed collaborators. This integration of visibility, accountability, and user-participation embodies HCAI principles and supports the RAF's overarching goal: to operationalize reliability as both a technical property and a lived human experience.

3.3.5 Deployment Layer

The RAF's domain- and deployment-level components ensure that the framework is not only theoretically sound but also practical, adaptable, and sustainable in real-world contexts. This layer governs how the RAF is instantiated, monitored, and maintained across diverse application domains, focusing on modularity, configurability, continuous monitoring, and alignment with domain-specific safety and regulatory standards.

3.3.5.1 Operational Extensibility and Configuration Management. The RAF's *plug-and-play architecture is* a defining characteristic of this modular design. Each major component—such as the model pipeline, reliability scoring modules, and user interface elements—can be enabled, disabled, or replaced without requiring extensive re-engineering of the surrounding system.

Equally important is *domain adaptation and configurability*. Every module can be tuned or customized to match the unique needs of its deployment context. Configuration files or user interface (UI)-based settings allow operators to adjust thresholds for masking, escalation, audit intervals, data privacy levels and other operational policies. In addition, ensemble configurations (for example, number of members or diversity settings) and reliability metric weightings can be tuned to match domain-specific risk tolerances and regulatory requirements. This approach supports rapid adaptation to new requirements, threat landscapes, or domain-specific standards and regulatory frameworks, such as ISO 13485 for medical devices or IEC 61508 for industrial safety.

Finally, the RAF is engineered for *future extensibility*. Its modular separation of concerns allows new reliability metrics, visualization tools, or integration with external safety protocols—to be incorporated with minimal disruption. Built-in version control and testing frameworks ensure that module updates or replacements do not introduce regressions or compromise reliability.

3.3.5.2 Safety-Critical Best Practices. The RAF is explicitly aligned with the risk management strategies and operational doctrines established in safety-critical domains such as aviation, healthcare, and industrial automation. In these industries, layered defense mechanisms, continuous auditability,

and formalized escalation pathways are essential for ensuring system safety and operational accountability.

Layered defenses form the backbone of the RAF's safety philosophy. Mirroring the Swiss cheese model, the RAF ensures that failures must pass through multiple independent barriers before causing harm. This is achieved through the combined use of diverse reliability metrics, ensemble redundancy, and multi-stage escalation protocols—each providing an additional safeguard against undetected or cascading errors.

Formalized escalation and HITL mechanisms further strengthen this defense structure. Automated detection of OOD conditions, persistent low reliability, or anomalous system states triggers predetermined escalation pathways, transferring decision authority to human operators or domain experts. This design parallels established procedures in high-stakes fields such as aviation (e.g., checklists and cockpit alerts) and healthcare (e.g., clinical override mechanisms or emergency escalation protocols such as "code blue" alerts), ensuring that responsibility for critical decisions remains with qualified human agents.

Continuous auditability completes this triad of best practices. All predictions, reliability assessments, system state changes, and user interventions are logged and retained in secure, time-stamped audit trails. These reports support post-incident analysis, regulatory audits, and iterative process improvement. By embedding these practices, the RAF enables organizations to align directly with existing safety management frameworks and regulatory standards, ensuring that both technical integrity and procedural accountability are maintained throughout the AI system's lifecycle.

3.3.5.3 Continuous Monitoring and Feedback. Sustained reliability requires ongoing observation, adaptation, and improvement once systems are deployed. The RAF therefore incorporates continuous self-assessment, real-time alerting, and feedback-driven refinement to maintain performance and trustworthiness over time.

Online monitoring provides the foundation for this capability. The framework continuously tracks key indicators such as reliability scores, input data distributions, model drift indicators, and operator interventions. Dashboards display live system status and trend analysis, enabling both real-time situational awareness and proactive system maintenance.

Real-time alarms and triggers automatically notify responsible operators when reliability or anomaly thresholds are exceeded. These alerts prompt immediate investigation, intervention, or fallback to predefined safe modes, depending on the severity of the detected issue.

Feedback loops ensure that all incidents, operator insights, and emergent data drift are systematically reviewed and integrated into the RAF's update cycle. Retraining, model-recalibration, and configuration adjustments are performed at defined intervals or after repeated triggers, allowing the system to evolve alongside its operational context.

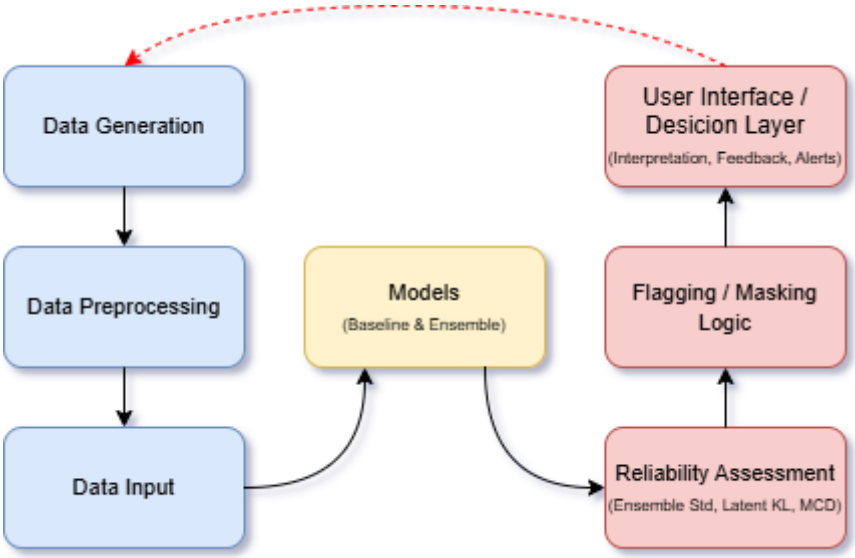
Post-deployment audits complement this process by verifying long-term compliance and performance stability. Conducted on a scheduled basis—or following major system updates—these audits leverage stored monitoring data and audit trails to identify weaknesses, document corrective actions, and guide ongoing system improvements.

Together, these mechanisms ensure that the RAF functions as a living framework rather than a static artifact. By embedding continuous monitoring, adaptation, and feedback, the framework remains capable of evolving in response to new operational challenges, regulatory developments, and emerging safety demands—supporting both automated and human-centered reliability management across all deployment settings.

3.4 RAF Conceptual Workflow

This section outlines the step-by-step workflow for applying the RAF in practice. The workflow serves to operationalize the theoretical principles established in the preceding sections, ensuring that reliability is systematically assessed, maintained, and continuously improved throughout the AI system lifecycle. Each phase corresponds to a critical stage of real-world AI development—from initial domain understanding through to post-deployment monitoring and iterative refinement. Importantly, the RAF workflow is not a static checklist but a dynamic, cyclic process. By following these stages, practitioners can embed reliability assessment seamlessly into both development and operational contexts, thereby strengthening technical robustness and fostering user trust.

Figure 3.4 *Conceptual Workflow of the RAF*



Note. The figure illustrates the cyclic application of the RAF across the AI system lifecycle and helps conceptualize its workflow. The workflow spans preconditions and context understanding, data validation, model output generation, reliability assessment, reliability-based action and communication, and feedback-driven system adaptation. Arrows indicate the logical progression and interdependence of workflow stages, while feedback pathways highlight the iterative nature of reliability management across development and deployment.

(0 ↔ 5) Preconditions: Understanding the "X-Space". In the RAF, the term X-space refers to the set of all relevant spaces that influence—or risk influencing—system reliability, including the problem definition, data characteristics, operational environment, and deployment context. Prior to data validation, model development, or reliability assessment, the problem domain, data characteristics, and deployment context must be clearly understood and defined.

Required prerequisites include: *problem definition and context mapping*—the operational domain must be specified, stakeholder needs must be identified, and explicit boundaries for acceptable system behavior and performance must be defined. *Data exploration and validation*—data sources must be examined, potential anomalies or sources of bias must be identified, and data relevance to the intended context must be verified. *Risk and requirement identification*—legal, ethical, operational, and user-centered requirements must be established, including constraints, acceptable failure modes, and escalation expectations.

These prerequisites must be established prior to applying subsequent workflow stages. Feedback from later workflow phases—particularly stage 5 ↔ 0—reflecting the RAF's cyclic and continuously adaptive structure, must be used to refine and update the X-space definition as system behavior, operational conditions, or deployment requirements evolve.

(1) Data Validation and Input Integrity. All input data used for reliability assessment must be validated for quality, representativeness, and relevance to the intended deployment context. Systematic data health checks must be performed, including identification and mitigation of anomalies or outliers, and verification that the statistical and structural properties of the data reflect real-world conditions. Validation must include assessing whether the available data adequately represents the range of conditions expected during deployment, allowing identification of potential coverage gaps or domain mismatches prior to model development. Controlled data generation may be employed to address coverage gaps or represent rare conditions.

The origin, preprocessing history, and version of all datasets must be documented and traceable. Dataset partitioning must be carefully managed to ensure proper separation between training, validation, and test sets.

(2) Model Architecture and Prediction Generation. Following data validation, model architectures should be selected and implemented to support reliable prediction generation and downstream assessment. Model architectures must be selected for appropriateness for the problem and data contexts. Outputs must be generated in a manner that supports downstream reliability assessment. Modular pipeline structures and independently trained model instances may be used where appropriate to support robustness and uncertainty evaluation.

Models should be developed and evaluated using clearly defined training, validation, and testing procedures to ensure transparent performance characterization. Baseline reference models should be preserved to provide stable comparison points. Careful model selection ensures that downstream reliability assessment and decision logic operate on well-characterized and appropriate model outputs.

Model versions, configurations, and training conditions must be documented and traceable to ensure reproducibility and enable auditability.

(3) Reliability Metrics Computation. Multiple complementary reliability metrics may be computed to achieve a comprehensive assessment of uncertainty and potential risk. Each metric should offer a distinct perspective on prediction reliability, capturing different aspects of uncertainty and variability. Their combined use provides a robust quantitative foundation for risk estimation and for identifying predictions that warrant masking, flagging, or escalation in subsequent workflow stages.

To ensure meaningful interpretation and integration, reliability metrics must be computed in a consistent and traceable manner, with each metric clearly associated with the corresponding prediction, model configuration, and input data. Where multiple metrics are used, their outputs should be normalized or otherwise structured to support comparison, aggregation, and threshold-based decision logic.

(4) Reliability-Based Action and Communication. Once reliability metrics are computed, reliability-based decision policies must be applied to determine appropriate system responses. These policies may be dynamically adjusted based on operational context, deployment requirements, or observed system behavior.

Domain-specific reliability thresholds must be defined and enforced to identify predictions that fall below acceptable reliability levels. Predictions that do not meet defined reliability criteria must be masked, flagged, or escalated according to established operational policies. Escalated predictions must be transferred to designated human operators or supervisory systems for review and resolution when required.

Reliability information must be communicated clearly and consistently to relevant system components and end-users. Reliability scores, alerts, and escalation indicators must be presented through appropriate interfaces. All decisions and resulting actions must be documented and traceable.

(5 ↔ 0) Feedback and Continuous Improvement. A defining feature of the RAF is its emphasis on continuous learning and adaptation. Insights gained during system deployment or through end-user interaction must be systematically fed back into earlier workflow stages. Without these mechanisms, reliability risks being eroded over time, as real-world conditions, data distributions, and operational contexts evolve. This feedback loop ensures that the RAF remains adaptive and continuously aligned with operational reality, supporting sustained reliability, risk reduction, and long-term system trustworthiness.

This process includes feedback derived from reliability assessments, flagged or escalated predictions, system performance monitoring, and human operator interactions. Identified reliability issues, performance degradations, or changes in operational conditions must trigger review and, where necessary, updates to system configuration, data validation procedures, or model architectures. Corrective actions may include updating datasets, retraining models, refining reliability thresholds, or modifying system policies. All updates must be validated through the appropriate workflow stages prior to deployment to ensure continued reliability and system integrity.

3.5 Chapter Summary

This chapter introduced the RAF as a modular, multi-layered framework for operationalizing reliability across the data, model, system, interaction, and deployment aspects of AI systems. It demonstrated how uncertainty quantification, reliability-based flagging and escalation, transparency and HITL oversight, and continuous monitoring and feedback processes can be integrated into a coherent architecture that supports sustained reliability and enables emergent system trustworthiness.

As discussed in Section 2.4, existing approaches to AI reliability and trustworthiness remain fragmented across several dimensions, with limited support for system-level integration and operational decision-making. The contribution of the RAF is therefore not the introduction of new safety principles or reliability techniques as such, but the synthesis and operationalization of established ideas within a unified framework that connects trustworthiness principles to actionable system behavior and user experience.

By combining technical reliability mechanisms with systemic safeguards and user-centric oversight, the RAF provides a coherent and auditable structure that can be adapted across domains and evolving deployment contexts. It functions both as a design guide for developing reliability-aware AI systems and as an evaluation framework for assessing and maintaining reliability in deployed systems.

To consolidate the RAF architecture and its conceptual foundations, Tables 3.2 and 3.3 summarize how RAF components support the three identified pivot points and how selected components align with HCAI-related design principles. These tables provide a representative overview of the framework's structure and component relationships, rather than an exhaustive specification of all possible implementation variants.

Table 3.2 *RAF Component–Pivot Point Mapping*

Component	Role in RAF	Examples	Supported pivot points
Overarching philosophy & principles	Define risk posture and human-centered goals	HCAI principles, ethical guidelines, governance mindset	Trustworthiness, user trust
Data health & validation	Ensure input integrity and representativeness	EDA, data quality checks, drift monitoring	Predictive performance, trustworthiness

Curriculum learning	Improve learning stability under staged complexity	Progressive exposure, staged difficulty	Predictive performance, Trustworthiness (via robustness)
Modular architectures	Enable robustness and redundancy	Autoencoders (e.g., VAE/DAE), ensembles	Predictive performance, trustworthiness
Reliability metrics (multi-metric)	Quantify uncertainty, risk flagging	$RS_{ensemble}$, RS_{latent}, RS_{MCD}	Trustworthiness
Thresholding & escalation policies	Translate reliability into operational action	Flagging, masking, review triggers	Trustworthiness, user trust
End-user/HITL/transparency	Enable oversight and calibrated trust	Reliability flags, uncertainty visualizations, explanations (XAI), human review mechanisms	User trust
Continuous monitoring & feedback	Sustain reliability over time	Drift monitoring, error reporting, retraining strategies	Predictive performance (via feedback), trustworthiness, user trust
Domain & deployment adaptation	Fit policies to context, compliance (regulatory, domain-specific)	Cost-sensitive thresholds, audit requirements, domain constraints	Trustworthiness, user trust

Note. Mapping of RAF components to their primary functional role within the framework, illustrative implementation examples, and the pivot points they support.

Table 3.3 *RAF Component–HCAI Alignment*

Component	**HCAI-aligned objective**	**Practical operationalization (examples)**
HITL mechanisms	Human oversight, accountability	Review triggers, override capability, audit trail
Transparency tools	Comprehensibility and explainability	Reliability flags, uncertainty visualizations, XAI explanations, user guidance, dashboards
Privacy & security safeguards	Privacy, harm minimization	Data minimization, access control, GDPR compliance, logging
Audit & logging	Accountability	Config and decision traceability, periodic audits
Decision guidance & safety-aligned system behavior	User support and safety	Conservative defaults, action guidance, alerting, safe automation boundaries
Risk mitigation & robust defaults	Resilience	Fallback modes, conservative policies, redundancy
User capability & operational understanding	User empowerment	Operational guidance, training materials, user support

Note. Mapping of RAF components to the HCAI-aligned design objectives and representative technical implementation examples.

Although the RAF intends to enable system trustworthiness, the term Reliability Assessment Framework is used deliberately. Rather than treating trustworthiness as an abstract or purely normative property, the RAF positions reliability assessment as the primary operational mechanism, together with human-centered design constraints, through which trustworthiness can emerge.

Following this conceptual and architectural foundation, Chapter 4 demonstrates the application of the RAF in a radar-based heartbeat detection case study, providing empirical insight into how reliability assessment mechanisms can be instantiated, evaluated, and analyzed in practice.

4 Practical Implementation

This chapter demonstrates how the RAF can be applied and evaluated in a concrete, end-to-end AI system. While Chapter 3 introduced the RAF as a conceptual and methodological architecture, its practical value depends on how its implemented components behave in practice: how the individual reliability metrics interact with model outputs, how the composite reliability scores reflect prediction quality, and how thresholding and flagging mechanisms perform under realistic conditions.

To investigate these questions, we implement the RAF within a radar-based heartbeat detection pipeline. This domain provides a controlled but meaningful testbed—the signals are noisy and highly variable, the prediction task is non-trivial, and uncertainty plays a central role in determining whether outputs are trustworthy. The use case therefore allows systematic evaluation of how well the RAF identifies unreliable predictions, isolates uncertainty and supports safer decision-making.

The scope of this study is deliberately limited, and not all components of the RAF will be evaluated. For example, no user studies will be performed to assess questions related to the interaction layer (UI, user comprehension, calibrated trust, and HITL workflow evaluation), and deployment layer components (such as regulation or policy questions) will not be implemented in this study and are therefore discussed conceptually in Chapter 5 rather than evaluated empirically here. However, Section 4.1 describes the data layer components, while Section 4.2 covers the model layer aspects, and Section 4.3 operationalizes the system layer by implementing reliability scoring and threshold-based flagging. Section 4.4 and 4.5 details the method and results of the study itself. Together, these sections provide a proof-of-concept of RAF operation in an end-to-end reliability-aware system.

*** insert diagram/image of RAF components used in the experiment?

Table 4.1 *RAF Component Coverage Instantiated in the Experiment*

RAF layer	Component	Definition section (Chapter 3)	Implementation section (Chapter 4)
Data layer	Synthetic data generation	3.3.1.2	4.1.2
	Dataset splitting and strict hold-out test evaluation	3.3.1.1	4.4.1
Model layer	Model pipeline (VAE→DAE cascade)	3.3.2.1	4.2, 4.2.1-4.2.2
	Ensemble construction and diversity	3.3.2.2	4.2.3, 4.4.2.2
	Curriculum learning (low, medium, high complexity stages)	3.3.2.3	4.1.3, 4.4.2

	Baseline/reference model benchmarking	3.3.2.4	4.5.1, 4.5.2.3
System layer	Architectural modularity (Replaceable models and assessment methods)	3.3.3.1	4.2
	Multi-metric reliability assessment ($RS_{ensemble}$, RS_{latent}, and RS_{MCD})	3.3.3.2	4.3.1
	Normalization (min-max, validation-derived)	3.3.3.2	4.3.1
	Composite reliability score (RS_{total})	3.3.3.2	4.3.1
	Thresholding policy and action logic (flagging)	3.3.3.3	4.3.2.1-4.3.2.3
Interaction layer	Conservative uncertainty communication policy ("No green signals")	3.3.4.2	4.3.2.2
Deployment layer	Not empirically evaluated	-	-

Note. Components not listed were not implemented in the experimental pipeline and are treated as out of scope for Chapter 4 empirical evaluation.

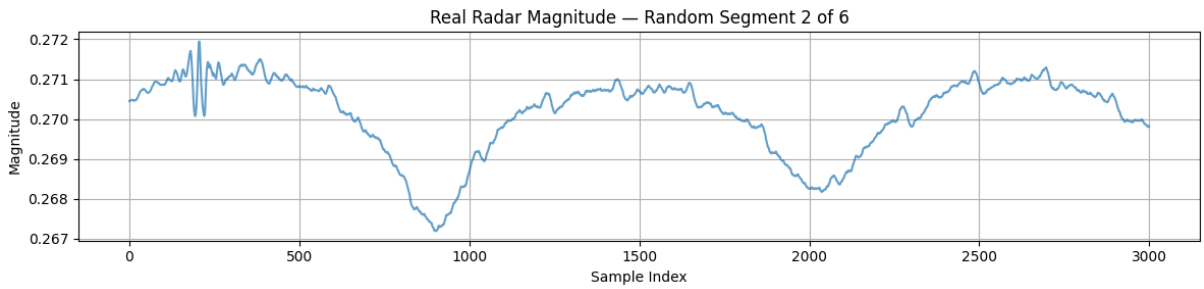
4.1 Data Layer: Synthetic Radar Signal Generation and Applicability

4.1.1 Rationale for Synthetic Data Use

In safety-critical domains such as healthcare, obtaining large and high-quality datasets for AI development is often limited by practical, ethical, and privacy constraints. Existing research highlights the increasing value of synthetic data generation for scenarios where real data is scarce, privacy-sensitive, or difficult to obtain (Frid-Adar et al., 2018; Kazeminia et al., 2020; Nikolenko, 2021). Approaches to address these challenges include both classical data augmentation methods and the creation of fully synthetic datasets tailored to a specific application domain.

For this thesis, a limited real-world radar dataset was available for reference, recorded in controlled automotive scenarios (engine off, idling, and driving). The original dataset consisted of four approximately one-hour long data streams, each segmented into ~15-second windows for initial data analysis and model development.

Figure 4.1 *Raw Radar Data Example*



Given the limited scope and controlled nature of this real data, we adopted a synthetic data generation approach guided by the statistical and temporal characteristics of the reference dataset. This approach enables full transparency and fine-grained control over data complexity, segmentation, and annotation, while maintaining sufficient realism for evaluating the modeling pipeline. Although the primary experiments rely on synthetic data, the characteristics (spectral, temporal, noise) were informed by the real reference signals, ensuring that findings retain both theoretical relevance and practical applicability.

4.1.2 Signal Generation Pipeline

The synthetic radar signal generation pipeline was designed to produce realistic one-dimensional time-series segments, each simulating a 15-second radar window, sampled at 200 Hz, resulting in 3,000 samples per segment. Each segment is constructed by combining multiple features observed in the reference data, including:

(1) Cardiac waveforms: modeled as sequences of amplitude-modulated spikes, with inter-beat intervals randomly sampled from a physiologically plausible range (55–95 bpm). Ground truth heartbeat locations are recorded for each segment.

(2) Breathing oscillations: a low-frequency sinusoidal component with randomized frequency (0.2–0.3 Hz) and amplitude, mimicking respiration-induced body motion.

(3) Body motion artifacts: sporadic, non-stationary additive disturbances, representing movement artifacts typical in automotive or healthcare radar data.

(4) Sensor and electronic noise: additive Gaussian white noise (standard deviation: 0.006), capturing the stochastic background noise present in raw radar recordings.

(5) Baseline drift and amplitude variability: slow-varying trends and heartbeat amplitude fluctuations, reflecting long-term signal drift and natural physiological variability observed in the reference dataset.

The result is a dataset that combines realistic variability with controlled ground-truth labeling, suitable for robust model development and systematic reliability assessment. While not a substitute for comprehensive real-world data, this approach provides critical benefits: full control over data complexity, exhaustive ground truth annotation, and the ability to ensure privacy and safety by design.

A key limitation of this dataset is its restriction to physiologically “normal” cardiac rates (55–95 bpm), mirroring the reference data. Pathological or extreme cardiac states (e.g., arrythmias, cardiac arrest) are therefore not represented, limiting the model’s ability to generalize to such cases. Addressing this gap will be essential for future work aimed at real-world deployment in healthcare monitoring.

4.1.3 Curriculum Learning Stages

To improve model robustness and facilitate gradual adaptation to increasingly realistic deployment conditions, we implemented a curriculum learning schedule inspired by the three complexity conditions present in the original reference radar data. Our synthetic data was generated at three

distinct complexity levels, each defined by varying combinations of heartbeat amplitude, noise, body motion, baseline drift, and breathing variability.

Table 4.2 *Curriculum Learning Stages: Signal Data Complexity Levels*

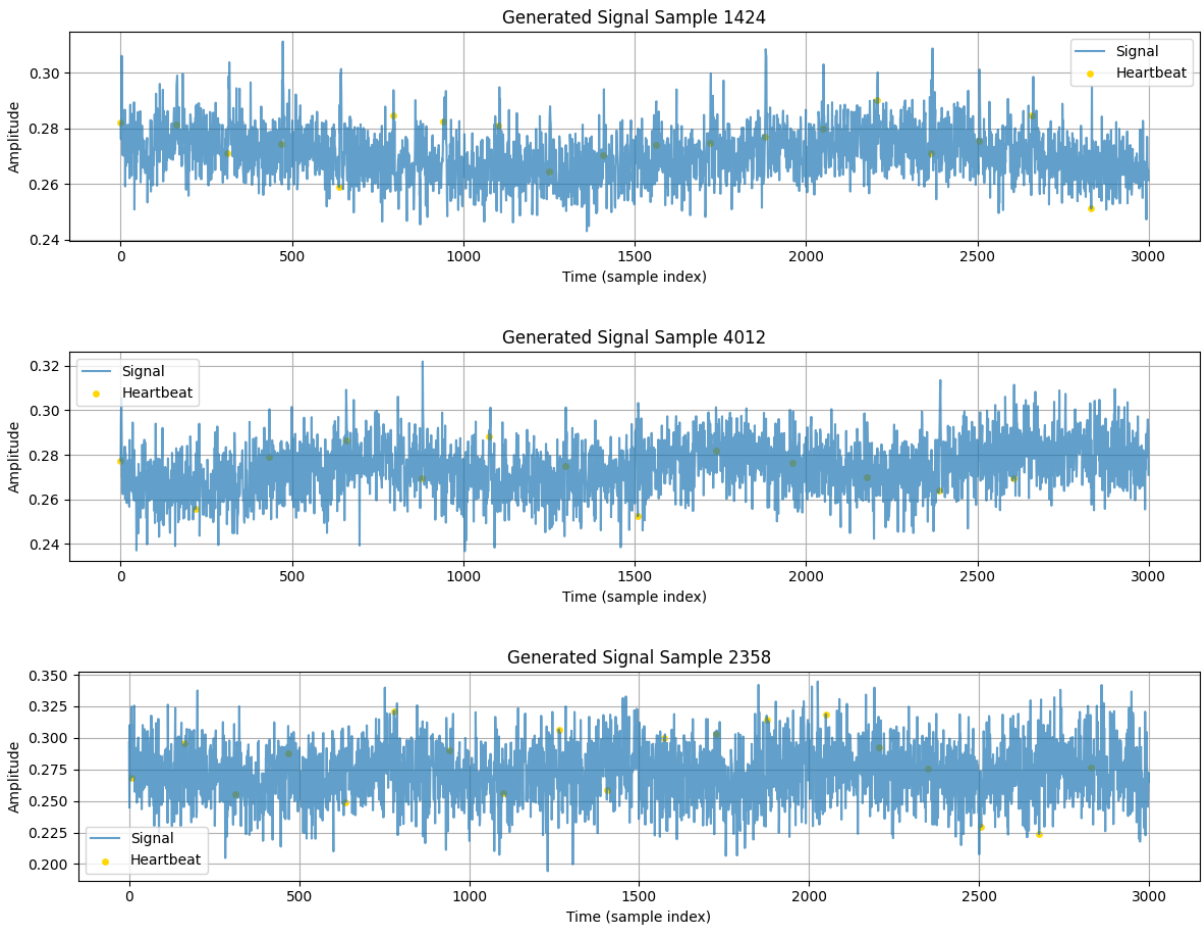
Complexity level	**Heartbeat amplitude**	**Noise level**	**Body motion artifacts**	**Baseline drift**	**Respiratory variability**	**Description**
Low	High	Low	None	Minimal	Minimal	Clear heartbeats, low interference
Medium	Moderate	Moderate	Moderate	Some	Some	More challenging: added body movements and mild drift
High	Low	High	Significant	Pronounced	High	Realistically challenging: strong noise, major artifacts, high variability

In the training pipeline, each model is initially trained on the low-complexity data with randomly initialized weights. After this stage, the optimized model state is saved and used to initialize training on the medium-complexity data. The same process is repeated for the high-complexity data, ensuring the model is progressively exposed to increasingly challenging data distributions. This sequential training approach is primarily used for baseline model optimization, and is a direct implementation of curriculum learning principles tailored to our scenario. While this schedule is theoretically motivated to improve model robustness, it was not formally benchmarked against alternative training strategies within this thesis.

Table 4.3 *Example Parameter Settings Per Stage*

Parameter	**Low-complexity**	**Medium-complexity**	**High-complexity**
Heartbeat amplitude	0.03	0.02	0.01
Additive noise (std)	0.005	0.015	0.025
Body motion (prob/freq)	0	0.2	0.4
Baseline drift	0	0.005	0.012
Breathing variability (%)	5	15	30

Figure 4.2, 4.3, 4.4 *Generated Synthetic Data (Low-, Medium-, High-Complexity Level)*



4.1.4 Real vs. Synthetic Data: Applicability and Limitations

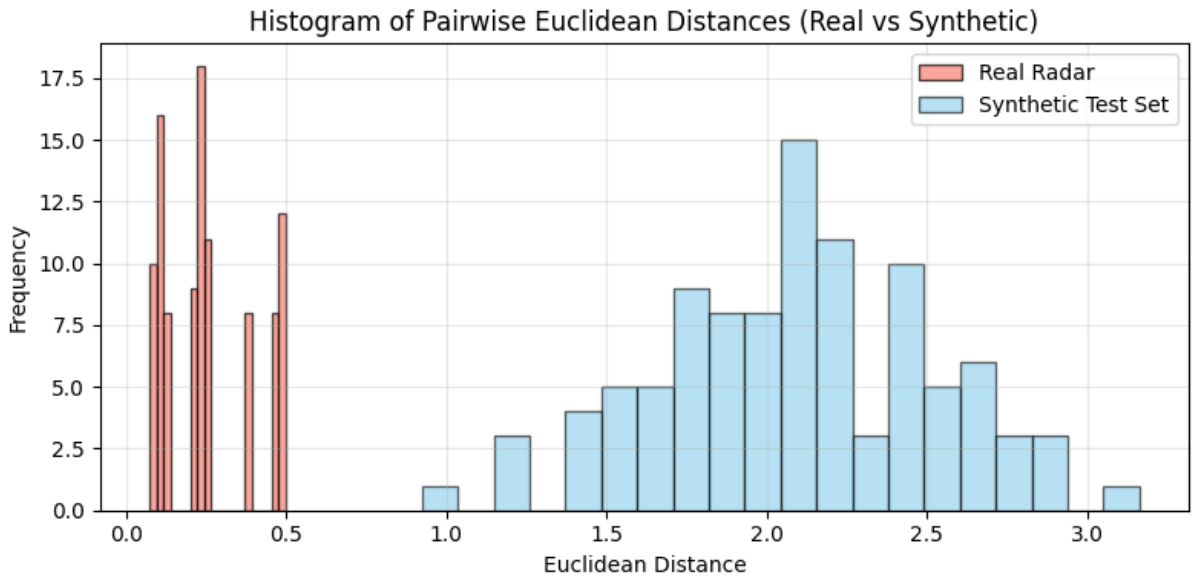
Synthetic data in this study enabled rigorous validation and comprehensive ground truth control for model development and reliability assessment. To verify the realism and applicability of our generation pipeline, we visually and statistically compared key features between the synthetic and reference data. This ensured that the generated signals captured the essential characteristics of the target application domain.

However, as our synthetic data was limited to physiologically "normal" cardiac activity and did not include pathological or extreme scenarios, the results are presented here should be interpreted as a proof-of-concept rather than direct clinical validation. The generalizability of findings to uncontrolled real-world data remains an open limitation and a target for future research.

Despite these constraints, a fully synthetic approach supports key HCAI principles by enabling transparency, user control, and ethical experimentation, while eliminating privacy concerns. It provides a flexible, risk-free platform for developing and validating reliability assessment frameworks in AI-driven contexts.

A pairwise distance analysis (Figure 4.5) showed that while the synthetic signals preserved structural characteristics observed in the reference data, they exhibited deliberately increased variability due to added noise, drift, and morion artifacts. This confirms their suitability for controlled testing but not equivalence to the real distribution.

Figure 4.5 *Pairwise Euclidean Distances Comparison (Real vs Synthetic)*



4.2 Model Layer: VAE→DAE Pipeline and Ensemble

The core objective of our practical use-case is to reliably predict cardiac activity from radar-based time-series signals—a task complicated by substantial measurement-irrelevant noise and artifacts. To address this, we adopted an autoencoder-based modeling strategy, leveraging the inherent denoising and signal reconstruction capabilities of these neural net architectures. In particular, the combination of VAEs and DAEs was identified as especially appropriate for this application, given their complementary properties. VAEs enable probabilistic latent representation learning, while DAEs support targeted denoising and reconstruction of clean signals from corrupted or noisy inputs.

Initial exploratory tests compared a range of architectures—including Long Short-Term Memory (LSTM) nets, standard autoencoders (AEs), VAEs, and two-stage combinations such as VAE→DAE and DAE→VAE—using our synthetic radar dataset. Although all models demonstrated similar initial results, the VAE→DAE cascade performed marginally better. But the main rationale for the use of this pairing was its inherent ability to clean and reconstruct noisy signals on its own, without further filtering or extensive manual preprocessing.

In the final implementation, the pipeline consists of three components. First, a VAE processes the input radar segment, learning a compressed probabilistic representation and produces an initial denoised reconstruction while supporting explicit uncertainty estimation. Second, a DAE receives the VAE output, and performs a focused denoising pass, further enhancing signal fidelity by suppressing residual noise and accentuating physiologically relevant features. Finally, to enhance reliability and robustness, we employ a deep ensemble approach: multiple independent VAE→DAE pipelines are trained and evaluated in parallel, and their outputs are aggregated to produce both a consensus prediction and a measure of model disagreement. Ensemble variance forms a key reliability signal in this chapter and directly connects to the RAF metrics described in Chapter 3.

It is important to note that the architecture selection was informed by preliminary experiments using an earlier version of the synthetic dataset and should not be interpreted as a definitive benchmark for radar-based physiological monitoring. More diverse and challenging datasets may reveal alternative architectures or improvements. Nevertheless, the present pipeline offers a principled, modular, and reliability-aware foundation for cardiac signal modeling in radar-based health monitoring.

4.2.1 Variational Autoencoder Architecture

The first stage in our radar signal processing pipeline employs a VAE (Kingma & Welling, 2014) for denoising and signal reconstruction.

In our setup, the VAE is implemented using a series of one-dimensional convolutional layers in both the encoder and decoder. The encoder compresses each input signal into a lower-dimensional latent representation that captures its essential structure. A latent vector is sampled from this distribution, allowing the model to learn a compact encoding of the latent representation. The decoder then reconstructs the original signal from the latent representation, and reconstruction quality is evaluated using mean squared error (MSE) loss. A KL divergence term regularizes the latent space by encouraging the representations to follow a consistent and structured distribution.

The VAE's probabilistic latent structure directly supports two key features of our pipeline:

(1) It enables calculation of per sample KL divergence, which we leverage as a component of the composite reliability score.

(2) It facilitates advanced OOD detection, as inputs producing large KL divergence are flagged as potentially unreliable.

4.2.2 Denoising Autoencoder Architecture

The second stage in the model pipeline is a DAE (Vincent et al., 2010), used to further suppress noise and enhance cardiac features in the radar-derived signals. While the VAE provides a probabilistic latent space that supports uncertainty modeling and OOD detection, the DAE is designed for denoising. It receives as input the reconstructed signal from the VAE and applies a deterministic, non-stochastic encoding and decoding process.

In this implementation, the DAE uses one-dimensional convolutional layers in both the encoder and decoder, mapping the VAE output into a compact latent representation and reconstructing a refined signal. The loss function is the MSE between the DAE output and the clean ground truth signal, encouraging effective noise removal and focused enhancement of heartbeat-related structure. Because the DAE is deterministic, each input produces a single, consistent reconstruction, which is important for highlighting subtle physiological features relevant to heartbeat detection.

In this pipeline, the DAE effectively "polishes" the VAE output, improving overall signal quality. Together, the VAE→DAE cascade leverages the strengths of both probabilistic modeling (uncertainty-aware denoising) and deterministic reconstruction (precise feature refinement), making the approach well-suited for radar-based physiological monitoring.

4.2.3 Deep Ensemble Methods and Implementation

To further increase reliability and quantify model uncertainty, we employ deep ensembles—a method for uncertainty estimation in neural networks. In this setup, multiple independent instances of the full VAE→DAE pipeline are trained separately. Each ensemble member uses identical architecture and hyperparameters but is initialized with a different random seed and trained on a uniquely generated synthetic dataset. This independence promotes diversity among members and prevents convergence to identical solutions, which is essential for meaningful uncertainty estimation.

At inference time, all members process the same input segment. Their predictions are averaged to produce the consensus reconstruction, while the variance (or standard deviation) across outputs forms the ensemble-based component of the $RS_{ensemble}$ reliability score. Higher ensemble variance indicates greater disagreement among members and therefore elevated epistemic uncertainty, signaling that the prediction may be less reliable.

4.3 System Layer: Reliability Scoring, Thresholding, and Flagging

A core contribution of this thesis is the integration of the RAF into an AI-based radar heartbeat detection pipeline. The aim is to move beyond predictive performance as a sufficient model evaluation metric and systematically provide quantitative, interpretable, and actionable reliability scores for each prediction. This not only supports safer deployment in safety-critical domains, but also increases transparency and enables user-centered decision support.

The reliability assessment-layer in this thesis operates post-prediction: after model inference and the extraction of 3 different uncertainty signals, a composite reliability score is computed for each data sample. The resulting overall reliability score is then used to support thresholding and flagging. This design separates model-level uncertainty estimation from system-level reliability decision logic.

4.3.1 Reliability Metrics and Calculation

In this experiment, reliability was quantified using three complementary uncertainty metrics, each capturing a different aspect of model stability: ensemble standard deviation, VAE latent space divergence, and MCD variance.

Ensemble standard deviation (*$\mathbf{RS_{ensemble}}$*) was computed by evaluating a pooled set of reconstructed outputs produced for each input segment by the VAE→DAE ensemble and quantifying their variability. All ensemble members were trained independently but evaluated using identical preprocessing, inference, and post-processing settings, ensuring that observed differences in output reflect model-level uncertainty.

Ensemble standard deviation was computed as:

$$RS_{\text{ensemble}}(x) = \text{std}\left(\{f_i(x)\}_{i=1}^{N}\right)$$

where $f_i(x)$ denotes the i-th reconstructed output for input segment x. The index i ranges over the pooled set of reconstructions generated across all ensemble members and MCD passes, such that N denotes the total number of reconstructions considered per segment. The resulting value was averaged across timepoints to provide a single scalar reliability score per segment. Higher values of $RS_{ensemble}$ indicate greater disagreement among reconstructions and correspond to reduced prediction reliability.

VAE latent space divergence **(**$\mathbf{RS_{latent}}$**)** was computed directly from the VAE encoder by evaluating the KL divergence between the encoder's posterior distribution and the VAE's latent prior. For a given input segment x, the encoder outputs a posterior distribution $q(z \mid x)$ parameterized by a mean $\mu(x)$ and variance $\sigma^2(x)$. The latent divergence was computed as:

$$RS_{\text{latent}}(x) = D_{KL}\left(q(\mathbf{z} \mid x) \parallel p(\mathbf{z})\right)$$

where $p(z) = N(0,I)$ denotes the standard normal prior. This divergence provides a sample-level measure of how atypical an inputs latent representation is relative to the models learned latent structure. KL divergence values were extracted during inference for each ensemble member and averaged per segment to obtain a single latent divergence score. Higher values of RS_{latent} indicate greater deviation from the learned distribution and correspond to reduced prediction reliability for the given segment.

MCD variance **(**$\mathbf{RS_{MCD}}$**)** was computed by enabling dropout layers in the VAE during inference and performing M stochastic forward passes for each input segment ($M = 20$ in this experiment). This

procedure provides a Bayesian approximation of epistemic uncertainty by exposing variability in the model's predictions under stochastic perturbations of its internal representations.

For a given input segment x, the model produced M reconstructions (one per forward pass), indexed by j:

$$\left\{ f_{\text{drop}}^{(j)}(x) \right\}_{j=1}^{M}$$

MCD uncertainty was then computed as the variance across these outputs:

$$RS_{\text{MCD}}(x) = \text{Var}\left(\left\{ f_{\text{drop}}^{(j)}(x) \right\}_{j=1}^{M} \right)$$

where higher values of RS_{MCD} indicate greater variability in the model's outputs and correspond to reduced prediction reliability for the given segment.

Normalization was applied to the three reliability metrics using min–max scaling. For each metric RS_k , raw values were normalized to the [0, 1] range according to:

$$RS_{k,\,\text{norm}}(x) = \frac{RS_k(x) - \min(RS_k)}{\max(RS_k) - \min(RS_k)}$$

where $RS_k(x)$ is the raw value of metric k for prediction x, and $\min(RS_k)$ and $\max(RS_k)$ correspond to the minimum and maximum values observed for that metric on the validation set. This process was applied independently to each reliability metric.

The resulting normalization parameters were fixed after validation and applied unchanged during test-set evaluation. As a result, all reliability metrics lie on a common scale and are directly comparable for subsequent aggregation into the composite reliability score.

Composite reliability score **(RS_{total})** was computed for each test segment by aggregating the three separate reliability metrics into a single scalar value. The composite score was defined as a weighted sum of the normalized metrics, using equal weights ($w_1 = w_2 = w_3 = 1/3$):

$$RS_{\text{total}} = w_1 \cdot RS_{\text{ensemble}} + w_2 \cdot RS_{\text{latent}} + w_3 \cdot RS_{\text{MCD}}$$

where w_1, w_2, and w_3 are configurable weights (equal by default) that can be tuned to reflect domain-specific priorities, to elevate or reduce the contribution or importance of any metric.

The resulting composite score provides a unified indicator of prediction reliability for each segment, with higher values corresponding to more reliable model output stability and lower values indicating elevated uncertainty. RS_{total} was subsequently used for thresholding and reliability-based flagging during evaluation.

4.3.2 Thresholding and Flagging

In this experiment, reliability scores were not only computed for analysis purposes but were actively used to determine how each prediction should be treated. After calculating and normalizing the reliability metrics for every test segment, these values were translated into operational actions through a thresholding procedure that determined whether predictions were flagged as uncertain or not. This

ensured that uncertainty analysis was explicitly incorporated into the evaluation pipeline, in accordance with the RAF principles introduced in Chapter 3.

4.3.2.1 Thresholding Policy and Configuration.

An *empirical threshold* setting was based on the distribution of RS_{total} values on the validation set. In this experiment, the threshold was set at the 10th percentile, such that the lowest scoring 10% of predictions were flagged as uncertain.

Domain-specific policy configurations were not varied in this controlled study, but the implementation supports stricter or more lenient policies. For instance, a safety-critical deployment could flag a prediction if any individual metric falls below threshold, whereas this experiment relied solely on the composite RS_{total}.

Situational and adaptive thresholding was not activated here, as the synthetic data and stable evaluation environment made a static threshold sufficient. However, the framework supports dynamic threshold adjustments when operating in settings where data characteristics evolve over time.

Transparency and auditability were maintained by recording the threshold value, reliability score distributions, and per-segment reliability metrics. This ensures that every decision can be traced and inspected retrospectively.

In this experiment, the thresholding mechanism was intentionally kept simple, without adaptive logic or UI components, to ensure controlled evaluation of the RAF's core reliability metrics.

4.3.2.2 Action Logic: Flagging and HITL. Once thresholding has been applied, the reliability scores were translated into operational labels for downstream logic and structured analysis. The primary mechanism in this process was flagging.

Flagging was used to separate predictions who's composite RS_{total} fell below the configured threshold. These predictions were labelled as unreliable in all analysis outputs and visualizations. In this experiment, flagging does not suppress or remove predictions, rather, it allows downstream processes—or a human reviewer—to immediately identify low-confidence cases.

HITL and automation were not operationalized in this experiment but is supported conceptually by the RAF. Flagged predictions could trigger escalation to human review, be masked (excluded from automated downstream decision processes), or be combined with secondary data sources (e.g., ECG) for validation before use. Here, they served only to allow structured comparison between flagged (low confidence) and retained predictions.

A "*no green signals" policy* was applied, meaning that predictions were never explicitly marked as "reliable", or color-coded green. Only unreliable predictions were highlighted, ensuring that unflagged predictions were interpreted only as not falling below the threshold, rather than as positively validated. For readability, however, in some instances the non-flagged subset is referred to as "trusted". This denotes only "not flagged by the reliability threshold" and is not a positive reliability guarantee.

4.3.2.3 Evaluation of Reliability Mechanisms. To assess the practical effect of reliability-based thresholding in this experiment, predictions were analyzed separately for flagged and non-flagged subsets. For each dataset (validation and test), every prediction was assigned to one of these two groups based on the configured RS_{total} threshold. Predictions falling below the threshold were labelled as uncertain. Importantly, no predictions were marked as reliable, in accordance with the RAF's caution-first design.

Metrics and performance evaluation—performance evaluation was carried out by computing precision, recall, and F1-score independently for the flagged and non-flagged subsets. Accuracy was also reported for completeness, although it is less meaningful given the strong class imbalance

inherent in time-series event detection. Analyzing these two subsets separately made it possible to determine whether uncertain or erroneous predictions were indeed concentrated in the flagged group, as expected for an effective thresholding mechanism.

The analysis was intentionally descriptive rather than inferential. No formal statistical hypothesis tests (e.g., t-tests or chi-square tests) were applied, as the experiment uses synthetic data, aggregated event counts, and does not involve repeated paired measurements. Instead, the separation of metrics across flagged and non-flagged subsets served to evaluate whether the reliability threshold functioned as intended—isolating uncertain predictions without requiring formal statistical significance testing.

Error types and analysis—beyond basic metrics, deeper insight is obtained by categorizing and quantifying specific prediction error types occurring within flagged versus non-flagged predictions. The error categories are: false positives (FP), false negatives (FN), and OOD errors—predictions while encountering significantly unfamiliar data (detected by VAE latent space divergence).

We explicitly evaluate the distribution of these errors across both subsets. An effective reliability assessment tool is expected to flag uncertain predictions preferentially, resulting in a higher error concentration among flagged compared to non-flagged predictions.

Visualizations and interpretability—for clarity and transparency, the metrics above are presented alongside intuitive visualizations, such as: Reliability histograms (illustrating the distribution of RS_{total} across flagged and non-flagged predictions), Error bar plots (comparing accuracy, precision, recall and F1-score metrics explicitly for both subsets, and Confusion matrices (visualizing detailed classification errors in predictions, facilitating interpretability).

Coverage and safety trade-off analysis—implementing reliability thresholding inherently involves trade-offs between prediction coverage and overall system safety. To explicitly characterize these trade-offs, we report:

(1) Coverage: Percentage of total predictions marked as unreliable versus not.

(2) Safety Improvement: Reduction in false positives or critical errors within the trusted subset compared to the entire prediction set.

Quantifying this trade-off provides valuable operational guidance and demonstrates clear benefits of reliability-driven prediction management, particularly in safety-critical environments.

4.4 Method: Experiment Design and Evaluation Protocol

This section details the experimental setup used to validate the RAF in the context of radar-based heartbeat detection. It covers the data split strategy, curriculum learning implementation, ensemble training procedure, and the evaluation protocol applied to both performance and reliability metrics.

4.4.1 Dataset Generation and Splitting Strategy

For each experimental run, a total of nine unique datasets were created, each comprising 10,000 samples of 15-second synthetic radar segments (sampled at 200 Hz, yielding 3,000 samples per segment). The datasets are structured as follows:

4.4.1.1 Test Set.

A single, independent test set was generated prior to any model training and strictly isolated from all models until final evaluation. This test set remains the same for all models, ensuring that every baseline and member is evaluated on identical unseen data. This strict holdout prevents any possibility of information leakage or overfitting to test samples, thereby providing a fair assessment of

performance and reliability. The test set consisted of 10,000 samples, each 15 seconds long, generated as described in Section 4.1.2.

4.4.1.2 Baseline Curriculum Set.

For the baseline model, three distinct datasets were generated—one for each stage of increasing data complexity (low, medium, high). Each dataset was randomly generated and split 80/20 for training and validation within that stage. The curriculum learning procedure begins with training on the low-complexity set, with model weights successively fine-tuned on the medium, and the high-complexity sets, each with their own unique data. All datasets generated with identical dimensions, each comprising 10,000 samples of 15-second-long time series windows.

4.4.1.3 Ensemble Training Splits.

For the ensemble, five model instances were created, with architecture and parameters inherited from the final baseline model. For each member, its own dataset was generated, based on the high-complexity level baseline dataset, 10,000 samples each, and split 80/20. This procedure guarantees diversity among members—critical for robust uncertainty estimation—while preserving comparability, as all members are evaluated on the exact same held-out test set. The dataset generated for each member shares dimensions and complexity with the last stage of baseline training, but contains unique randomly generated samples for each member.

4.4.2 Curriculum Learning and Model Training

To enhance model robustness to noise and signal variability, we employed a curriculum learning strategy in training the baseline model. As described in Section 3.3.2.3, this approach progressively exposes the model to increasingly complex synthetic radar data, mirroring real-world signal challenges. Each complexity stage is trained with a freshly generated dataset. The model weights are transferred between stages, but datasets remain strictly non-overlapping.

4.4.2.1 Curriculum Schedule. The following training stages were employed:

(1) Stage 1 (low complexity): high signal-to-noise ratio, minimal motion artifacts.

(2) Stage 2 (medium complexity): moderate noise and motion artifacts introduced.

(3) Stage 3 (high complexity): full noise profile, including baseline drift, variable breathing frequency, and sporadic movement artifacts.

In practice, a model was randomly initialized and trained on the low-complexity data (80/20 split). The resulting weights were used to initialize training on the medium-complexity set. The final model, after high-complexity training, is referred to as the baseline model and forms the starting point for all ensemble members.

4.4.2.2 Ensemble Fine-Tuning. Each member was initialized using the architecture and hyperparameters inherited from the curriculum-trained baseline model. Members reinitialized with unique random seeds and trained independently on their own respective high-complexity synthetic dataset. No weights were shared between members during this phase. After training, all ensemble members were evaluated on the identical, strictly isolated test set, enabling direct comparison of ensemble performance and uncertainty estimates.

4.4.2.3 Notes. The specific complexity configurations for each stage were inspired by patterns in the real radar reference data (engine off, idling, driving), but are ultimately determined subjectively for this synthetic scenario. All data generations were fully controlled and logged for reproducibility.

4.4.3 Evaluation Protocol and Performance Metrics

Model evaluation is performed using a standardized pipeline:

(1) Input Processing: Each test segment is processed through the VAE→DAE pipeline.

(2) Heartbeat Detection: Heartbeats are detected using an optimized peak detection algorithm, with grid-searched parameters on validation data.

(3) Reliability Scoring: For each prediction, three reliability metrics are computed—ensemble disagreement (standard deviation across ensemble outputs), VAE latent space divergence (distance from training latent distribution), and MCD variance (variance across multiple stochastic forward passes). Each metric is normalized (min-max over validation set). A composite reliability score (RS_{total}) is then calculated as a weighted sum of the individual reliability scores ($RS_{ensemble}$, RS_{latent}, RS_{MCD}).

(4) Thresholding and Flagging: A single global threshold for RS_{total} was set to the 10th percentile. All test-set predictions with RS_{total} below this cutoff were flagged as uncertain, and those above it were kept as non-flagged predictions for separate evaluation.

(5) Metrics: For both flagged and unflagged subsets, the following metrics were computed: accuracy, precision, recall, F1-score, and coverage.

(6) Diagnostics and Visualization: Diagnostic metrics such as reliability score histograms, FP/FN rates, and ensemble agreement are logged and visualized for analysis.

Given the rare-event and safety-critical nature of the task, no single evaluation metric was sufficient to capture detection correctness, error trade-offs, and pointwise separability between heartbeat-labeled and background-labeled time samples. Therefore, a combination of event-based and pointwise metrics was used.

As for event-based evaluation, precision, recall, and F1-score were computed based on detected heartbeat events relative to ground-truth annotations. These metrics quantify the correctness of detected heartbeats, the proportion of missed events, and the tradeoff between false positives and false negatives. In addition, pointwise accuracy was computed over all time samples for completeness, although it is not used as a primary indicator of detection performance, due to the significant class-imbalance in the specific task. Coverage was reported for reliability-filtered results to quantify the proportion of predictions retained after thresholding. ROC/AUC (Receiver Operating Characteristic / Area Under the Curve) was computed as a supplementary waveform separability diagnostic. Using the heartbeat label mask as a pointwise ground truth (y_true(t)) and reconstructed signal amplitude (y_score(t)) as a continuous score, ROC curves were obtained by sweeping a threshold over the scores to compute the corresponding true positive rate (TPR) and false positive rate (FPR) across pooled time samples, and AUC summarizes the resulting ROC curve. This diagnostic reflects pointwise separability between heartbeat-labeled and background-labeled time samples and should not be interpreted as event-level peak detection performance. All metrics were computed using identical procedures for the baseline model, individual ensemble members, and ensemble-aggregated outputs.

4.5 Results

4.5.1 Baseline Model Performance

The baseline model consists of a single VAE→DAE pipeline trained using a curriculum learning schedule on synthetic radar data. It serves as the reference system against which all enhancements—ensemble methods, uncertainty estimation, and reliability thresholding—are evaluated.

4.5.1.1 Evaluation Protocol. The final baseline model was obtained by sequentially training on the low-, medium-, and high-complexity synthetic datasets described in Section 4.4.2.1. All evaluations use the final weights from the high-complexity stage.

Evaluation was performed exclusively on the strictly isolated high-complexity test set (N=10,000 samples), which was never used during training or validation. Each segment was processed through the VAE→DAE pipeline, and heartbeat locations were identified in the DAE output using the peak detection parameters (prominence, distance) optimized on the validation data.

No reliability masking or thresholding was applied. All predictions are reported directly, forming the unfiltered benchmark for later comparisons.

4.5.1.2 Results Summary.

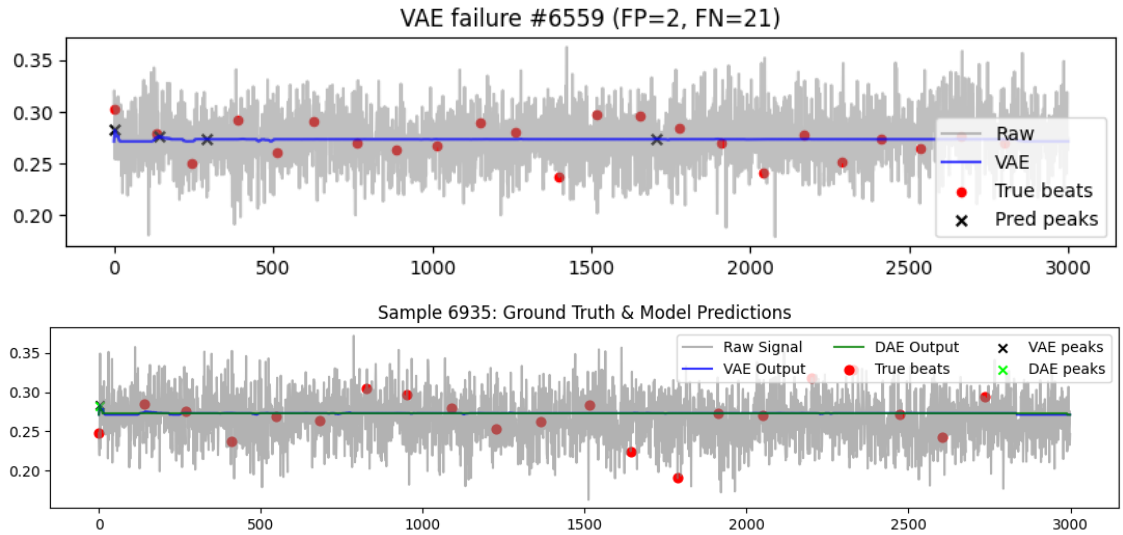
Table 4.4 summarizes baseline heartbeat-detection performance on the held-out test set. Metrics include event-based precision, recall, F1-score, false-positive and false-negative rates, and the average number of correctly detected heartbeats per 15-second segment. For completeness, results are shown for both the VAE reconstruction and the final DAE-denoised signal, highlighting how each stage affects heartbeat detectability.

Table 4.4 *Baseline Model Performance Metrics*

Metric	**VAE**	**DAE**
True positives (TP)	43,094	10,000
False positives (FP)	28,904	0
False negatives (FN)	139,988	173,082
Precision	59.85%	100%
Recall	23.54%	5.46%
F1-score (macro)	33.26%	10.61%
F1-score (strict)	33.79%	10.36%
False positive rate	40.15%	0.00%
False negative rate	76.46%	94.54%
ROC/AUC	0.5540	0.5266
Average heartbeats correctly detected per segment	7.2 (39.3%)	1.0 (5.5%)

Note. Pointwise accuracy is not reported here because the evaluated system outputs discrete heartbeat events (peak locations) rather than an explicit per-sample class decision. While a pointwise heartbeat mask is available for diagnostic analyses (ROC/AUC), reporting accuracy would require introducing an additional per sample binarization rule that is not part of the event-detection pipeline, and would be dominated by class imbalance.

Figure 4.6, 4.7 *Baseline VAE & DAE (Final Output) Signal Plot: Predicted vs Ground Truth*



4.5.2 Ensemble Performance and RAF Metrics

The ensemble model consists of five independently trained VAE→DAE pipelines, each initialized with a unique random seed and trained on its own high-complexity synthetic dataset. For each test segment, the reconstructed outputs from all members were averaged to form the ensemble signal. Reliability metrics were computed for each segment and later used for reliability-based filtering via the composite score RS_{total}.

4.5.2.1 Evaluation Protocol. The ensemble system was evaluated on the isolated high-complexity test set (N = 10,000 samples). For each test segment, all five members processed the same input independently, producing their own reconstructed outputs. The outputs of all members were averaged to generate a consensus ensemble signal. Heartbeat detection was then applied to this ensemble-averaged signal using a peak detection algorithm, with parameters (prominence, distance) optimized on the validation set. Reliability metrics ($RS_{ensemble}$, RS_{latent}, and RS_{MCD}) were computed for each segment, normalized, and combined into the composite RS_{total} which was used for threshold-filtering.

The performance metrics reported below reflect post-RAF evaluation, i.e., after applying the reliability threshold to isolate the higher-confidence predictions.

4.5.2.2 Results Summary.

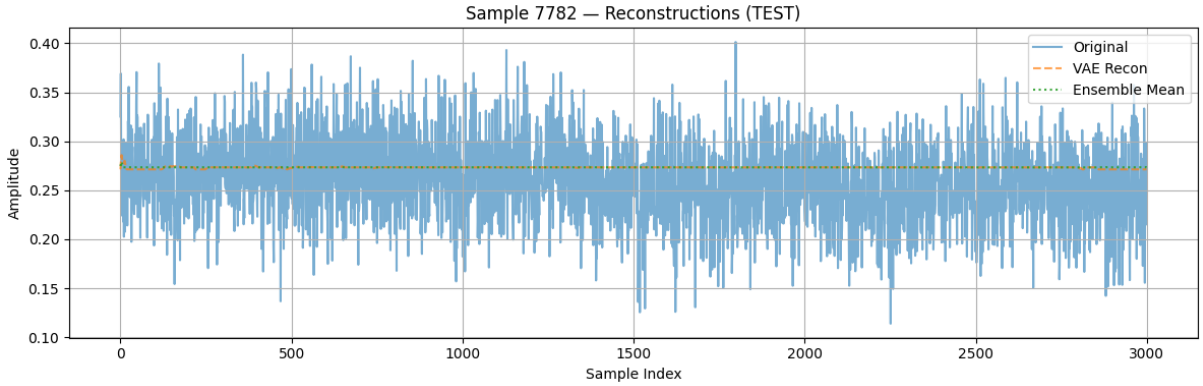
The post-RAF performance of the ensemble model is shown in Table 4.5. Applying reliability filtering retained 90% of predictions (unflagged) while substantially reducing false positives relative to the unfiltered VAE baseline, and maintaining a low false-positive rate overall. Recall remained low due to the difficulty of the high-complexity test data, but the ensemble achieved higher consistently high precision and improved false-positive behavior without inflating recall.

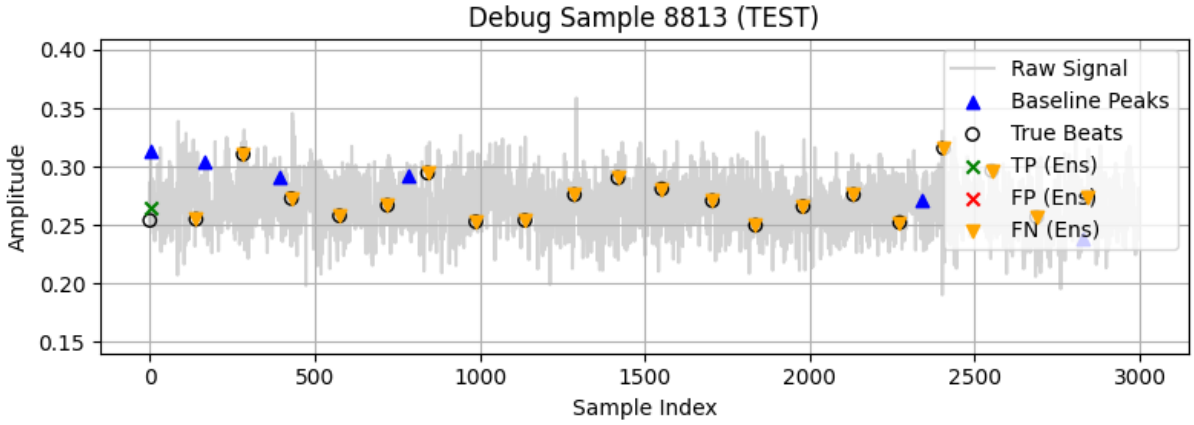
Table 4.5 *Post RAF Ensemble Performance*

Metric	Ensemble (post-RAF)
True positives (TP)	10,009
False positives (FP)	35
False negatives (FN)	173,073
Precision	99.65%
Recall	5.47%
F1-score (macro)	10.61%

F1-score (strict)	10.37%
False positive rate	0.35%
False negative rate	94.53%
ROC/AUC	0.5219
Average heartbeats correctly detected per segment	7.2 (39.3%)
Composite reliability threshold	0.6491
Coverage (unmasked)	90.00%

Figure 4.8, 4.9 *Ensemble (Final Output), Reconstruction & Heartbeat Predictions*





4.5.2.3 Comparison to Baseline.

Table 4.6 illustrates the effects of post-assessment reliability filtering, by comparing baseline metrics to post ensemble metrics. The delta-column shows the absolute change in each metric after applying the RAF reliability assessment techniques. These deltas provide a direct measure of the practical impact of the proposed approach.

Table 4.6 *Performance Deltas Pre/Post Reliability Assessment*

Metric	**Baseline**	**Ensemble**	**Δ (Ensemble - baseline)**
Precision	98.51%	99.65%	+1.15
Recall	5.76%	5.47%	-0.29
F1-score (strict)	10.89%	10.37%	-0.52
False positive rate	1.49%	34.85%	+33.35
False negative rate	94.24%	94.53%	+0.29
ROC/AUC	0.554	0.522	-0.03

Coverage (unmasked)	100%	90%	-10
Predictions masked	0	1000	N/A

4.5.3 Reliability Assessment: Statistics and Visualizations

This section analyzes the reliability metrics produced during ensemble evaluation. For each test segment, the three core RAF metrics ($RS_{ensemble}$, RS_{latent} and RS_{MCD}) were computed and normalized using validation-set statistics, and then combined into the composite RS_{total} score. These results describe the distribution and behavior of the reliability measures prior to thresholding.

4.5.3.1 Score Distributions and Normalization. The following statistics were obtained over the test set:

Table 4.7 *Summary Statistics for Reliability Scores*

Score type	Mean	Std dev	Min	Max
$RS_{ensemble}$	0.692	0.129	0.000	1.000
RS_{latent}	0.737	0.329	0.000	1.000
RS_{MCD}	0.867	0.080	0.190	0.998
RS_{total}	0.775	0.096	0.223	0.984

Note. Higher values indicate greater model agreement or confidence according to each metric.

4.5.3.2 Correlation with Prediction Quality. To assess whether higher reliability metrics corresponded to more accurate predictions, we calculated the correlation between each score and a binary correctness label for each heartbeat (1 = correct, 0 = incorrect or missed).

Table 4.8 *Correlation: Reliability Scores – Prediction Correctness*

Reliability score	Correlation (r)
$RS_{ensemble}$	0.53
RS_{latent}	0.47
RS_{MCD}	0.39
RS_{total}	0.58

Note. Higher values indicate a stronger positive association between the reliability score and the likelihood of a correct heartbeat detection.

4.5.4 Reliability Thresholding and Flagging Outcomes

This section evaluates the effect of applying the composite reliability score to separate high-confidence predictions from low-confidence ones. The goal is to determine whether reliability-based thresholding can isolate uncertain predictions—thereby improving safety and interpretability—without excessively reducing usable coverage.

4.5.4.1 Thresholding Procedure. A global threshold for RS_{total} was selected based on the distribution of reliability scores in the validation set. For this experiment, the 10th percentile of RS_{total} was used as the cutoff, meaning that the lowest-scoring 10% of predictions were flagged as unreliable. Thresholding was applied post hoc after ensemble prediction, ensuring that the evaluation reflects the final model behavior.

4.5.4.2 Flagged vs non-flagged Prediction Analysis.

Predictions with RS_{total} below the threshold form the flagged subset, while those above form the retained subset. This separation enables a direct comparison between the two groups to assess whether the reliability metric effectively concentrates uncertain or error-prone predictions into the flagged category.

4.5.4.3 Outcome Metrics and Coverage.

Table 4.9 summarizes performance for the trusted subset, the full test set, and the flagged subset. Coverage denotes the proportion of predictions retained after thresholding

Table 4.9 *Reliability Flagging Metrics*

Metric	Trusted only	Full test set	Flagged only
Accuracy	0.994	0.994	0.994
Precision	0.044	0.045	0.055
Recall	0.002	0.002	0.003
F1-score	0.005	0.005	0.006
Coverage (% non-flagged)	90%	100%	10%
Error rate (non-flagged)	0.6%	0.6%	0.6%
Error enrichment factor	N/A	N/A	1.00x

Note. Accuracy is reported pointwise across all time samples, however, due to class imbalance it is not a meaningful indicator of heartbeat detection quality. Event-based metrics (precision, recall, F1) are the primary evaluation indicators.

5 Discussion

5.1 Overview of Findings

This thesis set out to define, develop and rigorously evaluate the RAF for AI-driven radar-based heartbeat detection use-case. The goal was to address the critical need for trustworthy, risk-aware signal analysis in safety-critical applications by integrating uncertainty estimation, ensemble modeling, and post-hoc reliability scoring into the detection pipeline.

Through a comprehensive series of experiments using curriculum-trained deep learning models on synthetic radar data, we assessed event-level detection performance metrics (precision, recall, and F1-score) together with reliability-based metrics. The baseline and ensemble models were subjected to a multi-stage evaluation protocol, including reliability flagging and correlation analyses.

The key findings indicate that, under current model and data configurations, event-detection performance remains limited, with all tested approaches achieving high pointwise accuracy but very low precision, recall, and F1-score for heartbeat detection events. The application of reliability flagging via a composite reliability score (RS_{total}) did, in this implementation, show limited enrichment of errors in the flagged subset, and it did not significantly improve predictive performance in the trusted subset. Correlation analyses revealed that while reliability scores are modestly predictive of detection correctness, further refinement is required for robust risk filtering.

Overall, the work establishes a robust experimental and analytical framework for reliability assessment in AI-driven systems, while also highlighting key challenges—such as extreme class imbalance in the evaluated scenario, and the need for improved event-detection sensitivity—that must be addressed in future research and real-world deployment.

5.2 Interpretation of Evaluation Metrics

Throughout this chapter results are reported under two distinct evaluation protocols: (1) event-level evaluation, where detected heartbeat peaks are matched to ground-truth events and assessed using precision, recall, and F1-score, and (2) pointwise evaluation, where individual time samples are labeled as heartbeat or background and assessed using precision, recall, and ROC/AUC. These metrics capture different aspects of system behavior and are therefore not directly comparable.

Interpreting the results of the heartbeat detection experiment requires particular care due to the extreme class imbalance of the task. Each 15-second segment contains only a small number of true heartbeat events, relative to background signal (non-heartbeat). Because of this, pointwise accuracy remains high even when event-detection performance is poor and should not be interpreted as evidence of effective heartbeat detection.

The event-based metrics reported—precision, recall, and F1-score—reveal that detection performance was severely constrained under the evaluated conditions. In particular, the denoising stage of the pipeline (the DAE) over-cleaned the raw radar signal, reducing the amplitude/prominence of heartbeat-related peaks. Under the chosen peak detection thresholds, this produced few predicted events, which lowered false positives (high precision) but missed the majority of true heartbeats (low recall). The resulting low F1-score reflects this imbalance.

ROC/AUC was computed as a supplementary diagnostic measure, but its interpretation differs fundamentally from event-based metrics in this study. Here, ROC/AUC is computed pointwise using reconstructed signal amplitudes as a continuous score against the heartbeat label mask, and therefore reflects separability between heartbeat-labeled and background-labeled time-samples rather than event-level peak detection performance. The observed ROC/AUC values close to 0.5 indicate near chance-level amplitude separability, which is consistent with weak heartbeat signatures in the reconstructed signals. This explains the seemingly counterintuitive combination of high precision and low ROC/AUC: precision is driven by conservative peak detection logic (few predicted events), while ROC/AUC reflects poor separability across pooled time samples. These metrics capture different aspects of system behavior and are not expected to covary in these conditions.

Rather than necessarily indicating overfitting or implementation errors, the results highlight the limitations of amplitude-based separability and thresholded peak detection in rare-event scenarios, and are in this scenario consistent with the observed over-cleaning after the DAE stage.

5.3 Baseline Performance Analysis

The baseline model provided a foundational reference point for heartbeat detection performance. Evaluation results revealed high pointwise accuracy (~0.994), however, this metric is primarily a reflection of the overwhelming proportion of non-event (background) samples in the dataset and does not meaningfully indicate the model's true detection ability. In contrast, event-based metrics—precision (~4.4%), recall (~0.2%), and F1-score (~0.005)—were all markedly low, highlighting the intrinsic difficulty of reliable heartbeat detection in noisy radar data and underscoring the severe class imbalance inherent to the problem.

These findings were broadly anticipated given the nature of the synthetic data generation process and the complexity of the detection task. Still, the dramatic gap between overall accuracy and event-wise performance was striking. The low recall confirms that the baseline model missed the vast majority of true heartbeats, while the low precision indicates that most detected events were actually false positives.

A closer examination of model architecture further clarifies this trade-off. At the event level, under the selected peak detection thresholds, the DAE stage exhibited near-zero false positives but very low recall, reflecting a highly conservative operating regime in which most true heartbeats were missed. Conversely, the VAE stage detected a larger number of true events, but at the cost of a much higher false positive rate. The macro F1-scores capture this sensitivity-specificity trade-off at the event-level, while ROC/AUC reflects pointwise amplitude separability between heartbeat-labeled and background signal samples rather than event-level detection performance. Notably, the average number of heartbeats detected per segment by the DAE corresponded to only 5.5% of the true average, further illustrating the system's tendency toward under-detection.

The baseline model lacked any uncertainty quantification or reliability assessment; as such, it provided no mechanisms for flagging potentially unreliable predictions. This limitation makes the baseline ill-suited for safety-critical scenarios, where transparency, reliability, and risk-aware decision support are essential. The absence of per-sample reliability measures further restricts its practical utility in high-risk environments. These limitations collectively confirm the necessity for architectural enhancements and additional reliability mechanisms—such as ensemble-based redundancy and explicit reliability scoring—to improve both the sensitivity and trustworthiness of heartbeat detection in this application.

5.4 Ensemble and Post-Assessment Analysis

The use of an ensemble architecture and the RAF was intended to improve detection robustness and enable risk-aware prediction filtering. By aggregating outputs from five independently trained VAE→DAE pipelines, the ensemble aimed to capture a broader range of data variations and uncertainties, while composite reliability scores provided a principled basis for flagging low-confidence predictions.

Despite these enhancements, post-assessment analysis revealed that the ensemble approach—whether evaluated across the full test set or after flagging unreliable predictions—did not yield substantial improvements over the baseline in event-detection performance. Core metrics such as precision (~4.5%), recall (~0.2%), and F1-score (~0.005) remained virtually unchanged, and the application of reliability-based flagging did not significantly isolate errors in the flagged subset or improve the performance of the trusted set. Both subsets exhibited similarly low F1-scores and event detection rates, with an error enrichment factor near unity.

Interestingly, while high event-level precision was achieved among retained predictions—confirming that almost all trusted outputs corresponded to true heartbeats—this came at the cost of a further reduction in recall. The substantial increase in reported false positive rate and the minimal change in overall detection metrics suggest that, under the current synthetic data and model setup, ensemble diversity and reliability flagging were insufficient to overcome the underlying challenges of rare-event detection.

These results were somewhat unexpected, as ensemble methods are typically associated with gains in robustness and uncertainty quantification. The findings underscore the need for further methodological refinement, potentially including real data augmentation, improved model architectures, or alternative reliability metrics, to fully realize the potential benefits of ensemble-based reliability assessment for event detection in noisy physiological data.

5.5 Reliability Assessment Performance

In this study, the RAF provided a multi-faceted approach to quantifying model confidence by integrating ensemble disagreement ($RS_{ensemble}$), latent space divergence (RS_{latent}), and MCD variance (RS_{MCD}), as well as the weighted composite reliability score (RS_{total}). All reliability metrics were normalized against the distribution of validation set values, enabling meaningful interpretation, potential threshold selection, and direct comparison.

Analysis of reliability score distributions across the test set revealed that most predictions were assigned moderate to high reliability, with mean RS values ranging from approximately 0.69 to 0.87 for the different components. The composite RS_{total} showed a mean of 0.775 (std = 0.096), indicating that the framework generally assigned higher confidence to its outputs.

To assess whether the reliability metrics reflect prediction quality, point-biserial correlation coefficients were computed between each score and a binary indicator of correct event detection (Table 4.8). All reliability measures showed positive but moderate association with event-level prediction correctness (e.g., $r = 0.53$ for $RS_{ensemble}$, $r = 0.58$ for RS_{total}), demonstrating that higher

reliability scores tended to align with correct heartbeat detections. Summary statistics further showed that reliability scores were generally concentrated at moderate to high values across the test set (Table 4.7). However, despite these tendencies, substantial overlap between correct and incorrect predictions remained. Consistent with this, thresholding analysis using a 10th-percentile cutoff did not meaningfully separate outcomes or enrich errors in the flagged subset (Table 4.9).

Taken together, these results indicate that the reliability metrics used are interpretable and show a moderate positive association with prediction correctness. However, applying the reliability threshold did not make the retained predictions meaningfully more correct than the full set or concentrate errors in the flagged subset, motivating further refinement of both the reliability scoring and threshold selection or calibration.

5.6 Reliability Flagging Outcomes

The practical impact of reliability flagging was assessed by applying a global threshold (at the 10th percentile) to the composite reliability score (RS_{total}), partitioning predictions into trusted (non-flagged) and flagged subsets. The intention was to determine whether flagging low-confidence predictions could enhance safety and interpretability by isolating the most uncertain outputs, while retaining high coverage for downstream use.

Reliability-based flagging in this study was evaluated primarily by using event-based performance metrics (precision, recall, and F1-score), as these directly reflect the correctness of heartbeat detections. ROC/AUC was not used to assess filtering effectiveness, since it is computed pointwise on raw signal amplitude and does not capture event-level detection quality.

The results, however, indicated that reliability-based flagging did not achieve the anticipated enrichment of errors within the flagged subset. Core event-detection metrics remained nearly identical across trusted, flagged, and full test set predictions. The error enrichment factor for the flagged subset was approximately 1.00×, meaning that flagged predictions were no more likely to be incorrect than the trusted ones. In addition, pointwise accuracy and error rate were essentially unchanged across subsets, reflecting that the thresholding did not materially alter overall correctness under the present context.

In practical terms, this outcome underscores both the promises and the limitations of the evaluated instantiation of the RAF. While the framework enables flexible, post-hoc filtering based on reliability estimates, its effectiveness depends fundamentally on the discriminative power of the underlying scores. As presently configured, neither the ensemble architecture nor the composite reliability measure yielded the desired separation of reliable and unreliable predictions. This highlights the need for further research, potentially including model improvements, enhanced reliability metrics, or alternative post-processing strategies.

5.7 Key Trends and Takeaways

Across the series of experiments, several consistent trends and lessons emerged. The integration of an ensemble modeling approach and a RAF allowed for the systematic exploration of uncertainty and risk-aware prediction in the context of radar-based heartbeat detection. While the absolute event-detection performance remained modest, some reliability scores demonstrated a moderate positive association with event-level prediction correctness (e.g., $r = 0.58$ for RS_{total}). This, though modest, supports the conceptual value of reliability scoring as a component in future risk management and decision support, although detection performance was not sufficient for deployment in this setup.

A second key takeaway is that evaluation outcomes in rare-event heartbeat detection are strongly shaped by task characteristics such as extreme class imbalance and the divergence between pointwise reconstruction quality and event-level detection performance. As discussed in Section 5.2, this can

lead to counterintuitive metric combinations—such as high pointwise accuracy coexisting with poor event-level precision, recall, and F1-score—underscoring the importance of interpreting results in relation to the intended operational task and evaluation protocol.

5.8 Limitations and Threats to Validity

In this study, several limitations were identified that constrain the interpretability and external validity of the findings. Threshold selection for reliability flagging relied on a fixed heuristic, however, more domain-informed calibration and cost-sensitive thresholding strategies may be required for robust deployment. Thresholds should ideally be chosen based on cost trade-offs (false reassurance versus false alarm) and coverage trade-offs (flagging rate versus performance), or other domain-specific factors, not just percentile heuristics.

A further limitation concerns interpretability and human oversight. Although the RAF provides post-hoc reliability scores, the current implementation used in the above experiment does not directly expose, or present, actionable explanations for low or high reliability states. This constrains immediate usability for HITL review and makes it difficult to translate reliability outputs into transparent operational guidance.

Finally, the empirical scope of the study is limited. No experiments were conducted on real or clinical datasets, and no user studies were performed to assess how reliability information affects perceived trust, decision-making, or usability in practice. Moreover, the detection models were not optimized for peak performance or broad generalizability, and additional contextual data sources (such as accelerometer signals for example) were not incorporated. These limitations do not undermine the conceptual motivation of the RAF, but they define the boundaries of its current implementation and empirical evaluation and motivate specific next steps.

5.9 Implications for Practice

At the level of the present experiment, the observed limited event-detection performance (likely due to excessive denoising) together with reliability scores that showed a moderate positive correlation with event-detection correctness, highlights a crucial fact: metrics meant to indicate system performance or reliability must be interpreted in context and supported by human oversight, and must not be treated as standalone guarantors of correctness but should be assessed together with complementary metrics and analysis. This also makes clear why reliability scores should be interpreted, and used as decision-support signals, not as autonomous decision triggers—the need for robust HITL-integration remains. This reinforces the RAF's call for a broad, multi-layered safety architecture, where mechanisms and metrics overlap to achieve fail-acceptable system behavior.

Another design requirement that was exemplified is the need for clear system–user communication and integrated interpretability mechanisms. For reliability information to be operationally meaningful, users must understand what the scores represent, what actions they are intended to support, and their limitations. This will likely not only be important for regulatory compliance and system usability, but also to support appropriate user trust.

5.10 Future Work and Recommendations

Building on this foundation, several avenues for future research emerge. First, while the RAF is intended to be modality-agnostic, validating the present radar-based instantiation on real-world and clinical radar datasets will be critical to assess generalizability under practical, domain-specific conditions. Longitudinal evaluation would further test the generalizability under varying environmental and user characteristics.

Further optimization of both detection models and reliability metrics is also necessary to improve event-detection sensitivity and enhance the ability of the framework to discriminate between high- and low-risk predictions. Improvements could include optimized noise-reduction models, richer feature extraction, and further model-combination experimentation. More advanced noise-reduction strategies were investigated during the preparatory work for this study but not implemented. This, however, remains an interesting avenue for future work, especially using various pre-filtering techniques based on likely amplitude ranges for example.

Expanding the input data to include additional sources (such as accelerometer- or other contextual signals) could enable a more precise and context-aware approach to reliability assessment, more suited for real-world deployment. For example, accelerometer data could be used for filtering out components of the signal produced by car movements. In parallel, user studies and interpretability research are recommended to investigate how reliability information impacts human trust, workload, and decision-making, and to determine how to best communicate reliability outputs to support safety.

Finally, translating reliability scores into real-time user-facing alerts, or automated system controls, remains a crucial practical direction to explore. Here, an informative user interface would likely have to be developed and integrated, as described by the RAF. Proposed mechanisms include reliability-aware filtering, automated fail-safes, and HITL-triggers that respond appropriately and automatically to uncertainty. This could provide dual benefits—improved functionality of the system, and increased transparency—which together may improve both practical utility and societal value by supporting more appropriately calibrated trust.

5.11 Ethical Considerations and Deployment Risks

This thesis investigates a reliability assessment framework using AI-enabled radar-based heartbeat detection as an example use case. Ethical considerations therefore arise at two levels: the conduct and interpretation of the experiment itself, and implications for responsible use if a RAF-guided system were operationalized in safety-critical settings.

The experimental work in this thesis was conducted primarily using synthetic data. As such, no personal data were processed within the main experimental pipeline, and risks related to privacy, consent, or participant harm were minimal. The synthetic signals were inspired by characteristics observed in a real-world radar dataset recorded from human subjects. However, this source material was accessed only in anonymized form, with no identifiable information available, and no user studies were conducted. Therefore, formal ethical approval was not required for the present study.

The deployment of AI raises well-documented ethical concerns, including the risk of harm from unreliable predictions, the potential for algorithmic bias, and the opacity of complex models. A key ethical risk associated even with reliability-aware AI systems is mis-calibrated trust—reliability scores or confidence indicators presented without careful design and guidance may be misinterpreted as precise guarantees of correctness, increasing the likelihood of inappropriate reliance and trust.

The RAF explicitly motivates system design principles intended to mitigate such risks, including transparent uncertainty communication, structured flagging of low-confidence states, and support for HITL oversight and escalation. However, the framework itself does not guarantee ethical deployment. Its responsible use depends on integration within appropriate governance structures, including auditability mechanisms, human oversight, and domain-specific operational safeguards. Overall, this thesis recognizes that increased reliability, trustworthiness, and transparency are not merely technical objectives, but ethical requirements.

5.12 Conclusion of Discussion

In summary, this study provides a systematic investigation into reliability assessment for AI-enabled radar heartbeat detection, combining ensemble modeling and multi-layered uncertainty quantification within a unified experimental pipeline. The experimental findings should be interpreted primarily as an initial validation of the RAF's conceptual and architectural viability, and a limited showcasing of the framework "in action", rather than as a performance-optimized evaluation of heartbeat detection tasks. While the immediate improvements in event-detection performance and error isolation were modest, the framework demonstrated a pathway for risk-aware and reliability-driven system design, and revealed both limitations in this instantiation and critical amendments for further evaluation and experimentation. The lessons learned regarding data constraints, metric calibration, and interpretability point directly to future research priorities. Ultimately, the findings underscore the importance—and challenge—of developing reliable, transparent, and generalizable AI solutions for safety-critical domains, and lay the groundwork (albeit modest) for continued advancement in this area.

6 Conclusion

Chapter 6 concludes the thesis by addressing the research questions, synthesizing their implications in relation to the three pivot points, and summarizing the core contributions of the proposed framework.

6.1 Addressing the Research Questions

This section answers the research questions posed in Chapter 1, drawing on the theoretical foundations presented in Chapter 3 and the empirical analyses in Chapters 4 and 5.

6.1.1 Primary Research Question

PRQ: *In what ways, and to what extent, can a modular, multi-layered reliability assessment framework maintain or improve AI system performance while systematically increasing both system-level trustworthiness and end-user perceived trust in safety-critical domains?*

Conclusion: The results show that the proposed RAF did not improve event-based heartbeat detection performance under the evaluated conditions, but maintained predictive behavior compared to baseline. The primary empirical contribution of this work is therefore not improved detection performance, but the implementation and analysis of a modular reliability-assessment layer that computes uncertainty metrics (RS_{latent}, $RS_{ensemble}$ and RS_{MCD}), aggregates them into a composite score (RS_{total}), and applies a defined threshold-based flagging policy, without compromising with predictive performance.

At the framework level, the thesis demonstrates that reliability assessment can be integrated as a modular layer within an end-to-end AI processing pipeline without disrupting core predictive functionality. This supports the architectural feasibility of combining performance-oriented modeling components with reliability-oriented monitoring and control layers within a unified system design. The results therefore indicate that a multi-layered reliability framework can be realized in practice without introducing structural conflicts between performance optimization and reliability-oriented system design goals.

With respect to end-user perceived trust, no user studies were conducted within the scope of this work. The thesis therefore does not provide empirical evidence regarding how reliability outputs affect user trust or decision-making. However, the work demonstrates that user-relevant reliability information can be generated, communicated, and logged systematically, providing a technical foundation for future research evaluating how such information influences user trust and decision behavior in real-world contexts.

Taken together, these findings indicate that the RAF is a technically feasible architectural approach for integrating predictive modeling, reliability assessment, and user-facing reliability communication within a single system design. While not all relevant dimensions were empirically validated within the scope of this work, the combined empirical and theoretical contributions demonstrates that the framework can support the coordinated treatment of predictive performance, system trustworthiness, and user-facing reliability transparency in safety-critical AI system design, meaning that the RAF aligns with the proposed requirements associated with the three pivot points for viable AI system deployment.

6.1.2 Secondary Research Questions

SRQ1: Limitations of Performance-Only Evaluation. To what extent does predictive performance alone suffice for successful AI deployment in safety-critical domains, and what are the limitations of performance-driven evaluation without reliability assessment or human-centered principles?

Conclusion: When evaluated in isolation, pointwise accuracy was shown to be insufficient for fully describing model behavior in the experimental rare-event detection setting. Although overall accuracy remained high (due to class imbalance), event-based metrics (precision, recall, and F1-score) revealed that the system failed to detect the majority of true heartbeat events. This illustrates that accuracy-driven evaluation can obscure operationally relevant failures in imbalanced event-detection scenarios.

Performance-only evaluation also lacks mechanisms for identifying unreliable individual predictions, communicating uncertainty, or supporting human oversight. Without complementary reliability assessment or human-centered evaluation principles, such failures may remain unnoticed, increasing the risk that system outputs are inappropriately trusted or dismissed.

The results therefore support the necessity of a more holistic evaluation approach that incorporates uncertainty- and reliability-aware analyses. Taken together, these findings support the RAF design principle that predictive performance alone cannot ensure safe, trustworthy or robust real-world deployment, and must be complemented by reliability assessment and human-centered safeguards, reinforcing the role of the first pivot point as a foundational dimension in successful AI system deployment.

SRQ2: System Trustworthiness via Reliability Assessment. How does the integration of layered reliability assessment methods (such as ensemble modeling, uncertainty quantification, and redundancy mechanisms) affect the statistical performance and operational safety of the system? To what extent do these methods enhance system trustworthiness?

Conclusion: The integration of layered reliability assessment methods (ensemble disagreement, latent space divergence, and MCD variance) did not improve event-detection performance under the evaluated conditions. However, these methods enabled structured uncertainty assessment via reliability scores and allowed a threshold-based flagging policy to be applied. In the evaluated configuration, the reliability scores showed a positive, moderate association with correctness, while flagging did not meaningfully separate incorrect from correct predictions nor improve event-level performance in the non-flagged subset. Importantly, the inclusion of reliability assessment did not reduce predictive performance, demonstrating that uncertainty quantification and reliability scoring can be applied without negatively impacting detection performance.

From a system design and theoretical perspective, the thesis supports the claim that layered reliability assessment contributes to system trustworthiness by enabling explicit uncertainty communication, structured flagging of low-confidence outputs, auditability, and mechanisms for human-centered oversight. These properties are argued throughout the thesis as core requirements for trustworthy AI systems, particularly in safety-critical contexts. While the empirical study did not directly evaluate changes in operational safety or trustworthiness outcomes, this thesis shows by theoretical argumentation that reliability assessment can be implemented as an integral part of AI systems to support trustworthy behavior.

Taken together, these findings support the feasibility of the RAF as an architectural approach for addressing the system trustworthiness pivot point, by enabling structured reliability assessment and communication without degrading predictive performance.

SRQ3: End-User Perceived Trust. How do user-facing reliability outputs (e.g., per-sample confidence scores, actionable visual alerts) influence perceived trust and support informed decision-making by end-users in real-world scenarios?

Conclusion: No user studies were conducted within the scope of this work, and this thesis therefore does not provide empirical evidence regarding how reliability outputs affect perceived trust or real world decision-making.

What this work establishes is the technical feasibility of generating and integrating user-relevant reliability information—including reliability scores, flagging of uncertain predictions or system states, together with conservative communication principles—directly into system outputs and logging structures. This provides an implementation-level basis for future empirical evaluation of how reliability information should be presented, interpreted, and acted on by end-users.

From a theoretical and human-centered design perspective, the thesis further argues that transparent communication of model uncertainty, explicit signaling of low-confidence system states, and traceable system behavior are foundational preconditions of calibrated user trust. By facilitating these capabilities at the system design level, the RAF provides the structural conditions required for trustworthy human–AI interaction, even though the behavioral effects on user trust were not empirically evaluated within this study.

6.2 Conclusion on the RAF and the Critical Pivot Points

This thesis set out to develop a modular and extensible reliability assessment framework for AI-based systems in safety-critical contexts and to assess its feasibility. The RAF was formalized conceptually, instantiated in a working pipeline, and evaluated through a controlled radar-based case study, showing that reliability-oriented components can be integrated into AI systems without disrupting core predictive functionality.

Central to the thesis is the proposal that successful AI system deployment depends on addressing three critical pivot points: predictive performance, system-level trustworthiness, and end-user perceived trust. Across these dimensions, the combined theoretical and empirical findings indicate that the RAF provides a feasible architectural and methodological approach for addressing all three within a unified system design. The case study demonstrates technical and practical feasibility of integrating reliability assessment alongside predictive modeling, while the theoretical contribution establishes the conceptual and human-centered rationale for why the proposed components of the RAF are not only viable, but argued in this thesis to be necessary conditions for trustworthy AI operation.

Taken together, the conclusions on the research questions reflect the three critical pivot points proposed in this thesis. SRQ1 establishes the limitations of predictive performance as a standalone success criterion, supporting the need for reliability-aware evaluation and multi-dimensional system assessment. SRQ2 demonstrates the feasibility of integrating layered reliability assessment as a mechanism contributing to system-level trustworthiness without degrading predictive performance. Finally, SRQ3 addresses the role of user-facing reliability communication as a structural prerequisite for informed human interaction and calibrated user trust.

As such, the RAF represents a principled and feasible foundation for reliability-aware AI system design in safety-critical domains.

6.3 Summary of RAF Contributions

To consolidate the contributions of the RAF, Table 6.1 summarizes the conceptual, architectural, and operational principles of the framework together with the system capabilities they enable. The table is intended to show how the RAF translates high-level requirements—such as reliability, trustworthiness, and user trust—into concrete design principles and behaviors that support viable AI deployment.

Table 6.1 *Core Contributions of the RAF*

RAF contribution	Resulting capability
Reliability as a first-class system property (distinct from	Enables explicit modeling, monitoring, and control of reliability as a system

predictive performance)	concern rather than an implicit by-product of performance metrics
Formalization of three critical pivot points for viable AI deployment	Enables a structured requirements model that guides AI system design and evaluation by ensuring that predictive performance, system trustworthiness, and end-user trust are jointly addressed from the outset, reducing the risk that any necessary condition for real-world viability is overlooked
Multi-metric reliability assessment	Enables holistic estimation of reliability by capturing complementary uncertainty signals and avoiding single-metric blind spots
Composite reliability scoring as an operational signal	Enables translation of heterogeneous reliability metrics into a unified, actionable reliability signal
Reliability-based system actions	Enables dynamic system responses based on assessed reliability rather than raw predictions
Fail-acceptable system behavior	Enables safety-oriented system behavior in which outcomes remain acceptable to relevant stakeholders despite the probabilistic nature of AI, without implying guarantees of absolute safety
Layered (defense-in-depth) reliability architecture	Enables redundancy and resilience through overlapping reliability mechanisms that reduce the likelihood of undetected or cascading failures
Modular framework design	Enables component-level replacement with maintained system architecture, clear responsibility boundaries, and extensibility across models, metrics, domains, and deployment contexts

Lifecycle-aware reliability management	Enables continuous reliability monitoring, adaptation, and post-deployment assessment across the AI lifecycle
Human-centered reliability integration (HITL-driven)	Enables structured human oversight, intervention, and feedback triggered by reliability signals
Reliability-aware uncertainty communication as an explicit system function	Enables transparent, actionable communication of uncertainty to support calibrated trust and informed human decision-making
Context-aware reliability assessment (X-space framing)	Enables adaptation of reliability evaluation and thresholds to task-, domain-, and situation-specific risk contexts
Integrated trustworthiness-enabling architecture	Enables coordinated treatment of technical reliability, human trust, and governance requirements within a single system architecture
Explicit linkage between reliability, trustworthiness, and viability	Enables reasoning about long-term sustainability and responsible deployment of AI systems beyond performance-only evaluation criteria

6.4 Closing Remarks

This thesis argues for a shift in how reliability is understood and operationalized in safety-critical AI systems. Rather than treating reliability as an emergent by-product of predictive performance, the work positions it as a crucial system property that must be explicitly designed, assessed, and communicated throughout the AI lifecycle.

By framing successful AI deployment around predictive performance, system-level trustworthiness, and end-user perceived trust, the thesis highlights how failures can occur even when individual models appear statistically strong. The RAF responds to this challenge by providing a structured way to integrate uncertainty awareness, human oversight, and transparency into AI system design, supporting risk-aware operation even under imperfect or evolving conditions.

While the empirical case study was limited in scope and did not evaluate downstream safety outcomes or user trust directly, the framework establishes a principled foundation for future implementations and evaluations. As such, this work contributes a conceptual and architectural basis for developing AI systems that are not only performant, but also responsible, accountable, transparent, aligned with human needs, and appropriately trusted and used.

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8 Appendix

8.1 Glossary of Key Terms and Concepts

Action (reliability-based)	In this thesis, a predefined system response (e.g., masking, flagging, or escalation) initiated when a reliability score falls below a threshold, indicating elevated uncertainty or risk. Designed to mitigate risk by limiting automation, alerting users, or transferring control to human operators.
Adversarial training	A robustness-enhancing training technique in which deliberately perturbed or misleading inputs (adversarial examples) are included during training to reduce model sensitivity to subtle input manipulations. Contributes to system reliability within the RAF by mitigating vulnerability to adversarial or unexpected inputs.
AI system lifecycle	The end-to-end process of development, deployment, operation, monitoring, and iterative updating of an AI system, including feedback loops that support continuous adaptation and reliability over time.
AI winter	Periods of reduced funding, public interest, and research in AI, historically resulting from unmet expectations and overpromised capabilities. Sustained public and institutional trust, this work argues, depends on maintaining focus across all three pivot points: system performance, system trustworthiness, and end-user trust.
Auditability	The capacity of an AI system to record, trace, and reconstruct its processes, decisions, and outcomes through systematic logging and documentation. Auditability enables external verification, regulatory compliance, and accountability by allowing system behavior to be examined post hoc.
BNNs (Bayesian Neural Networks)	A class of neural networks in which model parameters are represented as probability distributions rather than fixed values, enabling estimation of uncertainty through Bayesian inference. BNNs provide well-founded uncertainty estimates but are also often computationally expensive and difficult to scale.
Calibrated trust	The alignment between a user's level of trust in an AI system and the system's actual reliability and limitations. Calibrated trust is achieved when users rely on the system appropriately—neither over-trusting unreliable outputs nor under-utilizing reliable capabilities. In the RAF, calibrated trust is supported through transparent reliability signals, uncertainty communication, and HITL mechanisms that help users form accurate mental models of system behavior.

Composite reliability score (RS_{total})	A reliability indicator in the RAF formed by aggregating multiple complementary reliability signals into one combined measure, and it functions as the operational trigger for reliability-based actions. The aggregation method in this thesis uses a weighted sum of the individual metrics—part of the multi-metric design regime of the RAF—but can be adjusted per domain- or context requirements.
Concept drift	A change over time in the relationship between input features and the target variable. As real-world conditions evolve, patterns learned during training may no longer hold, even if the input data distribution remains stable. Concept drift can lead to systematic performance degradation and must be detected and mitigated to maintain long-term system reliability.
Distributional shift	A mismatch between the input data distribution encountered during model training and that observed during deployment. Distributional shift may arise from environmental changes, sensor degradation, or operational context shifts, and can cause unreliable predictions even when the underlying task remains unchanged. Simply put, distributional shift changes the data, whereas concept drift changes the problem.
Domain adaptation	A family of techniques that enable AI models trained in one data domain to maintain performance when applied to a related domain with different statistical characteristics. Methods include fine-tuning, feature alignment, and adversarial learning, and are commonly used to mitigate performance degradation under distributional shift. Within the RAF, domain adaptation supports robustness and cross-context reliability.
End-user perceived trust	The degree to which users subjectively judge an AI system to be reliable, understandable, and appropriate to rely on in a given context. In this thesis, end-user trust is shaped by transparency, uncertainty communication, explainability, and the presence of meaningful human oversight. It is treated as a distinct but interdependent dimension alongside predictive performance and system trustworthiness, and as a necessary condition for viable real-world deployment. Within this thesis, it is deliberately separated from the concept of trustworthiness.
Ensembling	A machine learning technique in which multiple independently trained models are combined to produce a single prediction, improving robustness, generalization, and uncertainty estimation. By aggregating outputs across diverse models (e.g., via averaging or voting), ensembling reduces sensitivity to individual

	model errors or biases. Disagreement between ensemble members can be used as an indicator of epistemic uncertainty, which is operationalized as the ensemble-based reliability score ($RS_{ensemble}$) in the experimental evaluation in this thesis.
Epistemic uncertainty	Uncertainty arising from limited knowledge about the model or the data-generating process, typically due to insufficient, biased, or unrepresentative training data. Epistemic uncertainty reflects what the model does not know and can, in principle, be reduced through additional data, improved modeling, or architectural changes. In this thesis, it is a primary target of reliability assessment and is, in the experimental evaluation, approximated using ensemble disagreement, MCD variance, and latent-space divergence.
Escalation	A reliability-based action in which responsibility for decision-making is transferred from the AI system to a human operator when predictions fall below defined reliability thresholds or anomalous behavior is detected. In the RAF, escalation suspends or defers autonomous actions and routes flagged outputs to human review.
Fail-acceptable behavior	A thesis-specific system design principle for safety-critical AI systems in which uncertainty, degradation, or partial failure is treated as expected, and system behavior is engineered to default to conservative and controlled outcomes (e.g., withholding outputs, triggering fallback modes, or escalating to HITL review), rather than allowing unqualified automated behavior. Rather than signaling absolute safety (as implied by "fail-safe"), fail-acceptable behavior aims to ensure that outcomes remain acceptable to all relevant stakeholders, while acknowledging the probabilistic nature of AI.
Failure mode	The characteristic ways in which a system can fail, degrade, or produce incorrect or unsafe outcomes. In AI systems, failure modes may include phenomena such as distributional shift, calibration loss, overconfidence, or silent failure, where errors occur without clear indication. Identifying and analyzing failure modes is central to reliability engineering, as it enables the design of layered defenses that reduce the impact of individual weaknesses. Within the RAF, multiple failure modes are addressed through modular, multi-layered reliability assessment and control mechanisms.
Feedback loop	A system-level mechanism through which information about model performance, reliability, uncertainty, or user interaction is

	continuously fed back into earlier stages of the AI lifecycle or architecture. In the RAF, feedback loops enable ongoing adaptation and learning after deployment, supporting long-term reliability and preventing degradation due to distributional shift or changing operational contexts.
Flagging	A reliability-based action in which predictions or system states are explicitly marked as uncertain, low-confidence, or potentially unsafe based on computed reliability metrics. Flagging is non-blocking: the output remains visible to users but is accompanied by explicit warnings (e.g., visual indicators, alerts, or severity cues) to support informed human judgement. Within the RAF, flagging may be rendered with varying levels of urgency (e.g., "red-flagged" cases) and operates alongside masking and escalation as part of reliability-aware system behavior.
HCAI (human-centered AI)	A design and development approach that places human values, capabilities, and oversight at the center of AI systems across their lifecycle. HCAI emphasizes transparency, accountability, meaningful human control, and alignment with user needs. In this thesis, HCAI principles inform the RAF's emphasis on explainability, HITL mechanisms, and calibrated trust, ensuring that reliability and trustworthiness extends beyond technical performance to real-world human use.
HITL (Human-in-the-loop)	A system design paradigm in which human judgement and intervention are explicitly integrated into AI decision processes. HITL mechanisms allow authorized users to review, confirm, or override automated outputs, providing safeguards against model error and unintended consequences. HITL can be operationalized through reliability-based escalation, operator alerts, and manual override pathways, supporting accountability and safe operation.
Hold-for-review pathway	A reliability-based action in which the system temporarily suspends automated use of a prediction until it has been reviewed by a human operator. In the RAF, this mechanism supports fail-acceptable behavior and meaningful human oversight by preventing uncertain outputs from propagating into downstream processes.
Layered framework architecture	The structural organization of the RAF into five distinct layers—data, model, system, interaction, and deployment—each responsible for different aspects or reliability, oversight, and lifecycle management. This architectural separation clarifies responsibilities and supports modular design. It is, however, distinct from the defense-

	in-depth notion of multi-layeredness used elsewhere in the thesis.
Masking	A reliability-based action in which AI predictions that fall below a defined reliability threshold are selectively suppressed or withheld from automated downstream processed. Masked predictions are prevented from influencing system behavior, thereby containing uncertain or potentially unsafe outputs. Within the RAF, masking operates alongside flagging, hold-for-review, and escalation as part of the system's reliability-based action logic.
Model aging	The gradual degradation of a model's performance or reliability over time due to changes in data distributions, user behavior, or environmental conditions. If unmanaged, model aging may result in reduced performance, miscalibration, or increased failure risk. Within the RAF, model aging is addressed through continuous monitoring, feedback loops, and corrective actions such as retraining, recalibration, or adjustment of reliability thresholds.
Model confidence	A quantitative estimate produced by an AI model indicating the strength or certainty of a specific prediction, typically expressed as a probability or score. Model confidence reflects the model's internal assessment of likelihood, but does not by itself guarantee correctness or reliability as it may be miscalibrated or misleading under distributional shift or uncertainty.
Modularity	A system design principle in which a framework is composed of discrete, interchangeable components with clearly defined responsibilities and interfaces. In the RAF, modularity enables flexibility, extensibility, and resilience by allowing individual components—such as reliability scoring modules, data validation processes, or user interaction mechanisms—to be updated, replaced, or reconfigured without disrupting the overall system. This supports long-term adaptability to new methods, evolving regulatory requirements, and diverse deployment contexts.
Multi-layeredness	A defense-in-depth design principle in which reliability and safety are supported through multiple, overlapping layers of protection, each addressing different sources of uncertainty, failure, or risk. In this thesis, multi-layeredness refers to the integration of complementary reliability mechanisms (for example the three different uncertainty assessment techniques that form the composite reliability score) that together form redundant safety barriers inspired

	by the Swiss cheese model of accident prevention. This approach ensures that weaknesses in any single mechanism are compensated by others, substantially reducing the likelihood of undetected or catastrophic system failure.
Multi-metric reliability assessment	A reliability evaluation strategy that combines multiple quantitative metrics to capture different dimensions of uncertainty and failure risk. Rather than relying on a single confidence or uncertainty measure, the RAF integrates three signals—such as ensemble disagreement, latent-space divergence, and MCD variance—to reduce metric-specific blind spots and improve robustness.
OOD detection (Out-of-distribution detection)	Techniques for identifying inputs that differ substantially from the data distribution encountered during model training. OOD detection helps mitigate unreliable predictions by flagging samples for which model behavior is uncertain or unvalidated.
Pivot points	A thesis-specific conceptual construct describing three critical dimensions that jointly determine whether an AI system can be considered viable for real-world deployment: (1) predictive performance, (2) system trustworthiness, and (3) end-user perceived trust. These dimensions function as evaluative "gates", all of which must be sufficiently satisfied. The RAF is specifically designed to support and operationalize progress across all three pivot points.
Predictive performance	A broad term referring to how well an AI system performs its intended predictive task, typically evaluated using statistical metrics such as accuracy, precision, recall, or F1-score. In this thesis, predictive performance is treated as a necessary but insufficient condition for real-world AI deployment: strong predictive metrics alone do not guarantee system trustworthiness, robustness under uncertainty, or appropriate user trust. Predictive performance constitutes one of three pivot points that must be jointly satisfied for viable AI systems.
Probability calibration	The alignment between a model's predicted probabilities and the true empirical frequencies of outcomes. A well-calibrated model that predicts 70% confidence is correct approximately 70% of the time. Calibration techniques—such as temperature scaling or isotonic regression—aim to correct systematic over- or underconfidence. Within the RAF, maintaining calibration is essential for ensuring that uncertainty estimates and reliability scores remain meaningful for decision-making and trust calibration.

RAF (Reliability Assessment Framework)	A modular, multi-layered framework proposed in this thesis for assessing and operationalizing AI reliability across the system lifecycle. The RAF is explicitly designed to address the three critical pivot points identified in this work—predictive performance, system trustworthiness, and end-user perceived trust—by structuring how uncertainty, risk, and system behavior are assessed, acted upon, and communicated. Rather than treating trustworthiness as a property that can be directly implemented, the framework positions it as an emergent condition arising from reliable technical model behavior, reliability-aware system behavior, and meaningful human-centered interaction and oversight. (See Chapter 3.)
Reliability (in this thesis)	The degree to which an AI system, assuming a minimum level of predictive validity, produces safe, stable, and robust outputs under both expected and unexpected operating conditions. In this thesis, reliability concerns the system's behavioral stability, robustness to noise and distributional shifts, failure characteristics, and its capacity to detect, signal, and respond when its own predictions may be unreliable. Reliability serves as the technical foundation on which system trustworthiness and end-user trust are built within the RAF, but is not sufficient on its own to guarantee either.
Reliability assessment	The systematic evaluation of an AI system's ability to produce safe and robust outputs, including the detection, quantification, and communication of unreliability. In the RAF, reliability assessment is not treated as a purely diagnostic activity, but as an enabling mechanism for reliability-aware system actions—such as thresholding, masking, flagging, and escalation—that supports safe operational decision-making, human oversight, and lifecycle management beyond standalone performance reporting.
Reliability-aware system	An AI system designed to explicitly assess, monitor, and act on uncertainty and failure risk during operation, rather than assuming reliability from performance alone. In the RAF, this is supported by reliability assessment mechanisms, and can lead to reliability-aware decisions and actions during operation.
Reliability score	A set of complementary quantitative signals used within the RAF to represent the assessed reliability of model outputs or system behavior. Reliability scores aggregate one or more uncertainty- or stability-related measures and serve as the primary inputs to reliability-aware system actions. Importantly, the RAF is metric-

	agnostic: while three specific reliability scores are used in the empirical case study of this thesis (e.g., $RS_{ensemble}$, RS_{latent}, and RS_{MCD}), these represent one possible implementation rather than a fixed or mandatory set of metrics. The role of reliability scores within the RAF is to support operational decision-making and human oversight, not standalone performance reporting.
Reliable vs. unreliable prediction	A distinction used in this thesis to characterize AI model outputs based on their assessed reliability rather than solely on correctness. A reliable prediction is one for which the system has sufficient evidence—based on reliability scores, uncertainty estimates, and contextual checks—that the output can be safely acted upon or presented without additional safeguards. An unreliable prediction is one for which uncertainty, instability, or contextual risks exceed acceptable thresholds, or where insufficient reliability information is available, triggering reliability-based actions. Importantly, reliability is treated as a property of the system's confidence and operating conditions, not a guarantee of correctness—a prediction may be correct but unreliable, or incorrect but assessed as reliable under limited information.
Robustness	The ability of an AI system to maintain stable, safe, and predictable behavior under variability, uncertainty, and partial failure. In this thesis, robustness is treated as a system-level property that emerges from the coordinated interaction of robust models, uncertainty-aware reliability assessment, reliability-based actions, and human oversight. While model-level robustness-techniques—such as ensembling or adversarial training—contribute to robustness, they are not sufficient on their own. Within the RAF, robustness is achieved by ensuring that the system can detect when its predictions may be unreliable and respond with controlled, risk-aware behavior.
Safety-critical domain	An operational context in which failures of an AI system could directly or indirectly result in harm to human life, health, the environment, or significant economic or societal disruption. Examples include healthcare, aviation, transportation, and industrial automation, or large-scale information systems such as social media platforms. In such domains, reliability, transparency, and human oversight are critical design requirements. The RAF is developed with these high-stakes contexts in mind, emphasizing risk mitigation and fail-acceptable system behavior.

Silent failure	A failure mode in which an AI system produces incorrect, misleading, or unsafe outputs without emitting warnings or uncertainty signals to users or downstream systems. Silent failures can be particularly consequential because they can propagate unnoticed and thereby compound over time. The RAF explicitly targets the prevention of silent failures through layered reliability assessment, reliability-based flagging, and HITL escalation mechanisms.
Swiss cheese model (of accident causation)	A safety engineering concept describing how accidents or failures occur when multiple layers of defense—each containing inherent weaknesses, or "holes"—align, allowing a hazard to pass through all safeguards. In the RAF, the Swiss cheese model underpins the RAF's multi-layered reliability philosophy—by combining diverse and partially independent reliability mechanisms, the likelihood of undetected or catastrophic failure is reduced.
Thresholding	The process of applying predefined or adaptive cut-off values to reliability metrics in order to determine whether model predictions will be accepted, masked, flagged, or escalated for review. Within the RAF, thresholding operationalizes reliability scores by translating quantitative uncertainty into concrete safety actions. This mechanism ensures that predictions falling below acceptable reliability levels are withheld or routed for human inspection, supporting fail-acceptable system behavior.
Trustworthiness (in this thesis)	An emergent system-level property describing the extent to which an AI system can be relied on to operate safely and appropriately in real-world contexts. In this thesis, trustworthiness is a primary objective of the RAF, but not a property that can be directly implemented or guaranteed. Within the RAF, trustworthiness emerges through the successful interaction of three foundational elements (predictive performance is an assumed, but separate factor for successful systems): (1) assessment and monitored reliability, (2) implemented transparency and human-centered interaction mechanisms, and (3) designed governance and accountability structures. Reliability assessment provides the technical foundation, transparency enables meaningful human oversight, and governance constrains system behavior within acceptable limits. While trustworthiness is necessary for safe deployment, it is not sufficient on its own for real-world viability. End-user perceived trust constitutes a distinct and complementary

	requirement, together with trustworthiness forming the conditions under which AI systems become viable, effective, and socially acceptable in practice.
Uncertainty communication	The process of conveying model uncertainty and reliability information to human stakeholders in a clear, interpretable, and actionable manner. Rather than exposing raw probabilistic outputs, uncertainty communication translates technical estimates into meaningful signals such as warnings, reliability flags, visual indicators, or explanatory messages. Within the RAF, uncertainty communication is a core human-centered mechanism that supports calibrated trust, informed decision-making, and effective human oversight.
Uncertainty quantification	The set of methods used to estimate how reliable or uncertain an AI system's predictions are. In the RAF, uncertainty quantification is primarily realized through epistemic uncertainty indicators for operational reliability assessment, which reflect model uncertainty and lack of knowledge. These quantitative estimates form the technical foundation of the RAF's reliability scoring mechanism, enabling the system to detect when predictions may be unreliable and to trigger appropriate reliability-based actions. While aleatoric (data-driven) uncertainty assessment is compatible with the framework, the mechanisms implemented to evaluate the experimental use-case in this thesis focus on epistemic uncertainty.
Viable AI system	An AI system that can be responsibly deployed, adopted, and sustained in real-world operation without unacceptable technical, human, or societal risk. In this thesis, viability is not equated with short-term performance or commercial success, but with the system's ability to function reliably and acceptably over time across multiple stakeholders. A viable AI system is one that jointly satisfies three necessary conditions (the pivot points identified in this thesis): (1) Sufficient predictive performance for its intended task, (2) System-level trustworthiness, grounded in reliability assessment, transparency, and governance mechanisms, and (3) Calibrated end-user perceived trust, enabling appropriate use, oversight and feedback, which may also translate to actual real-world use of the system.

	Viability is therefore treated as an emergent, system-level property arising from the interaction of technical robustness, human-centered design, and operational safeguards. The RAF is explicitly designed to support the development of viable AI systems by addressing these conditions within a single, integrated framework.
XAI (eXplainable AI)	A collection of methods and design principles aimed at making AI system behavior interpretable to human stakeholders. In this thesis, XAI is treated as a supporting mechanism rather than a standalone objective. Within the RAF, XAI techniques (e.g., SHAP, LIME, saliency mapping) are supported as optional and compatible components of the framework, enabling transparency, auditability, and human oversight of system behavior and reliability assessments. XAI therefore contributes to system trustworthiness and end-user trust, without replacing reliability assessment as the primary operational mechanism of the RAF.
X-space	A conceptual analysis space used in this thesis, referring to the full set of interrelated contextual domains that condition the planning, development, deployment, and real-world viability of an AI system. X-space encompasses data characteristics and availability, model assumptions, operational and deployment conditions, organizational constraints, and user-centered aspects. Mapping the X-space is treated as a prerequisite for meaningful system design and reliability assessment, as performance, reliability, and trustworthiness can only be evaluated relative to the conditions under which a system is intended to operate. Without such contextual grounding, downstream reliability metrics and safety mechanisms risk being misaligned, misleading, or operationally invalid.

8.2 Supporting Figures

Figure 8.1 *Baseline VAE Confusion Matrix (Per-Beat Detection)*

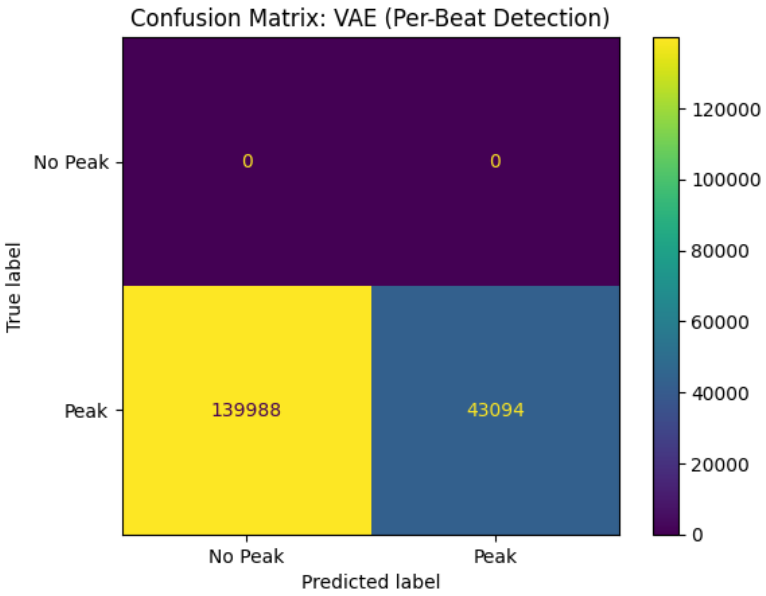

Figure 8.2 *Baseline DAE Confusion Matrix (Per-Beat Detection)*

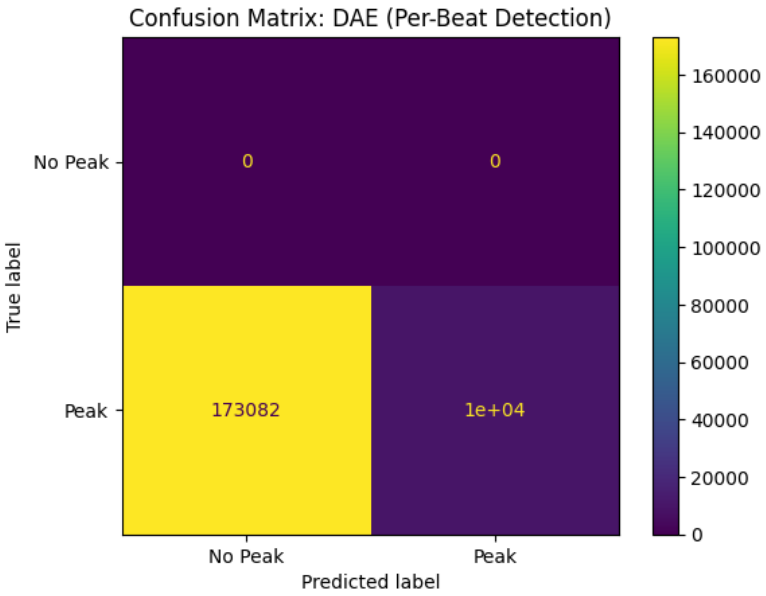

Figure 8.3 *Baseline Distribution of Detected Heartbeats (per Segment)*

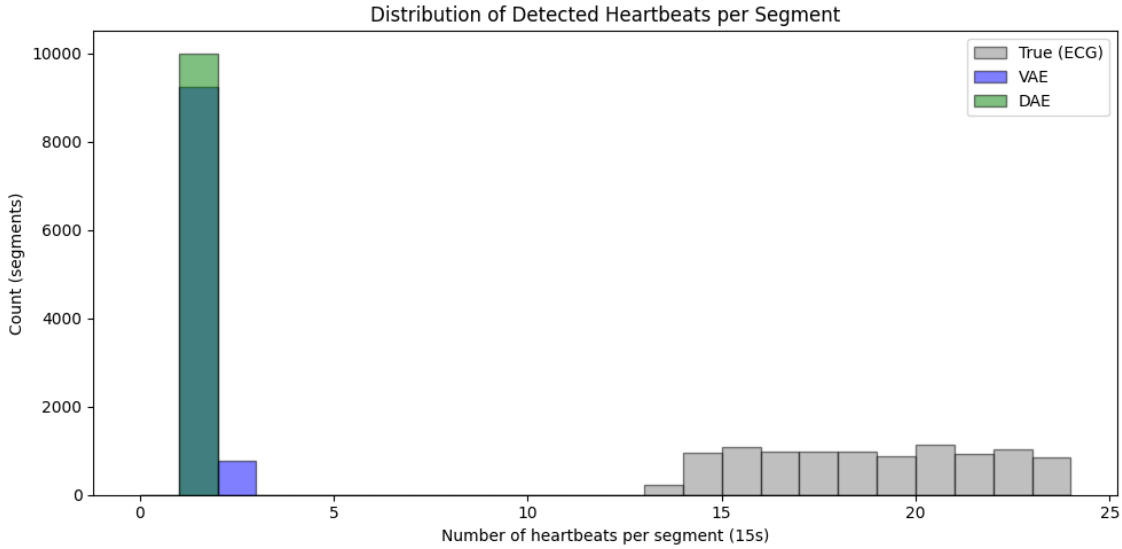

Figure 8.4 *Baseline Key Heartbeat Detection Metrics (Test Set)*

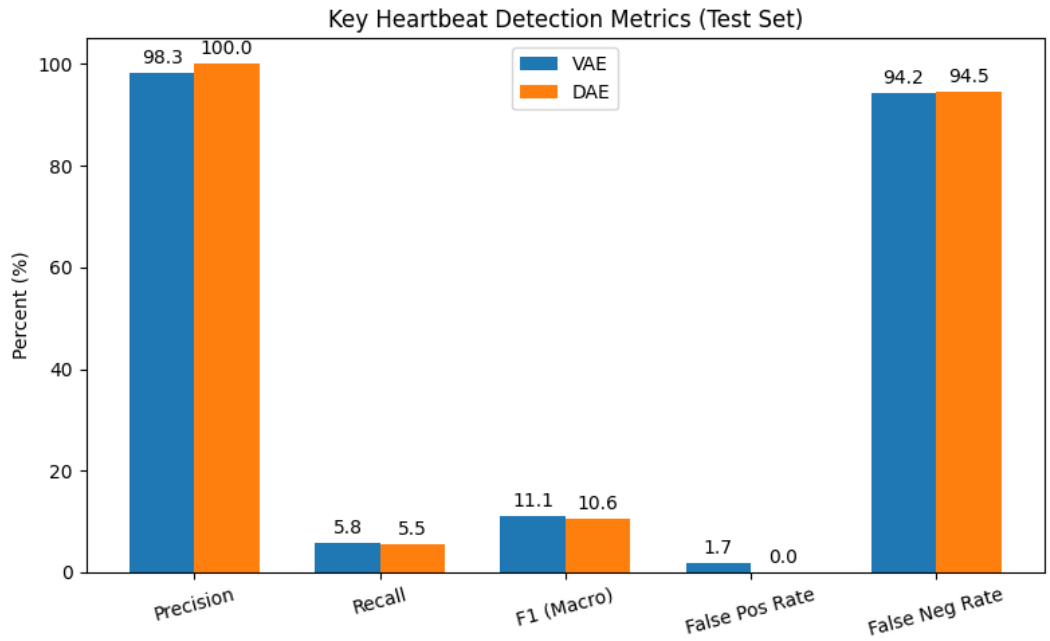

Figure 8.5 *Baseline Precision Recall Curve (Test Set)*

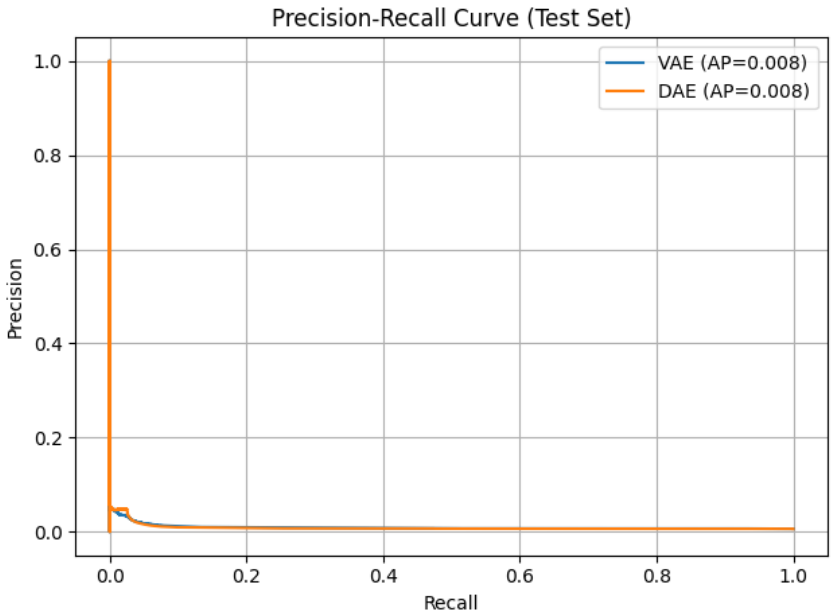

Figure 8.6 *Baseline ROC Curve on Test Set*

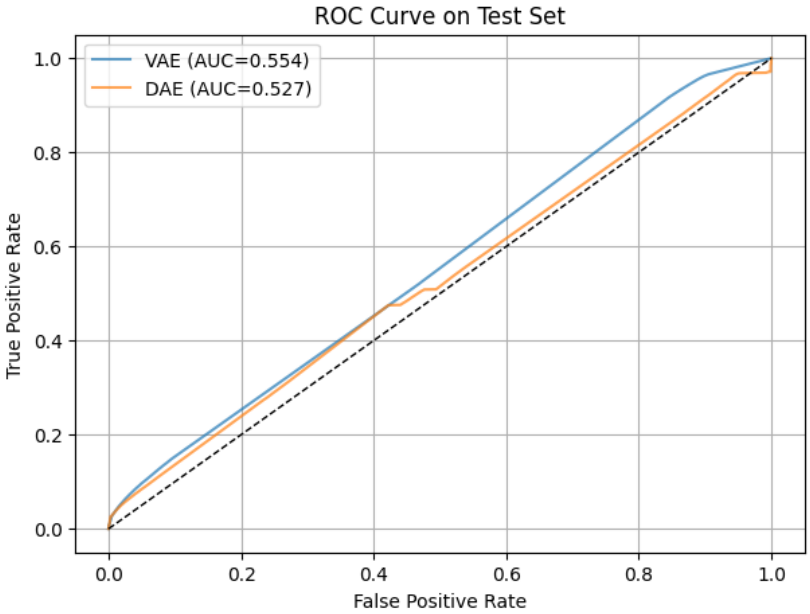

Figure 8.7 *Ensemble Confusion Matrix*

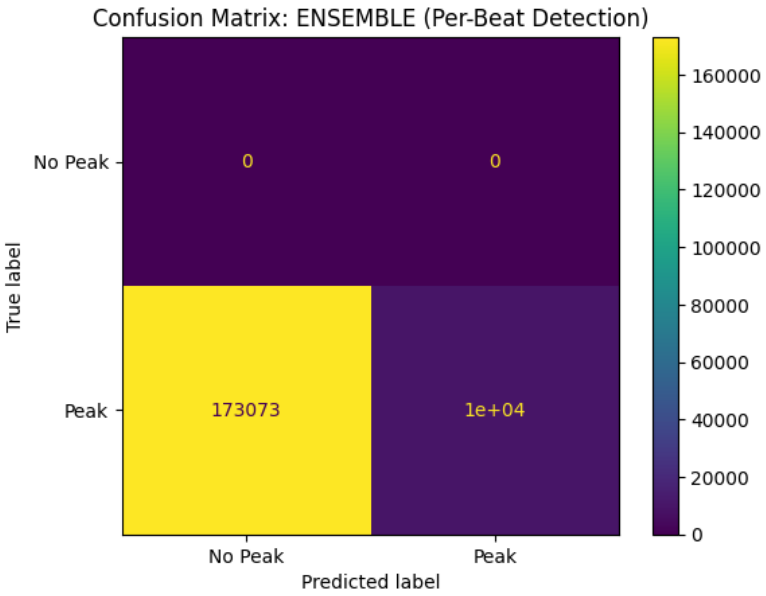

Figure 8.8 *Ensemble ROC Curve on Test Set*

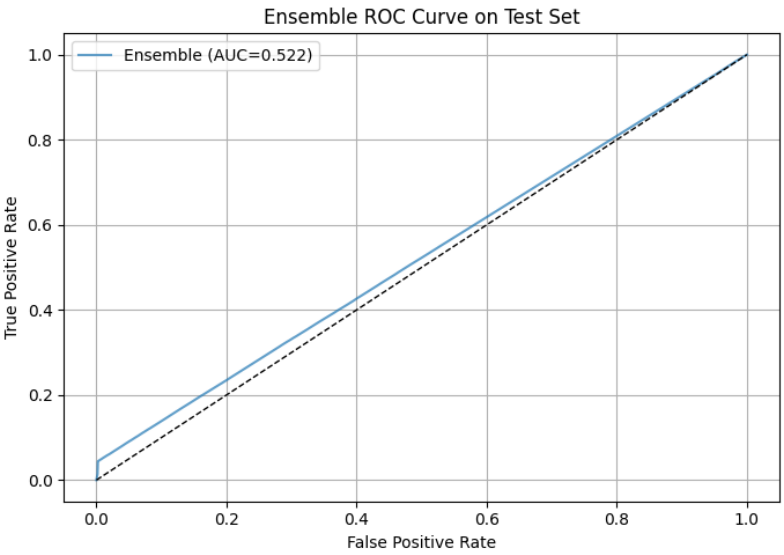

Figure 8.9 *Ensemble Key Heartbeat Detection Metrics*

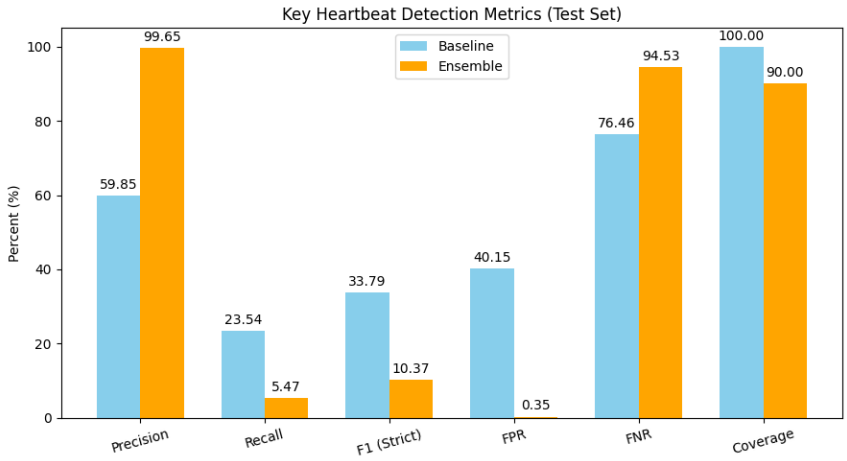

Figure 8.10 *Ensemble vs Baseline ROC / AUC*

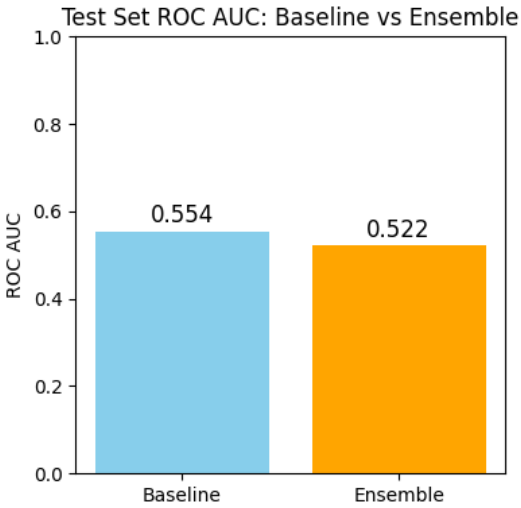

Figure 8.11 *Ensemble Peak Counts per Signal*

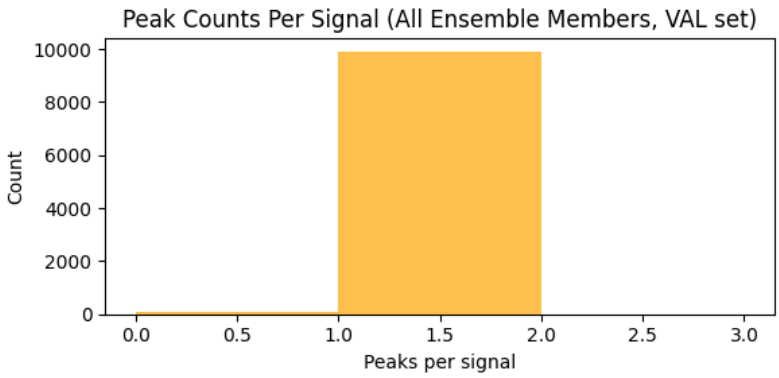

Figure 8.12 *Normalized Latent KL Reliability (Test Set)*

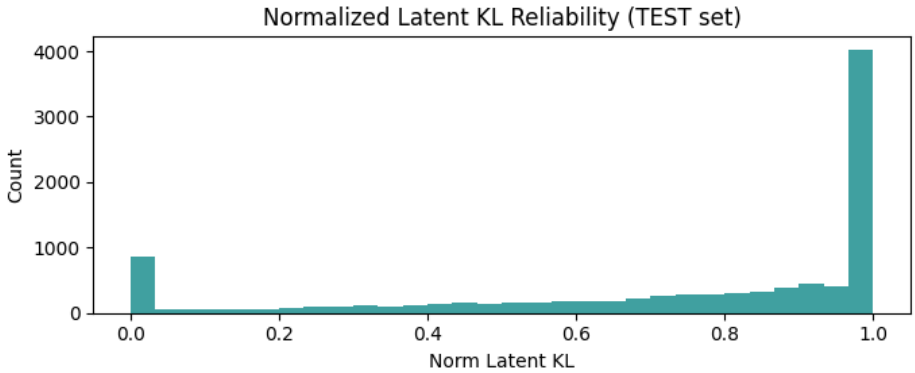

Figure 8.13 *Composite Reliability Score*

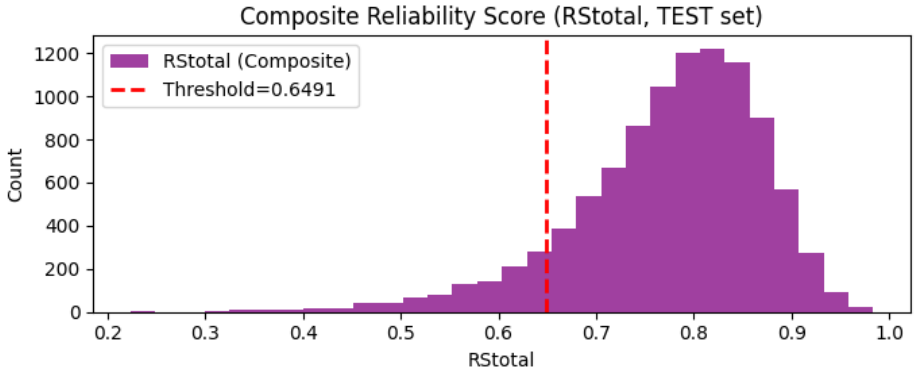

Figure 8.14 *Normalized Ensemble STD Reliability (Test Set)*

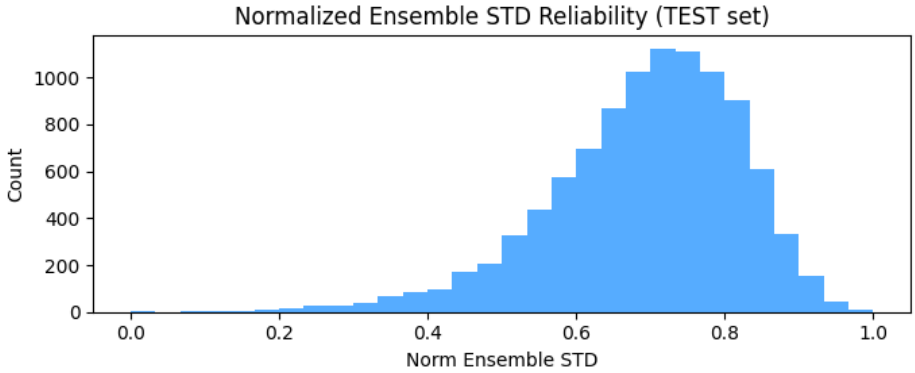

Figure 8.15 *Normalized MCD Reliability (Test Set)*

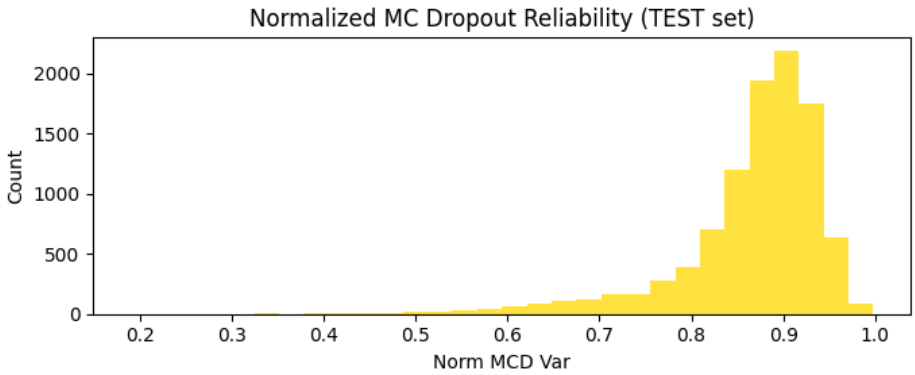

Figure 8.16 *Term Frequency Visualization*

Note. A qualitative overview of dominant concepts recurring throughout the thesis, reflecting the interdisciplinary nature of reliability assessment across technical and human-centered dimensions.

8.3 Supplementary Code Snippets***

Placeholder***.

8.4 Extended Results / Tables***

Placeholder***.